Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kwale County.

Draft Pongwe Kikoneni Ward Adaptation Plan

Consultant

Envasses Environmental Consultants Limited,
Nyerere Avenue, P.O. Box 2013-80100, Mombasa
Tel: +254 722347155; Email: info@envasses.org
Website:www.envasses.org

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1 Chapter 1: Background Information

1.1 Introduction

Plan international Kenya is implementing the Conservation and Sustainable Management of Marine and Coastal Ecosystems (COSME) Project in Kwale and Kilifi Counties. The two counties are endowed with rich coastal and marine ecosystems ranging from terrestrial forests, mangroves, seagrass beds, coral reefs, marine species and open sea marine resources among others. These natural assets not only underpin livelihoods but also drive key economic sectors including tourism, agriculture, fisheries, and maritime trade. Despite their economic and ecological value, the 2021 country and county climate risk profiles highlight that these resources are increasingly under threat from climate change impacts such as drought, flooding, sea level rise, ocean acidification, rising temperatures, and forest fires. The consequences of these risks exacerbate underlying socio-economic vulnerabilities, with women, youth, and Persons Abled Differently (PAD) experiencing disproportionate impacts due to structural inequalities. Loss of income, displacement, reduced productivity, and damage to infrastructure and markets are already affecting community well-being and resilience in the region.

In response, the COSME project aims to enhance women-led, climate-resilient livelihoods through ecosystem-based approaches. Hence, to support project implementation requires the development of Gender Responsive Adaptation and Preparedness Plans (GRAPPs) that integrate climate risk diagnostics with locally appropriate adaptation pathways. The Pongwe Kikoneni Ward Climate Change Adaptation Plan (WAP) is therefore prepared as part of the GRAPPs outputs. The ward which measures 153.6 km^2 is located in Kwale County at Kenya's southern coast with a population of 29,616 (14,729 male & 14,887 female) (KNBS, 2019) (Figure 1), faces a high vulnerability to climate change impacts. The ward's The climate is a tropical coastal characterized by bimodal rainfall and warm temperatures year-round. It's economy and livelihoods depend heavily on climate-sensitive sectors such as rain-fed agriculture, livestock, and coastal resources (Figure 1). However, recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode development gains and degrade critical natural resources upon which local livelihoods are dependent, making adaptation a critical priority for the ward.

Recurring droughts, erratic rainfall patterns, floods, and other climate-related stresses have already affected community well-being and local development. Climate change threatens to erode hard-won development gains, making adaptation a critical priority for the ward. In response, Kenya's policy framework has evolved to support local climate action, from the National Climate Change Response Strategy and Action Plans to the Climate Change Act 2023, which emphasizes devolved climate governance and resilience building at county and community levels. This Ward Adaptation Plan aligns with national and county climate change policies by mainstreaming adaptation into local planning and empowering the community to implement priority resilience measures.

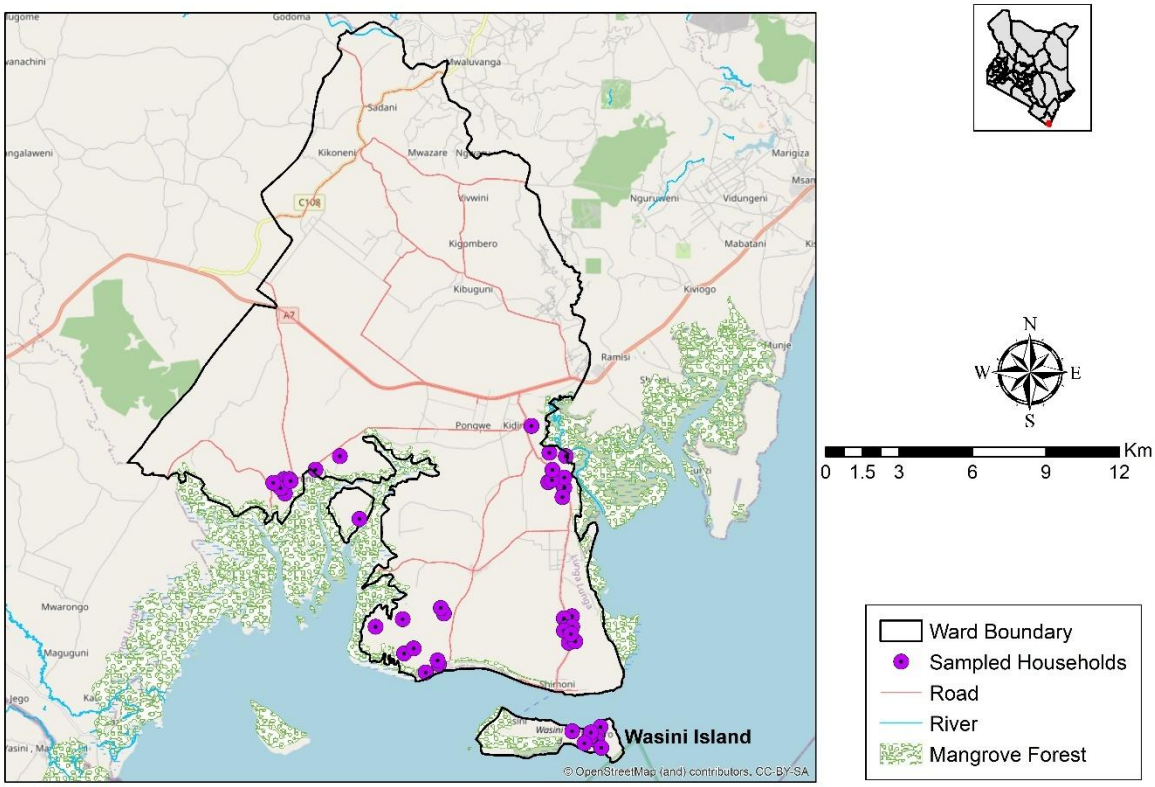

Figure 1: Location of Pongwe Kikoneni Ward and survey sites (Source: PCVA, 2025)

1.2 Plan development approach

The Ward Adaptation Plan for Pongwe Kikoneni was developed in a participatory, gender-responsive and livelihoods-based vulnerability framework that conceptualized climate risk as an outcome of the interaction between exposure, sensitivity and adaptive capacity (Figure 2). Within this analytical framing, vulnerability was understood as the degree of climate-induced damage experienced by socio-ecological systems, while resilience denoted the system's ability to maintain or rapidly regain functionality following climatic shocks. This approach acknowledged that climate risks in Pongwe Kikoneni, characterized predominantly by recurrent droughts, seasonal flooding, strong winds, sea-level rise, pests and diseases and human–wildlife conflicts arising from proximity to River Ramisi, do not affect population groups uniformly. Women, men, youth, elderly persons and Persons Abled Differently (PADs) face differentiated risks due to disparities in livelihood assets, labor burdens, mobility, access to resources, intra-household power dynamics and decision-making autonomy.

The plan was therefore designed using a process that combined experiential community knowledge with quantitative datasets, spatial analysis, ecological diagnostics and climatic modelling. This integrated methodology enabled the plan to link climatic hazards to specific livelihood systems; rainfed agriculture, artisanal fishing, seaweed farming, mangrove harvesting, small-scale trade and casual labour and to examine how these systems are shaped by gender norms, social vulnerabilities and institutional arrangements at ward level. The resulting insights provided a nuanced understanding of how climate extremes intersect with poverty, food insecurity, ecosystem degradation, weak infrastructure and limited institutional support, thereby shaping the context within which adaptation strategies must be developed.

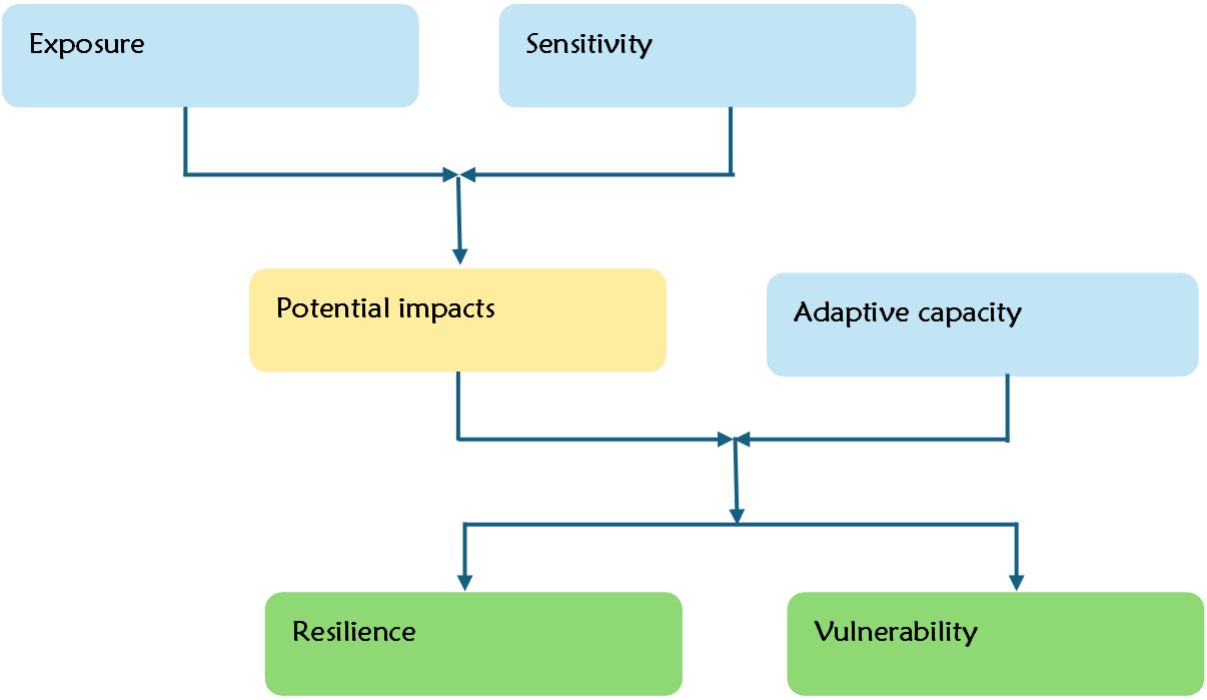

Figure 2: A model of factors that can be used to assess the vulnerability and resilience of systems (Source: Füssel and Klein, 2006, and Gauthier et al., 2014).

1.2.1 Inception Phase

The methodology was agreed upon through an inception phase that established a shared understanding among Plan International Kenya and the consultant on the objectives, scope and analytical expectations of the PCVA and WAP processes. An inception meeting held on 5th September 2025 brought together the COSME project teams to refine the methodological sequence, expected results, deliverable structure and timelines. During this phase, existing climate risk information for Kwale County was reviewed, data gaps specific to Pongwe Kikoneni Ward were identified, and a sequenced work plan was developed outlining participatory diagnostics, household surveys, spatial and climate analysis, and subsequent synthesis and validation activities.

Following institutional alignment, data collection instruments were developed and technically refined to ensure that they were methodologically sound, gender-responsive and culturally appropriate. The instruments included ward-level FGD guides segregated for women, men and youth; participatory mapping and historical timeline guides; seasonal calendar and daily clock templates; household decision-making pile-sorting tools; Leaky Bucket diagrams; impact chain protocols; vulnerability matrix templates; institutional Venn diagram formats; and adaptation pathway worksheets. Each tool integrated prompts on gender norms, intersectional vulnerability, unpaid care workloads, labour shifts during climate stress and disability inclusion.

Stakeholder mapping during the inception phase identified relevant actors within Pongwe Kikoneni Ward, including the Ward Climate Change Planning Committee, chiefs and assistant chiefs, village elders, leaders of women and youth groups, PAD representatives and support networks, seaweed farming groups, mangrove user groups, BMUs, WRUAs and community-based organizations. These actors facilitated community mobilization, venue preparation and dissemination of study information at ward level. The inception phase ensured that ward stakeholders were adequately oriented and positioned to meaningfully participate in the PCVA and adaptation planning process.

1.2.2 Participatory Climate Risk and Vulnerability Assessment (PCVA)

The Ward Adaptation Plan was informed by a comprehensive PCVA process implemented in five COSME target villages within Pongwe Kikoneni i.e., Anzuwani, Barabarani (Majoreni), Kibuyuni, Kiwambale and Mkwiro. The assessment followed a structured six-step participatory sequence designed to diagnose climate vulnerabilities, identify underlying drivers of sensitivity, assess existing coping and adaptation strategies, and translate diagnostic findings into actionable adaptation options suitable for inclusion in the WAP. The process applied the full suite of ten PCVA tools, each deployed to generate gender- and age-disaggregated evidence capable of capturing complex interactions between climate hazards, livelihoods, natural resources and social systems.

1.2.2.1 Step 1: Scoping

During scoping, the geographical scope of the ward was delineated using GIS shapefiles provided by Plan International and Kwale County Government. Within this boundary, the five COSME villages were purposively selected on the basis of ecological sensitivity, concentration of climate-sensitive livelihoods, recurrent exposure to droughts and floods, susceptibility to strong winds and sea-water intrusion, and documented human–wildlife conflict incidents, particularly crocodile attacks reported around Kiwambale near River Ramisi. Village profiles were then developed to summarize demographic characteristics, livelihood systems, existing infrastructure, ecosystem conditions, climate-related impacts, past interventions and vulnerability hotspots. These profiles served as the baseline data against which PCVA tools were deployed.

1.2.2.2 Step 2: Establishment of Core Participatory Analysis Teams

A ward-level core participatory analysis team was established for Pongwe–Kikoneni, drawing membership from Ward Climate Change Planning Committee (WCCPC) representatives, village elders, women leaders, youth leaders, PAD representatives and carers, NbS groups, GJJ groups, and other active CBOs and self-help groups. County technical officers from climate change, environment, forestry, agriculture, livestock and fisheries were engaged as resource persons.

The core team underwent a structured orientation on the PCVA framework, the ten PCVA tools, gender- and age-responsive facilitation, child safeguarding and ethical data collection. Training sessions used a combination of plenary presentations, group work and simulations of each PCVA tool, conducted primarily in Kiswahili to ensure comprehension. By the end of the training, the ward team had the capacity to co-facilitate hazard and resource mapping, historical timelines, seasonal calendars, daily clocks, household decision making and pile-sorting exercises, Leaky Bucket analyses, impact chain mapping, vulnerability matrix scoring, Venn diagrams and adaptation pathway development.

1.2.2.3 Step 3: Situation Analysis of Climate Vulnerability

The situation analysis integrated secondary datasets, primary field evidence and participatory analytics to generate a comprehensive picture of exposure, sensitivity and adaptive capacity across Pongwe Kikoneni Ward. This phase synthesised literature review, ward-level gender- and age-disaggregated Focus Group Discussions (FGDs), transect walks across the five COSME villages, and the structured application of PCVA Tools 1–7. Through triangulation, these methods produced a detailed, spatially nuanced and socially differentiated understanding of how climatic hazards interact with biophysical systems, livelihood structures and gendered social dynamics.

1.2.2.3.1 Literature Review

The literature review served as the technical backbone of the situation analysis, drawing from global, regional, national, county and historical ward-level evidence to contextualize observed

vulnerabilities. At the global scale, the review considered findings from IPCC AR6 Working Group I and II, particularly projections relevant to East African coastal systems, including accelerated warming rates, sea surface temperature rise, intensification of short-duration extreme rainfall, increased occurrence of coastal storms, saline intrusion risks and anticipated sea-level rise along the Western Indian Ocean. These findings were aligned with global adaptation and risk governance frameworks such as the Sendai Framework for Disaster Risk Reduction 2015–2030, the Paris Agreement adaptation cycle, and SDGs 2, 5, 6, 13 and 14, which collectively frame resilience building in coastal and marine socio-ecological systems.

At the regional scale, the review integrated IGAD Climate Prediction and Applications Centre (ICPAC) bulletins, the Africa Climate Policy Centre (ACPC) projections for Eastern Africa, and Western Indian Ocean Marine Science Association (WIOMSA) research on mangrove decline, seaweed vulnerability, fishery collapses and oceanographic shifts. This regional evidence positioned Pongwe Kikoneni within a broader coastal climate risk corridor characterised by rising temperatures, altered monsoon circulation, changing upwelling systems and degradation of nearshore ecosystems.

National climate policy and regulatory frameworks were examined comprehensively, including the Climate Change Act (2023), National Adaptation Plan (NAP 2015–2030), Kenya's Updated NDC (2023), the National Framework for Climate Services (NFCS), Kenya Climate-Smart Agriculture Strategy, National Disaster Management Policy, National Drought Early Warning System, and EMCA (2015) and sectoral regulations. These documents provided the lens for interpreting institutional mandates, priority hazards and adaptation responsibilities that cascade to county and ward levels.

At county scale, the review consolidated information from the Kwale County Integrated Development Plan (CIDP III, 2023–2027), the Kwale County Climate Change Act (2022), the County Climate Change Fund (CCCF) Regulations, the County Climate Change Action Plan (CCCAP 2023–2027), the County Spatial Plan, and sector strategies in agriculture, livestock, fisheries, water and coastal management. These county frameworks were triangulated with socio-economic and food security trends reported by the National Drought Management Authority (NDMA), historical rainfall data from the Kenya Meteorological Department (KMD), marine ecosystem studies from KMFRI, forestry and mangrove assessments by the Kenya Forest Service (KFS), fishery catch records from the State Department for Fisheries, and demographic data from the Kenya Population and Housing Census (2019).

The review also incorporated previous PCVAs conducted under COSME and other coastal adaptation programmes, which provided comparative vulnerability patterns, gender-responsive insights, and past lessons on nature-based solutions, community-based early warning systems, and local adaptation practices. This multilevel synthesis created a scientifically grounded baseline against which primary data from Pongwe Kikoneni could be interpreted.

1.2.2.3.2 Focus Group Discussions (FGDs)

Ward-level FGDs were conducted with segregated groups of women, men and youth from the five COSME villages; Anzuwani, Barabarani (Majoreni), Kibuyuni, Kiwambale and Mkwiro, to elicit gendered and generational insights into climate trends, livelihood impacts and socio-ecological vulnerability (Figure 3). The discussions explored perceived changes in rainfall reliability, including delayed onset and inconsistent cessation of rains, prolonged intra-seasonal dry spells, unprecedented periods of elevated temperatures, shifts in wind patterns and increased intensity of episodic storms and coastal surges. Participants described the material impacts of these climatic variations on agricultural productivity, seaweed quality and harvest

cycles, artisanal fisheries (including reduced catch per unit effort), mangrove health, livestock mortality and food security.

The FGDs also highlighted the deeply gendered nature of climate vulnerability. Women described increased walking distances and time burdens for water and fuel collection during droughts; heightened caregiving demands during disease outbreaks; increased exposure to GBV during water scarcity periods; and disruptions in women's informal trading activities. Youth reported loss of wage labour opportunities during climate shocks, stress-related migration, and lack of access to climate-resilient livelihood assets. Men described reduced fishing days caused by strong winds, damaged boats during storms and increased anxiety over household provisioning responsibilities. In addition, participants highlighted PAD-specific vulnerabilities, including mobility limitations during floods and exclusion from labour-intensive coping strategies.

These discussions produced climate calendars, gender-disaggregated vulnerability matrices and hazard trend narratives that deepened the temporal and social dimensionality of the vulnerability assessment.

Figure 3: FGDs in Pongwe Kikoneni Ward (Source: Field surveys, 2025)

1.2.2.3.3 Transect Walks

Transect walks undertaken across the five villages provided on-ground spatial verification of vulnerabilities identified in FGDs and documented in literature. These walks traversed rainfed agricultural plots, irrigated horticultural fields, mangrove stands and adjacent intertidal zones, seaweed farming lines, fish landing sites, riparian zones of River Ramisi, boreholes and shallow wells, public institutions (schools, health dispensaries), access roads, market areas, and known flood plains.

Structured observation sheets and GPS-enabled devices were used to capture geospatial attributes of key vulnerability hotspots. Evidence recorded included deterioration of feeder roads due to repeated flooding, heavily eroded gullies formed by runoff from bare catchments, saline intrusion into agricultural fields and domestic wells, siltation of wetlands, collapsed seaweed lines following strong winds, degraded mangrove stands exhibiting canopy thinning

or sediment stress, and infrastructure damage such as washed-out culverts and submerged pathways.

These observations enhanced the physical realism of the vulnerability analysis and provided geo-referenced evidence that later strengthened hazard–resource maps and the identification of priority zones for ecological restoration and climate-resilient infrastructure.

1.2.2.3.4 Tool 1: Hazard and Resource Mapping

The hazard and resource mapping tool enabled community members to visually represent their environmental and socio-economic landscape (Figure 4). Through participatory sketching, communities identified settlement clusters, farmlands, grazing areas, mangrove forests, seaweed farms, rivers, swamps, markets, schools, health facilities, water points and religious centres. Hazard features; such as recurrent flood zones, storm-surge-prone coastal segments, areas with saline intrusion, strong-wind corridors and crocodile-risk stretches along River Ramisi, were marked and discussed. This mapping exercise generated spatially explicit knowledge that later informed adaptation targeting and zoning.

Figure 4: Hazard and resource mapping in Kiwambale Village, Pongwe Kikoneni Ward (Source: PCVA, 2025)

1.2.2.3.5 Tool 2: Historical Timelines

Historical timelines reconstructed multi-decadal climatic and socio-economic events including major droughts, destructive floods, storm events, pest invasions (armyworms, whiteflies and rice pests), fishery fluctuations linked to monsoon variability, severe seaweed farm losses from temperature anomalies and episodes of heightened human–wildlife conflict. Each event was discussed in terms of magnitude, social and economic impact, community responses and institutional support. Timelines revealed a pattern of intensifying climatic shocks, shortened recovery cycles and cumulative stress on natural resource systems.

1.2.2.3.6 Tool 3: Seasonal Calendars

Seasonal calendars detailed intra-annual cycles of rainfall, temperature, wind regimes, tidal patterns, agricultural operations, fishing seasons, seaweed planting and harvesting, mangrove harvesting, disease outbreaks and social events. Comparisons between historical and current seasonal behaviour revealed declining predictability of rainfall, increased occurrence of off-season rainfall, persistent dry spells and shifts in traditional fishing and seaweed farming seasons. Calendars also captured periods of heightened GBV, school absenteeism, food scarcity and increased care burdens.

1.2.2.3.7 Tool 4: Daily Clock

The daily clock exercise revealed gendered labour allocation and how climate stress reconfigures time use. Under climate-normal conditions, tasks such as farming, seaweed farming, childcare, water collection and marketing followed relatively stable patterns. Under climate stress, women and girls reported significant increases in unpaid care work and water collection time, men noted prolonged periods of income loss due to weather-inhibited fishing, and youth described increased domestic workload substitution and reduced learning hours. These insights demonstrated how climate shocks deepen time-poverty and reduce adaptive capacity.

1.2.2.3.8 Tool 5: Household Decision-Making Pile Sorting

Pile sorting exercises clarified intra-household power/authority structures under normal and climate-stress scenarios. Under normal conditions, decision-making on crop choices, livestock sales and minor household investments tended to be joint or gender-specific depending on cultural norms. Under climate stress, however, the decision space often shifted toward male dominance, particularly in decisions involving financial risk, fishing activity, asset disposal or climate-response investment, which in turn constrained women's adaptive agency and reinforced gender-based vulnerabilities.

1.2.2.3.9 Tool 6: Leaky Bucket

The Leaky Bucket tool illustrated how climate shocks disrupt household financial stability. Communities mapped income streams such as farming, fishing, seaweed sales, petty trade, remittances and casual labour, alongside expenditure flows including food purchases, health costs, water, school fees, debt servicing and transport. Climate shocks were shown to reduce inflows drastically while simultaneously raising outflows through rising food prices, repair costs and medical expenses. This widening income–expenditure gap demonstrated the structural economic vulnerability of households.

1.2.2.3.10 Tool 7: Impact Chains

Impact chain analysis depicted the cascading consequences (direct and indirect impacts) of priority hazards. For instance, prolonged drought triggered crop losses, water scarcity, livestock mortality and reduced milk intake, which in turn contributed to household food insecurity, malnutrition risk among children, increased labour migration by men, rising care burdens for women, heightened GBV risk and reduced school attendance. Similarly, strong winds disrupted fishing and seaweed farming, leading to economic losses, debt accumulation and psychological stress. This tool exposed multi-dimensional vulnerability pathways and supported the design of integrated adaptation strategies.

1.2.2.4 Step 4: Analysis of Causes of Vulnerability (Tool 8 – Vulnerability Matrix)

The analysis of underlying causes of vulnerability was undertaken through the application of the vulnerability matrix (Tool 8), which provided a structured means of examining how different climatic hazards intersect with livelihood assets and socio-ecological systems in Pongwe–Kikoneni Ward. Hazards such as drought, floods, strong winds, sea-level rise, human–wildlife conflict and agricultural pests and diseases were arranged against critical assets including

rain-fed farms, irrigated plots, mangrove forests, seaweed farms, artisanal fisheries, livestock, domestic water sources, housing structures, schools, health facilities, feeder roads and market infrastructure. Community members assigned severity ratings for each hazard–asset interaction, supported by qualitative reasoning and lived examples.

The cumulative scoring process produced a ranking that demonstrated, for example, the high susceptibility of mangrove areas and seaweed farms to combined effects of temperature stress, strong winds and changing salinity regimes; the acute vulnerability of rain-fed agriculture to prolonged drought and unpredictable rainfall; and the differentiated impacts of floods on schools and feeder roads which disrupt mobility, market access and educational continuity. This step bridged descriptive hazard analysis with deeper socio-economic and institutional determinants of vulnerability, forming the foundation for identifying priority adaptation pathways.

1.2.2.5 Step 5: Analysis of Community Action (Tool 9 – Venn Diagrams)

Analysis of existing coping and adaptation practices revealed a range of coping strategies and externally supported interventions. The Venn diagram exercise (Tool 9) enabled communities to map the institutional ecosystem influencing adaptation in the ward. Institutions such as county line departments, NGOs, community-based organisations, religious institutions, savings and credit groups, fisheries and forestry officers, COSME project structures, and private sector actors were placed on diagrams according to their perceived influence and accessibility. The spatial arrangement revealed a significant presence of institutions but varying degrees of functional support. While some organisations maintained close, supportive relationships, such as community groups collaborating with NbS and seaweed farmer associations, others operated at a distance, limiting their impact. This analysis also identified critical gaps, including insufficient extension services for seaweed and fisheries, weak linkages to financial services, limited climate information dissemination, and inadequate institutional frameworks for systematic PAD inclusion. These findings provided essential insights for designing governance and institutional strengthening elements within the Ward Adaptation Plan.

1.2.2.6 Step 6: Drawing Action from Analysis (Tool 10 – Adaptation Pathways)

The final step in the participatory diagnostic process involved translating identified vulnerabilities, exposure pathways and institutional gaps into a coherent adaptation strategy through the Adaptation Pathways tool (Tool 10). Communities revisited impact chains and vulnerability dynamics to identify feasible response strategies across short-, medium- and long-term horizons. The exercise involved structured deliberations that considered gendered labour implications, the availability of financial and natural resources, household and community capacities, and alignment with ongoing COSME and county initiatives.

Resulting adaptation pathways articulated sequenced actions for key socio-ecological systems; agriculture, water resources, mangrove and terrestrial forests, seaweed farming and fisheries, community infrastructure, early warning systems and social protection mechanisms. This included immediate resilience-building measures such as water harvesting, reinforcement of seaweed lines, community drainage clearance and household-level livelihood diversification; medium-term actions such as improved coastal protection, establishment of climate-resilient seaweed strains, expansion of irrigation infrastructure and strengthening local governance structures; and long-term interventions such as large-scale mangrove rehabilitation, institutionalization of community-based climate monitoring systems, and improvement of rural infrastructure. These pathways formed the backbone of the Pongwe–Kikoneni Ward Adaptation Plan, linking diagnostic insights to strategic action.

1.2.3 Household Survey

A structured household survey was administered to 295 households in the study area to provide quantitative evidence complementing qualitative findings. Sampling was designed to include households across diverse livelihood portfolios and different vulnerability categories, including female-headed households, large households with high dependency ratios, households with PAD members, and those primarily reliant on climate-sensitive livelihoods such as rain-fed agriculture, artisanal fishing and seaweed farming. Primary respondents were predominantly women (approximately 65 %), ensuring robust capture of intra-household dynamics, care burdens, access to services and lived experiences of climate stress.

The survey collected data on demographic structure, education, disability status, land tenure and resource access, livelihood contributions, income diversification, exposure to recent climatic shocks, losses and damages, coping strategies, food security indicators, institutional linkages and participation in social protection and savings groups. Data were collected digitally using KoboToolbox to ensure consistency and reduce transcription errors. These data provided a statistically grounded basis for validating PCVA findings and understanding patterns of vulnerability at household level across gender, age and livelihood categories.

1.2.4 Current Mangrove and Forest Health Assessment

To assess the condition of key coastal and terrestrial ecosystems supporting nature-based adaptation in Pongwe–Kikoneni, a remote sensing analysis was conducted using Sentinel-2 imagery at 10-metre spatial resolution. Imagery underwent standard pre-processing steps including radiometric correction and cloud masking before the Normalised Difference Vegetation Index (NDVI) was computed using the formula.

$$\text{NDVI} = \frac{NIR - RED}{NIR + RED} \qquad \text{Equation 1}$$

Where RED is visible red reflectance, and NIR is near infrared reflectance.

NDVI values were categorised into low, moderate and high vegetation vigor classes. Low NDVI signatures corresponded to degraded or stressed vegetation, often reflecting factors such as saltwater intrusion, wood extraction pressure, sedimentation and hydrological alteration. High NDVI zones served as ecological benchmarks, indicating healthier vegetation stands. These spatial patterns were subsequently compared with PCVA hazard–resource maps and field observations to identify priority sites for ecological restoration, mangrove replanting, mangrove hydrological rehabilitation and sustainable forest management within the ward.

1.2.5 Current and Future Climate Projections

Climate projections relevant to Pongwe–Kikoneni were developed using downscaled county-level analyses supplemented by remote sensing and historical observational datasets. Daily rainfall Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS), Actual Evapotranspiration (AET), Potential Evapotranspiration (PET), (from ERA5-Ag) and temperature variables (minimum, mean, maximum and Land Surface Temperature (LST) from Landsat and ERA5-Ag) were extracted through Google Earth Engine after spatial boundaries had been validated in QGIS. The resulting datasets were aggregated into annual totals or means and subsequently analysed using Python (Colab) employing pandas, numpy, scikit-learn and matplotlib.

Linear regression models were fitted to each climatic variable to estimate historical trends and project future changes to 2035 and 2045. Temperature projections incorporated Intergovernmental Panel on Climate Change (IPCC) AR6-aligned warming increments for East

Africa's coastal belt, applying approximately +0.02°C per year for the moderate-emission scenario and +0.035°C per year for high-emission trajectories. Rainfall projections followed empirically derived trends, showing increasing variability, changes in seasonal distribution and heightened likelihood of intense episodic rainfall events. These climatic insights were interpreted in relation to Pongwe–Kikoneni's hazard profile, reinforcing the need for enhanced water security, climate-resilient agricultural practices, improved coastal and marine ecosystem management, stronger infrastructure planning and expanded early warning systems.

1.2.6 Data Analysis and Reporting

Quantitative data from the household surveys were cleaned, coded and analyzed using Microsoft Excel and Python. Analysis focused on frequencies, proportions, cross-tabulations and simple correlations across gender, age group, livelihood type and vulnerability status. These results provided empirical confirmation of patterns emerging from qualitative insights.

Qualitative data from FGDs, PCVA exercises and transect walks were transcribed and thematically coded to organise evidence under key domains such as hazard exposure, livelihood impacts, gender dynamics, coping strategies, institutional barriers and opportunities. Outputs from PCVA tools; maps, timelines, calendars, clocks, matrices and impact chains, were digitized and integrated with NDVI layers and climate trend visualizations to produce synthesized analytical outputs. Triangulation across all sources ensured internal consistency and improved the robustness of the final interpretation. Draft findings were compiled for review by Plan International Kenya in line with expected deliverables.

1.2.7 Validation

The draft findings were presented in county-level validation workshops held in Kilifi and Kwale on 20th and 21st November 2025 respectively. These sessions provided a platform for county technical departments, CCPC/WCCPC members, civil society, COSME partners and community representatives to review the PCVA results and refine the draft adaptation strategies. At ward level, stakeholders from Pongwe Kikoneni reviewed hazard rankings, adaptation priorities and institutional roles, and provided further refinement on feasibility, gender inclusion, sequencing and resource implications. Inputs from these processes were incorporated into the final Ward Adaptation Plan to ensure scientific accuracy, community legitimacy and institutional alignment.

1.3 Linkages to other planning processes

At both the National, County and Ward Levels, several other planning processes exist to which the WAP is linked. These include the National Adaptation Plan 2015-2030, the Kwale County Adaptation Plan 2023-2032 and Pongwe Kikoneni Ward level Participatory Climate Risk Vulnerability Assessment (PCVA) conducted in 2023. The NAP is Kenya's first plan on adaptation and demonstrates the commitment to operationalize the National Climate Change Action Plan by mainstreaming adaptation across all sectors in the national planning, budgeting and implementation processes. The mainstreaming approach recognizes that climate change is a cross-cutting sustainable development issues with economic, social and environmental impacts. Similarly, the CAP aligns with the NAP in terms of content and reporting focusing on Kwale County circumstances, climate hazards and adaptation actions which are largely derived from PCVAs for the 20 wards. The WAP adaptation plan is prepared considering both the NAP and CAP and aligning adaptation actions accordingly as informed by the village focused PCVAs.

2 Chapter 2: Coordination of Adaptation Actions

2.1 Introduction

Effective coordination converts adaptation intent into measurable resilience outcomes. In Pongwe Kikoneni Ward, climate risks affect multiple sectors including coastal ecosystem (fisheries and mangrove), agriculture, health, tourism, livestock, water security, wildlife and their impacts are not evenly distributed. Women and other vulnerable groups often face the greatest challenges. To translate the Ward Adaptation Plan into practical results, a clear governance framework is needed. This framework must connect local priorities with county-level programs and national policies, while ensuring communities have timely access to finance, technical expertise, and relevant climate information. This chapter outlines that coordination framework and demonstrates how it aligns with Kenya's National Adaptation Plan (NAP) 2015–2030. The approach promotes both horizontal and vertical coordination: horizontally, by encouraging collaboration among village institutions, civil society, private sector actors, and technical officers within the ward; and vertically, by integrating ward priorities into county planning and budgeting processes while aligning them with NAP strategic actions. These include coastal zone management, sustainable fisheries livelihoods, water resource management, climate-smart agriculture, disaster risk reduction and early warning, health system resilience, and ecosystem restoration. Ultimately, the coordination mechanism establishes a coherent institutional structure and process through which roles and responsibilities are clearly defined, duplication and gaps are reduced, and gender-responsive, locally led adaptation initiatives are adequately resourced, implemented, and monitored over time.

2.2 Current Institutional Arrangements

Coordination presently operates across three interlinked tiers. Kenya's national framework for climate change adaptation and resilience is anchored in the Climate Change Act, 2023, which provides the legal basis for coordinated action across all levels of government. Policy direction, technical guidance, and coordination are provided by the National Climate Change Council and the Climate Change Directorate, while operational priorities are set through the National Adaptation Plan (2015–2030) and the National Climate Change Action Plan (2023–2027). Specialized agencies reinforce this architecture: the Kenya Meteorological Department provides forecasts and impact-based early warnings essential for agriculture, coastal safety, and disaster preparedness; the National Drought Management Authority leads drought monitoring, early warning, and response, including social protection triggers; the Coast Development Authority coordinates integrated coastal zone development and sustainable management of marine and coastal resources; the Kenya Forest Service and Kenya Wildlife Service manage ecosystem restoration, forest conservation, and human–wildlife conflict mitigation; and the National Environment Management Authority enforces environmental safeguards for adaptation investments. These institutions collaborate closely with county governments and wards to translate national policies into locally relevant, gender-responsive, and climate-resilient actions, ensuring that strategic objectives reach communities while strengthening adaptive capacity across sectors.

At the county level, the Kwale County Government serves as the link between national policies and local implementation. The County Climate Change Act of 2022 created the County Climate Change Unit, which is responsible for coordinating climate-related initiatives and managing the County Climate Change Fund (CCCF). Efforts to mainstream climate considerations are increasingly reflected in the County Integrated Development Plan (CIDP) 2023–2027 and various sectoral plans, allowing departments such as Water; Agriculture, Livestock and Fisheries; Roads and Public Works; Health; and Environment and Forestry to incorporate resilience measures into their annual work plans. County-level coordination structures, including the County Steering Group and the Disaster Management Committee, oversee preparedness and response for multiple hazards and are progressively used to guide anticipatory actions based on

seasonal forecasts. Additionally, partnerships with NGOs and development partners are facilitated through county-led forums, ensuring that external initiatives in Pongwe Kikoneni align with public investments and minimize overlap.

At the ward and community level, the bulk of implementation efforts takes place. The Ward Administrator provides an administrative link to the county directorates, while the Ward Climate Change Planning Committee (WCCPC) serves as an inclusive platform for setting agendas, prioritizing investments, and reviewing progress. The WCCPC comprises the Ward Administrator, Community Development Officer, Agricultural and Public Health Officers, a representative for persons with disabilities, five community members, and delegates from local organizations. This composition reflects the National Adaptation Plan's (NAP) focus on locally led adaptation and social inclusion. Specialized groups such as water user committees, disaster volunteer teams, and environmental stewardship groups carry out day-to-day responsibilities. These include maintaining water resources, coordinating disaster preparedness, promoting sustainable livelihoods, overseeing ecosystem restoration, and managing communal resources. In the past, coordination among these groups often depended on informal personal networks. Today, the WCCPC provides a formalized structure for information sharing, joint planning, and accountability across villages and sectors.

2.3 Relevant Policies

The coordination framework for climate adaptation in Pongwe Kikoneni Ward is supported by a strong policy environment. At the national level, the Climate Change Act (2023), National Adaptation Plan (2015–2030) and National Climate Change Action Plan (NCCAP 2023–2027) provide the overarching structure for climate governance, emphasizing locally led adaptation and devolved financing. The National Adaptation Plan (NAP 2015–2030) identifies priority sectors relevant to Pongwe Kikoneni, including climate-smart agriculture, livestock development, water resource management, tourism and fisheries value chain enhancement, ecosystem restoration, disaster preparedness, gender, vulnerable groups and youth inclusivity and human health resilience. Programs such as the Conservation and Sustainable Management of Coastal and Marine Ecosystems (COSME Project), Restore Africa Project/ and Financing Locally Led Climate Action (FLLoCA) channel grants to support ward-level initiative.

Sub-nationally, the Kwale County Climate Change Act (2022) and the County Climate Change Action Plan (2023–2027) provide an enabling framework for ward proposals to access the County Climate Change Fund and align with county targets on water security, resilient agriculture, coastal and riparian restoration, sustainable livelihoods and climate-proofed infrastructure. Sector-specific policies reinforce these mandates: the Water Act and National Water Policy embed community-based water governance and integrated water resources management; the Agricultural Sector Transformation and Growth Strategy promotes diversified, climate-smart production and value addition; the National Disaster Risk Management policy architecture advances anticipatory action and local preparedness; the Wildlife Conservation and Management Act provides mechanisms for human–wildlife conflict mitigation and compensation; and national gender and social inclusion policies require equitable participation and benefit sharing. Together, these instruments establish the principles, standards, financing mechanisms, and safeguards that the Pongwe Kikoneni coordination system adopts, thereby ensuring consistency with the NAP while tailoring execution to place-based realities.

2.4 Proposed Coordination Arrangements

The proposed structure builds on existing strengths while introducing clear interfaces and standardized routines to align with NAP objectives for coherence, subsidiarity, and learning. At the ward–county level, a designated County Climate Change Officer will act as the formal

liaison to the WCCPC, attending scheduled meetings, sharing program updates, and translating ward priorities into departmental plans and CCCF pipelines. Quarterly progress reports, prepared by the Ward Administrator and endorsed by the WCCPC, will be submitted to the County Climate Change Unit. These reports will capture outputs, challenges, gender-disaggregated beneficiary data, safeguards compliance, and projections for the following quarter. This reporting routine fosters a performance-oriented culture compatible with FLLoCA and county M&E frameworks.

The WCCPC will convene a biannual Pongwe Kikoneni Climate Stakeholder Forum at the ward level, bringing together all active and potential actors, including county departments, national agencies operating locally, NGOs, CBOs, BMUs, private entities, research institutions, religious organizations, and microfinance partners. The forum will map who is doing what, where, and with which resources; establish technical standards; broker complementary roles across villages; and surface implementation bottlenecks for resolution. In the periods between these forums, the WCCPC will chair monthly coordination sessions that focus on near-term execution tasks. These sessions will cover the sequencing of procurements, community mobilization, safeguards check, and the integration of seasonal forecasts into operational plans.

To ensure effective project delivery, the WCCPC will form dedicated, task-focused teams for each initiative, combining local insight with technical expertise. An ecosystem conservation and restoration team, for example, will unite community representatives, local groups (tree nurseries), leaders, CFA, and, when needed, development partners, NGO or private-sector to tackle challenges on the ground. Further, an ecosystem restoration team will bring together Kenya Forest Service and BMUs (Mkwiro), local conservation advocates, and women's groups to restore and protect the natural environment. These teams will operate with clear timelines, maintain transparent records of activities and expenditures, and provide succinct monthly updates to the WCCPC, fostering accountability and steady progress.

Given Pongwe Kikoneni exposure to drought, sea level rise, pest and diseases, floods, strong winds and human wildlife conflict, adaptation efforts will be closely aligned with disaster risk management. The WCCPC will either function as, or closely coordinate with, the Ward Disaster Management Committee to ensure a seamless connection between long-term resilience initiatives and immediate emergency response. Clear protocols will guide the reception, interpretation, and dissemination of Kenya Meteorological Department advisories through village communication networks; outline the activation of anticipatory measures and social protection interventions when triggered by NDMA; and define coordination between ward teams and county emergency units for relief distribution, sensitization, awareness, evacuations, and health campaigns. This risk-informed strategy aligns with NAP priorities, emphasizing early warning systems and the integration of disaster risk reduction into development planning.

Flexibility and learning are designed into the system. An annual reflection, co-convened by the County Climate Change Unit and the WCCPC, will review results against NAP-aligned indicators, document lessons, and adjust institutional arrangements or operating procedures. As national guidance, data systems, or financing windows evolve, the ward will adapt its coordination protocols, ensuring continuity with national direction while preserving local ownership.

2.5 Implementation Roles

Effective implementation of the Ward Adaptation Plan requires clear role delineation among all stakeholders to ensure that adaptation measures are executed efficiently, inclusively, and in line with Kenya's National Adaptation Plan (2015–2030). Table 1 below outlines the mandates, decision-making authority, and accountability mechanisms of the key institutions and actors

engaged in rolling out adaptation interventions in Pongwe Kikoneni Ward. The WAP explicitly mainstreams Gender Equality and Inclusion (GEI) across all stages of planning and implementation and establishes strong coordination linkages with health, gender, and protection actors to guarantee continuity of essential Sexual and Reproductive Health and Rights (SRHR) and GBV prevention and response services during climate shocks.

Table 1: Institutional Roles in Implementation of Adaptation Actions

Institution/Actor	Key Roles and Responsibilities
Ward Climate Change Planning Committee (WCCPC)	Lead planning, coordination, and prioritization of ward-level adaptation actions; mobilize community participation; ensure gender and social inclusion in project design and implementation.
Kwale County Government Departments	Provide technical oversight, finance, and sectoral expertise (Water, Agriculture, Livestock, Fisheries, Roads, Health, Environment). Integrate ward priorities into departmental annual work plans and budgets.
National Drought Management Authority (NDMA)	Lead drought monitoring, early warning, and response; integrate Pongwe Kikoneni into national drought early warning systems and coordinate emergency relief when thresholds are reached.
Kenya Meteorological Department (KMD)	Provide localized weather and climate forecasts, early warnings, and advisories for decision-making in agriculture, fisheries, and disaster preparedness.
Kenya Forest Service (KFS)	Offer technical guidance on reforestation, afforestation, and mangrove restoration; support tree nursery establishment and monitoring.
Kenya Wildlife Service (KWS)	Manage and mitigate human–wildlife conflicts; support establishment of water points for wildlife and training of community wildlife scouts.
Kenya Fisheries Service (KeFS)	Protect, conserve, manage, and develop Kenya's fisheries and aquaculture resources.
Beach Managements Unit	Sustainable resource management of marine resources
National Environment Management Authority (NEMA)	Review and approve Environmental Impact Assessments (EIAs) for adaptation projects; ensure adherence to environmental safeguards.
Development Partners (World Bank, EU, FAO)	Provide technical assistance, capacity building, and climate finance to support implementation of local adaptation priorities; promote mainstreaming of climate resilience in development investments and infrastructure.
Non-Governmental Organizations (NGOs) and CSOs (Plan International, COMRED, Red Cross, Go Blue, KOMAZA, Kids Care Kenya, World Vision Kenya, Earthlungs)	Support implementation of community-based adaptation projects; promote gender-responsive climate resilience initiatives; facilitate training and awareness creation on nature-based solutions (NbS) and livelihood diversification.
Community-Based Organizations (CBOs) and Self-Help Groups	Implement localized adaptation initiatives such as tree planting, mangrove restoration, water harvesting, and livelihood diversification; mobilize local labor and indigenous knowledge; ensure equitable participation in project benefits.

Institution/Actor	Key Roles and Responsibilities
Faith Based Organizations (Muslim Community)	Promote social cohesion and community welfare
Private Sector and Cooperatives (Total Energies, Base Titanium Limited, East and Africa Company)	Promote investment in climate-smart agriculture, renewable energy,; integrate environmental sustainability in business operations; collaborate in corporate social responsibility (CSR) initiatives that support community resilience.
Academic and Research Institutions (KMFRI, KALRO, KEMFRI)	Generate and disseminate climate and ecological data; support participatory research and innovation for adaptation technologies; build capacity of local institutions on ecosystem monitoring and early warning systems.

3 Chapter 3: Situation analysis of climate vulnerability

3.1 Introduction

Pongwe Kikoneni Ward is increasingly exposed to multiple and interlinked climatic hazards that have intensified over recent years, undermining local livelihoods, ecosystems, and infrastructure. The Participatory Climate Vulnerability and Capacity Assessment (PCVA, 2025) established that the ward experiences recurrent droughts, floods, pest and disease infestations, strong winds, sea level rise, and escalating human–wildlife conflicts. The assessment combined community knowledge, historical timelines, and scientific projections to determine the frequency, magnitude, and spatial distribution of these hazards. Vulnerability was assessed across five representative villages Anzuwani, Barabarani (Majoreni), Kibuyuni, Kiwambale, and Mkwiro based on their exposure levels, sensitivity and adaptive capacity.

The findings showed that women and other vulnerable social groups, including the elderly, widows, persons abled differently (PADs), and low-income households, experience a disproportionate burden of climate-induced impacts. Their vulnerability is attributed to their heavy reliance on natural resources for subsistence, restricted access to financial and productive assets, and the intensified burden of unpaid care and domestic responsibilities during climate-induced crises. Spatial analysis of hazard hotspots indicates a clear clustering of vulnerabilities across the ward. Drought is a pervasive hazard affecting the entire ward, with Mabafueni, Mwandeo, Kiruku, Tswaka, and Mangawani villages being the most severely impacted. Pests, diseases, and floods are widespread, frequently damaging crops and infrastructure and exacerbating food insecurity. Human–wildlife conflicts are most prevalent in Kiwambale, Ganda, Mzizima, and Shimoni village units, driven by shrinking habitats and increased competition for water and forage resources.

3.2 Current climate trends

Pongwe Ward experiences a tropical coastal climate characterized by high humidity and warm temperatures throughout the year. The area receives bimodal rainfall, with long rains occurring from March to May and short rains from October to December. The current climatic conditions in Pongwe Ward, based on the baseline analysis under the SSP2–4.5 scenario, indicate a moderately wet environment with annual rainfall averaging 1,281.6 mm, against a potential evapotranspiration (PET) of 1,266.4 mm and an actual evapotranspiration (AET) of 1,614.1 mm (Figure 6). The pronounced fluctuations in the rainfall trend line indicate alternating wet periods and dry spells over time, reflecting the area's exposure to inter-annual rainfall variability. Peaks in rainfall correspond to potential wet years that enhance groundwater recharge, river flow, and soil moisture, while the sharp troughs suggest dry spells or drought periods that could stress crops, livestock, and water resources. The close alignment between rainfall and PET implies that most of the projected period will maintain a balanced hydrological condition, with rainfall generally sufficient to meet evapotranspiration demand. Meanwhile, consistently higher AET values signal adequate soil moisture and productive land cover, although prolonged dry spells may still lead to temporary moisture deficits. Overall, the projections suggest relatively stable but variable hydrological conditions, characterized by intermittent wet and dry periods, with a slight long-term improvement in water availability across Pongwe Ward.

Temperature analysis under the same scenario (Figure 7) shows a mean air temperature of 25.8°C and a land surface temperature (LST) of 34.1°C, signifying substantial surface heating across terrestrial landscapes. This temperature differential suggests that land surfaces absorb and retain more heat than the surrounding atmosphere, likely due to factors such as reduced vegetation cover, increased bare ground exposure, and enhanced solar radiation absorption. Such conditions contribute to higher evapotranspiration rates, soil moisture loss, and localized heat stress, which can adversely affect agricultural productivity, water resources, and overall ecosystem stability in Pongwe Ward.

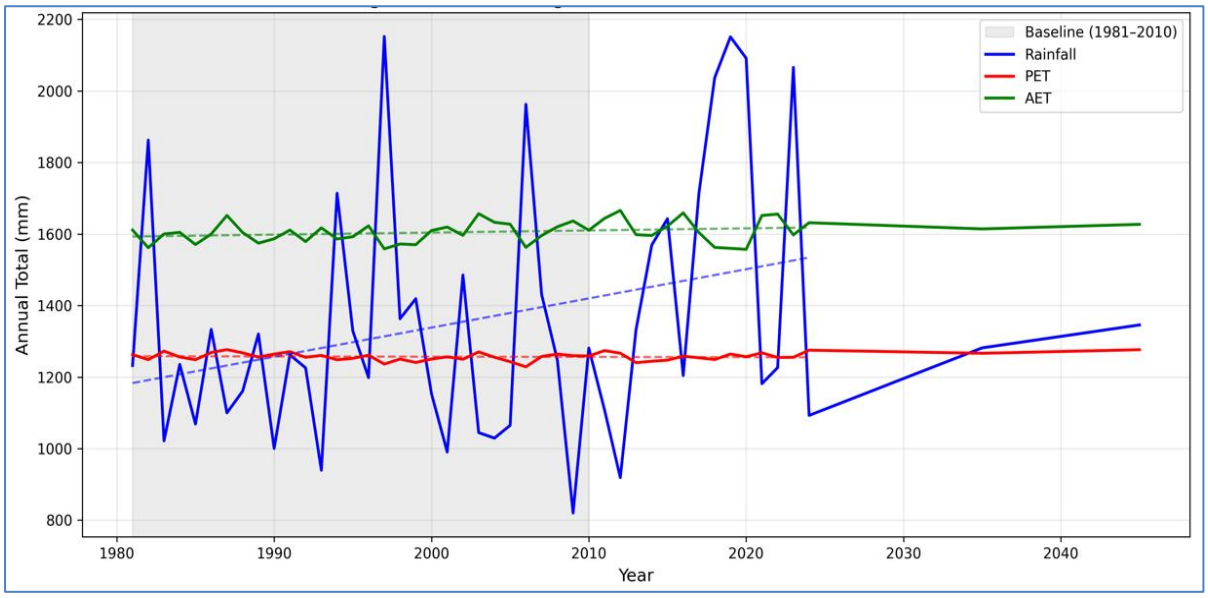

Figure 5: SSP2 4.5 Rainfall, AET and PET Projections (Source: Chirps database and ERA5-Ag, 2022)

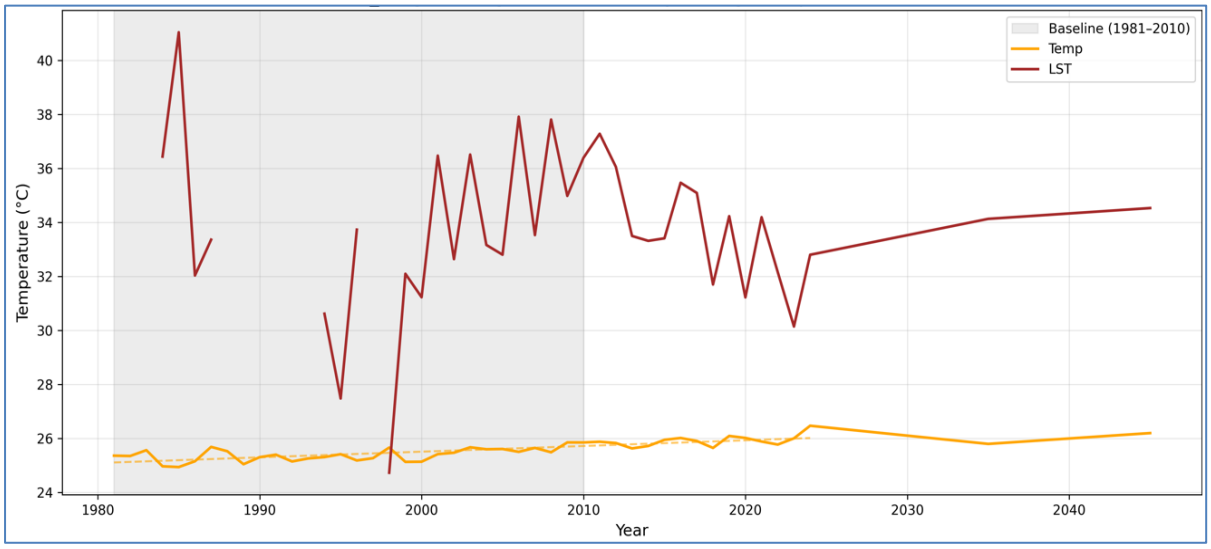

Figure 6: SSP2 4.5 Mean Temperature and Land Surface Temperature Projections (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.3 Future climate trends

Projected climate conditions under the high-emission scenario (SSP5–8.5) indicate that Pongwe Ward will experience slightly wetter but considerably warmer conditions relative to the baseline period. Rainfall is projected to increase from 1,320.0 mm to 1,409.7 mm, PET from 1,279.0 mm to 1,294.1 mm, and AET from 1,630.1 mm to 1,649.4 mm (Figure 8). The upward trend in rainfall, coupled with modest increases in PET and AET, indicates a relatively stable hydrological regime with a potential improvement in overall water availability. The projections suggest that future climatic conditions may support sustained soil moisture and vegetation growth, although short-term variations between wet and dry periods are still expected. Higher PET values also suggest increased atmospheric demand, possibly heightening dry-season stress despite wetter annual totals.

Future temperature projections further highlight a continued warming trend in Pongwe Ward. The mean air temperature is expected to rise from 26.0°C to 26.4°C, while the LST increases

significantly from 34.3°C to 34.7°C (Figure 9). The pattern underscores the intensification of surface heating across terrestrial areas, likely linked to factors such as declining vegetation cover, increased bare soil exposure and greater solar radiation absorption. The projected warming may elevate evapotranspiration rates, accelerate soil moisture loss, and exacerbate heat stress on crops, livestock, and human populations, thereby posing additional challenges to local water and food security.

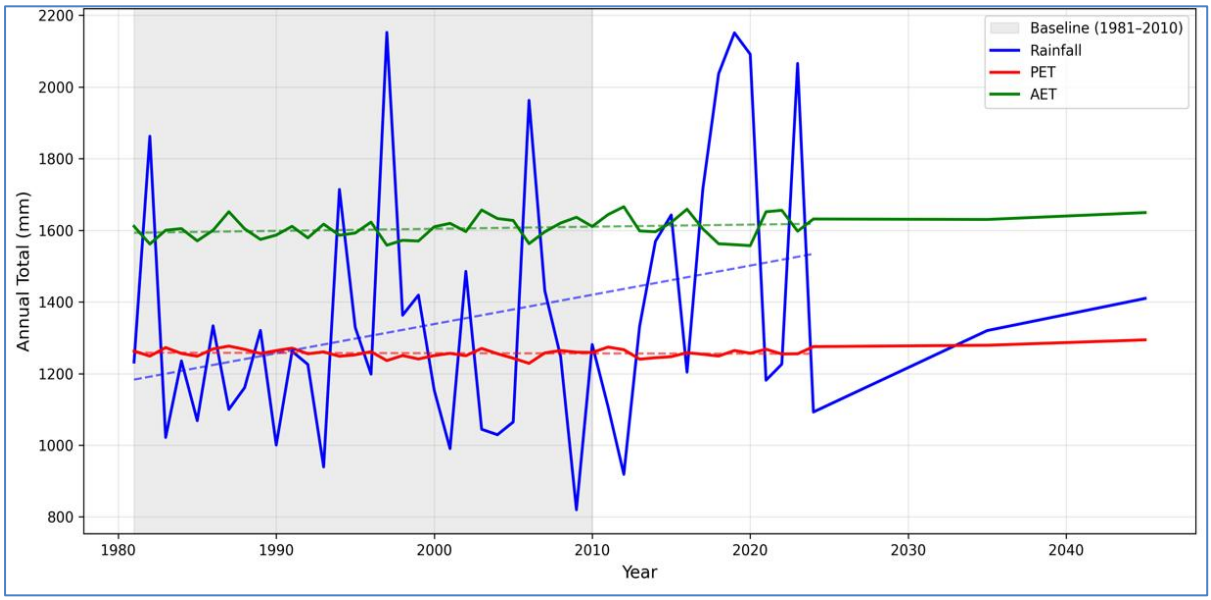

Figure 7: SSP5 8.5 Rainfall, AET and PET Projections. (Source: Chirps database and ERA5-Ag, 2022)

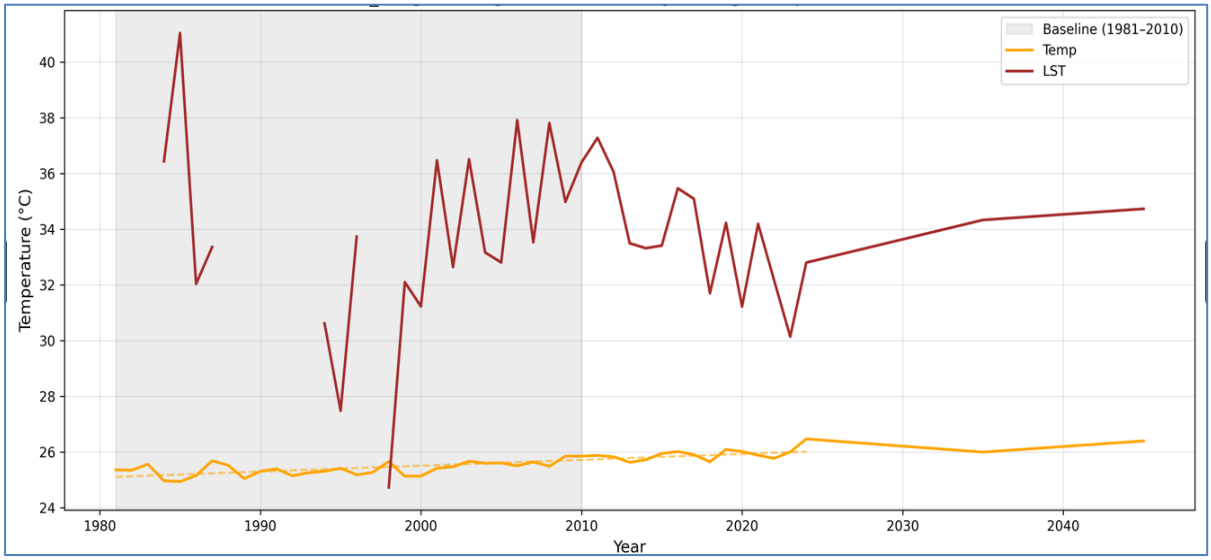

Figure 8: SSP5 8.5 Mean Temperature and Land Surface Temperature Projections. (Source: ERA5-Ag database and Landsat 5, 7, 8 & 9 SR ,2022)

3.4 Climate hazards and vulnerability

3.4.1 Drought

Drought remains the most recurrent and severe climatic hazard affecting Pongwe–Kikoneni Ward, occurring approximately once every three years. Historical analyses from 1960 to 2025 indicate a progressive increase in both frequency and intensity of drought events, with severe episodes documented in 1982 (Changilo), 1985 (Njaa ya Njenga), 1993, and 1999–2000 (Ukame wa Wasomali) (Figure 10). Each of these periods was marked by acute crop failure, depletion of water sources, reduced pasture productivity, and significant livelihood disruptions.

The pattern of recurrence underscores the combined influence of natural climate variability and anthropogenic drivers such as deforestation, poor land management, and encroachment into ecologically sensitive zones. During the 2010–2018 period, moderate dry spells persisted, yet their unpredictability emphasized an escalating vulnerability of local ecosystems and livelihoods to prolonged aridity.

Vulnerability analysis conducted under the 2025 Participatory Climate Vulnerability and Capacity Assessment (PCVA) assigned the highest drought vulnerability score to Kiwambale (70), followed by Mkwiro (58), Barabarani (Majoreni) (56), Anzuwani (46), and Kibuyuni (33) (Figure 11). These variations reflect differences in exposure, livelihood dependence, and adaptive capacity among the villages, particularly regarding access to alternative water sources, forest resources, and income diversification.

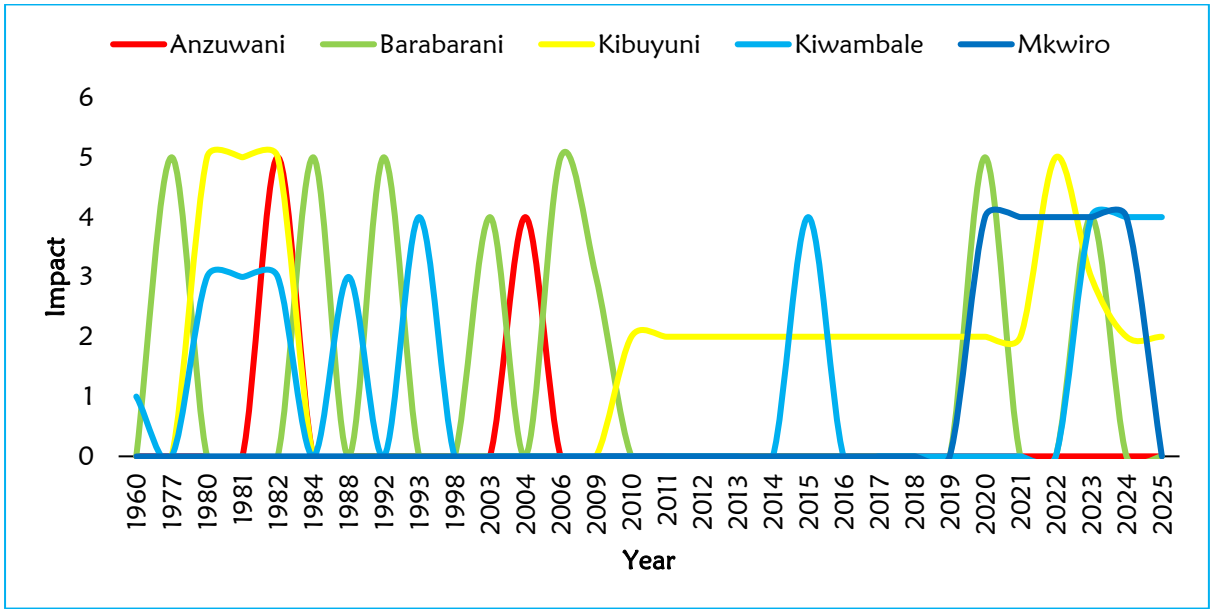

Figure 9: Drought prevalence historical trend (Source: PCVA,2025)

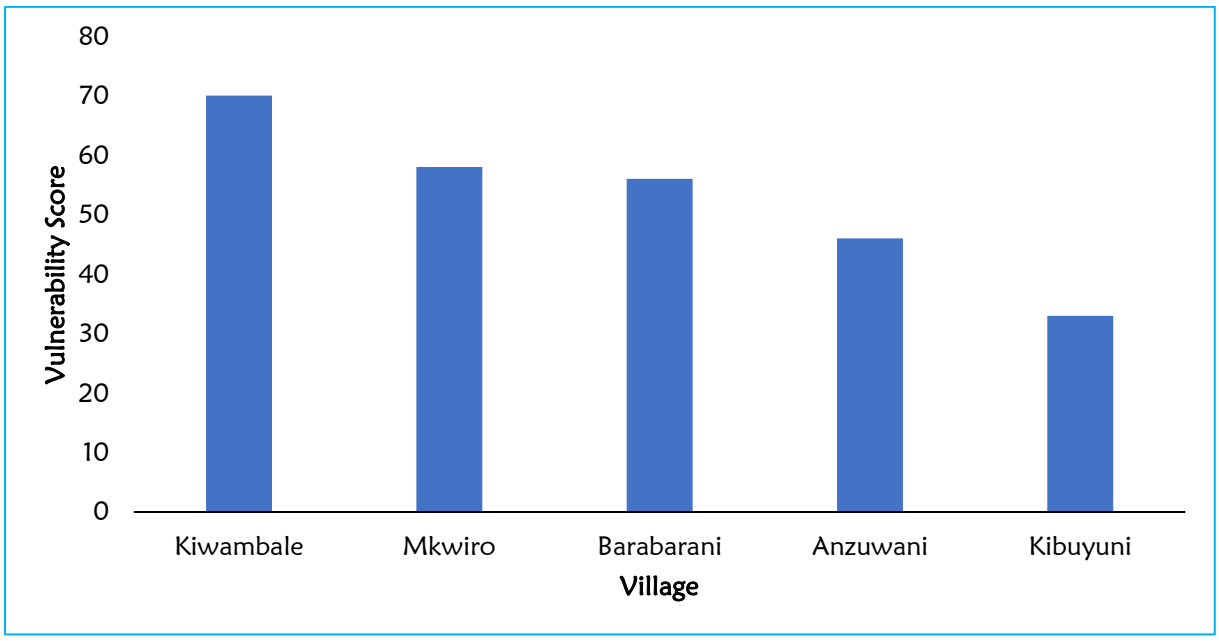

Figure 10: Drought vulnerability score in Pongwe Kikoneni Ward (Source: PCVA,2025)

The recurrent droughts have had profound ecological and socio-economic effects on key natural resources and community well-being. The Ganda and Kiriku forests, which provide critical ecosystem services such as microclimate regulation, fuelwood, and non-timber forest products, have experienced extensive vegetation stress and canopy thinning due to moisture loss and increased fire incidence. Similarly, mangrove forests along the coastal belt have shown visible signs of desiccation, reduced regeneration, and increased mortality of species such as *Rhizophora mucronata* and *Avicennia marina* due to heightened salinity intrusion and limited freshwater inflow. During the dry season, NDVI values range from 0.12 to 0.48, indicating moderate vegetation stress caused by high salinity, reduced freshwater inflow, and elevated evapotranspiration. In the wet season, NDVI values generally rise to 0.14–0.55, reflecting improved vegetation health and regeneration due to increased rainfall and nutrient availability (Figure 12). Nevertheless, lower NDVI values observed from Majoreni to Vanga during the wet season suggest localized vegetation stress caused by sedimentation and flooding, which can smother mangrove roots, reduce oxygen availability, and hinder regeneration. These patterns underscore the sensitivity of mangrove ecosystems to seasonal climatic fluctuations, where heightened rainfall intensity and sediment loads associated with climate change may accelerate degradation and reduce the resilience of mangroves along this coastal stretch.

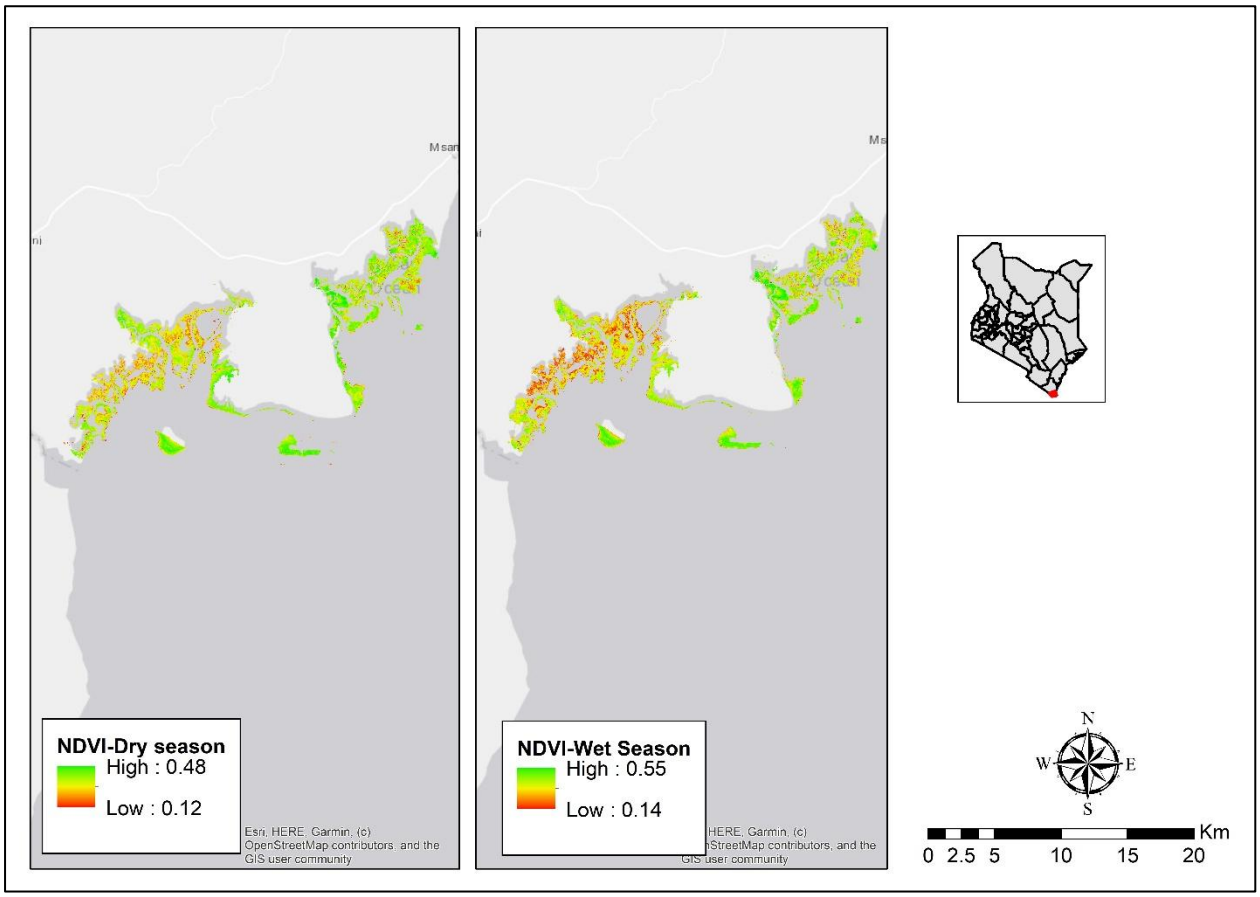

Figure 11: Mangrove Forest NDVI values across dry and wet seasons (Source: Source: Landsat 8/9 OLI and TIRS, 2025)

Seaweed farms in coastal villages like Kibuyuni and Mkwiro have also been adversely affected, with reduced productivity attributed to elevated sea surface temperatures and salinity fluctuations during prolonged dry spells. This has directly impacted women's cooperatives and youth groups who rely on seaweed farming as a primary income source, increasing economic vulnerability and food insecurity. In parallel, river systems including Ramisi, Mwachema, and Mwena have recorded diminishing base flows and extended dry riverbeds, constraining

irrigation, domestic water use, and small-scale aquaculture. The resulting water scarcity has exacerbated competition among households and between humans and wildlife, particularly in areas adjacent to forested catchments and farmlands.

3.4.1.1 Exposure to drought

Exposure to drought in Pongwe–Kikoneni Ward is widespread and multi-dimensional, affecting both natural ecosystems and socio-economic systems across the identified hotspots of Mabafueni, Mwandeo, Kiriku, Tswaka, and Mangawani. Drought events, which now occur approximately once every three years, have progressively intensified, exposing these areas to recurrent water scarcity, land degradation, and declining agricultural productivity. The extent of exposure is closely linked to ecosystem type, land-use practices, and the degree of dependence of local communities on climate-sensitive resources.

Mabafueni and Mwandeo, which form part of the ward's key agricultural zones, are acutely exposed due to their reliance on rain-fed farming and shallow groundwater systems. During extended dry spells, soil moisture depletion becomes severe, leading to crop wilting and near-total failure of staple crops such as maize, cassava, and cowpeas. These conditions are compounded by poor water retention capacity of the soils and limited access to irrigation infrastructure. As a result, most smallholder farmers experience recurrent food shortages and loss of household income, with women disproportionately affected due to their roles in food production and household water provision.

Kiriku area, which hosts part of the Kiriku Forest, exhibits significant ecological exposure. Prolonged droughts have led to drying of understorey vegetation, increased forest fires, and loss of biodiversity. Tree species such *as Afzelia quanzensis and Brachystegia spiciformis* show stunted regeneration due to moisture stress, while reduced canopy cover accelerates soil erosion and further desiccation of the forest floor. The forest's role in local microclimate regulation and carbon storage has consequently weakened, increasing the vulnerability of surrounding farmlands and settlements.

Tswaka and Mangawani, situated along the lower coastal belt, demonstrate dual exposure affecting both terrestrial and marine ecosystems. The adjacent mangrove forests and seaweed farming areas have been particularly sensitive to prolonged droughts. Reduced freshwater inflows from rivers Ramisi, and Mwena have increased saline intrusion, elevating soil and water salinity levels that hinder mangrove regeneration and reduce the productivity of seaweed farms managed predominantly by women's groups. This has translated into declining fish nursery habitats, reduced catches, and weakened coastal protection against tidal surges.

Furthermore, water sources across the drought-prone hotspots, including seasonal streams and shallow wells, experience drastic declines in recharge rates during prolonged dry periods. Groundwater tables recede significantly, leading to borehole failures and increased water-fetching distances. The situation often forces communities, particularly in Mwandeo and Mabafueni, to rely on unsafe or distant water sources, exacerbating health and sanitation risks. Livestock herders also face heightened exposure due to dwindling pasture and watering points, resulting in reduced livestock productivity, malnutrition, and economic losses.

3.4.1.2 Socio-Demographic Vulnerability to Drought

Socio-demographic vulnerability to drought in Pongwe–Kikoneni Ward is strongly influenced by population structure, livelihood dependency patterns, poverty levels, and access to adaptive resources such as water, land, and information. The ward's predominantly rural population relies heavily on climate-sensitive sectors, particularly smallholder rain-fed agriculture, livestock rearing, and marine-based livelihoods such as seaweed farming and artisanal fishing. As droughts

now recur approximately once every three years, the socio-demographic characteristics of the population play a critical role in determining the degree of exposure, sensitivity, and ability to cope with the impacts.

Women, who form the majority of the agricultural labour force, are among the most vulnerable groups. They are primarily responsible for food production, household water collection, and caregiving roles, which intensify during prolonged dry spells. The drying of rivers Ramisi and Mwena increases their workload, as they must travel longer distances to fetch water for domestic use and irrigation. Additionally, reduced yields from kitchen gardens and seaweed farms have led to declining household incomes, further constraining women's ability to purchase food and meet essential needs. Unpaid domestic and care work also increases significantly during droughts, limiting women's participation in economic and community decision-making activities.

Children and the elderly are equally vulnerable. Drought-related food shortages contribute to malnutrition and school absenteeism among children, particularly in Mabafueni and Mwandeo villages where subsistence farming dominates. The elderly, especially widows and persons abled differently, face reduced physical capacity to access scarce water and food resources. This increases their dependency on others, particularly during prolonged dry spells, thereby heightening household-level stress.

Poverty levels in the ward exacerbate vulnerability. Most households earn below KES 10,000 per month, leaving limited capacity to invest in adaptive technologies such as irrigation, water storage facilities, or drought-tolerant seeds. Informal employment and reliance on natural resource extraction (such as charcoal burning and fuelwood collection) also increase environmental degradation, perpetuating a vicious cycle of poverty and ecosystem depletion.

Education levels and access to early warning information further shape drought vulnerability. Households with limited education and access to information are less likely to receive or act on drought advisories disseminated by government and non-governmental institutions. Consequently, they fail to adopt preventive measures such as water harvesting, fodder preservation, or crop diversification. In contrast, communities with better access to extension services, mainly through county departments or NGOs, demonstrate relatively stronger coping mechanisms, though coverage remains uneven across the ward.

Gender disparities in resource ownership, particularly land and livestock, further compound vulnerability. Customary tenure systems limit women's control over productive assets, restricting their ability to make adaptive investments or benefit from formal credit and insurance schemes. In Kiriku and Mangawani, for instance, women's limited access to land and capital has hindered diversification away from drought-sensitive activities such as seaweed farming and small-scale horticulture.

3.4.2 Floods

Flooding in Pongwe–Kikoneni Ward occurs approximately once every four years and remains one of the most destructive yet cyclical hydrometeorological hazards in the area. Historical trends indicate that the most severe flood events were recorded during the 1970 and 1997 El Niño periods, both of which resulted in extensive property destruction, displacement of households, and widespread crop and livestock losses (Figure 13). Although large-scale floods have been episodic, the frequency of localized flash floods has increased in recent years, particularly along Rivers Ramisi and Mwena. The 2024 flood season, for instance, caused severe damage to farmlands and infrastructure in low-lying settlements of Mwakembe and Mabafueni, exposing the community to recurrent livelihood disruptions. These trends illustrate the growing

variability in rainfall patterns, where short, intense rainfall episodes increasingly trigger rapid runoff, soil erosion, and siltation of rivers and wetlands.

Vulnerability analysis shows that Barabarani (Majoreni) and Kiwambale recorded the highest flood vulnerability scores of 97, indicating severe exposure and sensitivity to flood impacts (Figure 14). Anzuwani scored moderately at 70, while Mkwiro, historically affected only during the major El Niño years, registered low vulnerability due to its relatively elevated topography and improved drainage infrastructure. The spatial distribution of vulnerability reflects the influence of terrain, settlement proximity to watercourses, and land-use practices, particularly in areas with degraded riparian buffers and encroachment into floodplains.

Floods have had pronounced ecological and resource-based impacts across the ward. In the Ganda and Kiriku forests, heavy rainfall and surface runoff have accelerated soil erosion, uprooted seedlings, and washed away topsoil, impeding forest regeneration. The loss of vegetation cover along forest margins has further destabilized slopes and increased sedimentation in adjacent rivers, degrading water quality and aquatic habitats. Similarly, mangrove forests fringing the coastline have experienced flooding-induced stress through sediment overload and altered salinity regimes. Species such as *Rhizophora mucronata and Ceriops tagal* show dieback in areas where stagnant floodwaters persist, reducing the forests' ecological productivity and their role in shoreline stabilization.

Floods also affect seaweed farming activities, especially in Kibuyuni, Mwandeo, and Mangawani, where excess freshwater runoff and sediment inflow from rivers Ramisi and Mwena disrupt salinity balance and water clarity. These conditions inhibit seaweed growth and trigger disease outbreaks, leading to significant production losses and income decline for women's groups and youth cooperatives dependent on seaweed farming. Moreover, repeated inundation along the lower reaches of the three rivers damages irrigation infrastructure, contaminates shallow wells, and increases the prevalence of waterborne diseases such as cholera and typhoid.

The community-level impacts of flooding are multifaceted. Agricultural fields are often submerged for prolonged periods, destroying crops and leading to post-harvest losses, while roads and bridges become impassable, disrupting market access and service delivery. Livestock deaths due to drowning and loss of grazing lands are common, particularly in low-lying areas such as Barabarani (Majoreni). Households experience recurrent displacement, damage to housing structures, and contamination of water sources, further heightening food insecurity and public health risks. Women and children bear the greatest burden during these crises, as they must manage relocation, caregiving, and recovery under limited resources.

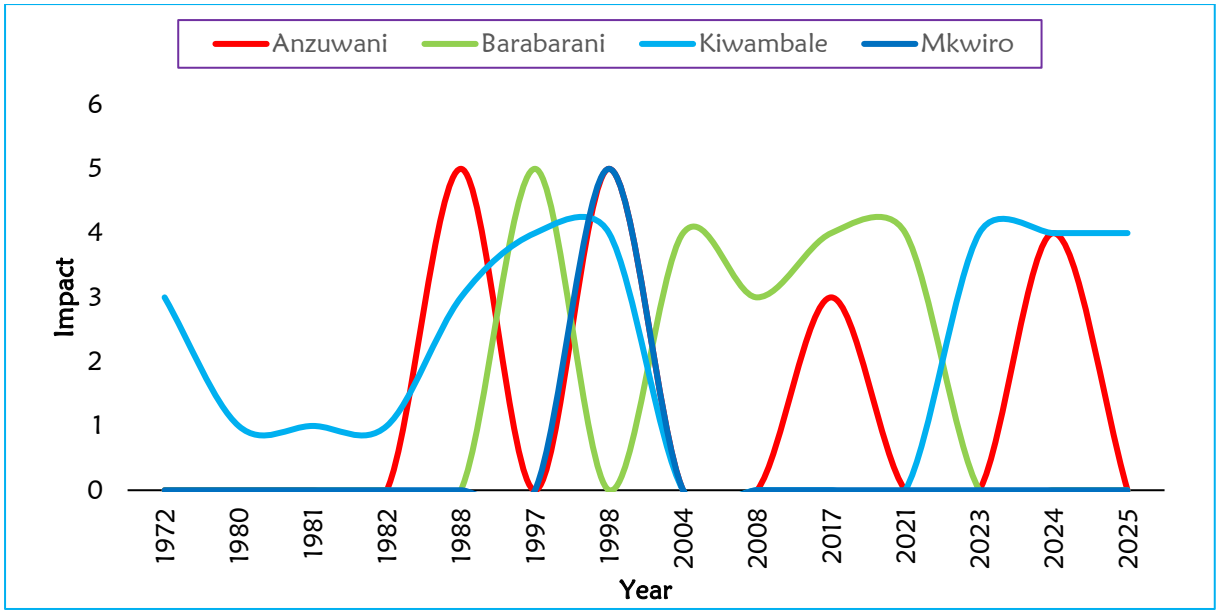

Figure 12: Flood prevalence historical trend (Source: PCVA,2025)

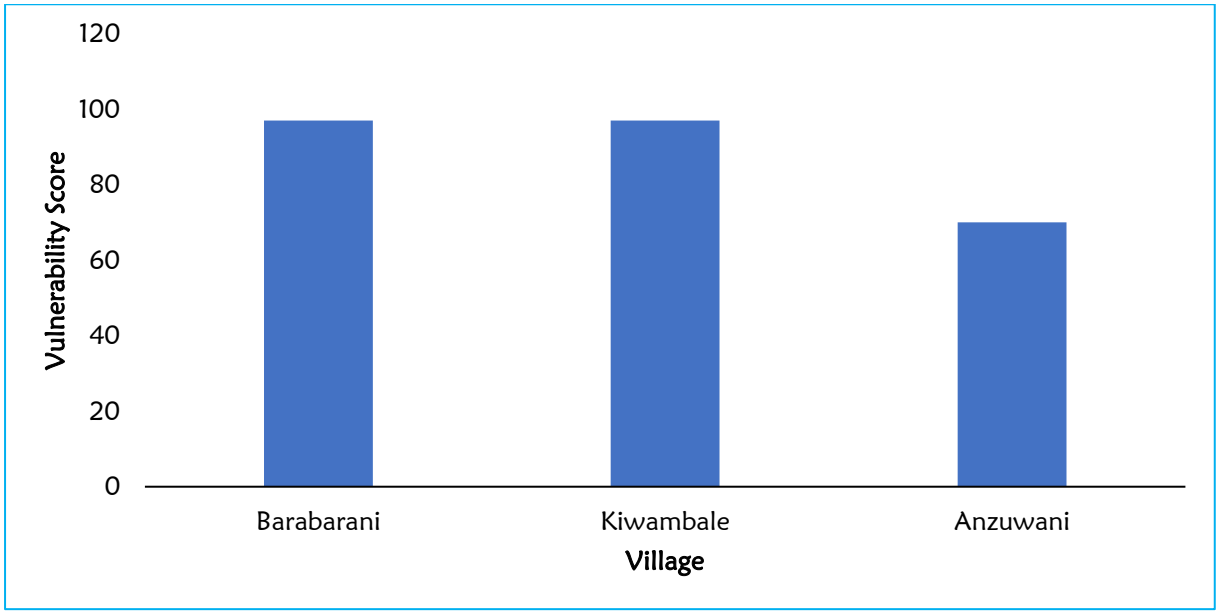

Figure 13: Floods vulnerability score in Pongwe Kikoneni Ward (Source: PCVA, 2025)

3.4.2.1 Exposure to flooding

Exposure to flooding in the ward is spatially concentrated within the key hotspots of Mabafueni, Mwandeo, Kiriku, Tswaka, and Mangawani, where the interaction of climatic, hydrological, and land-use factors amplifies flood risk. The ward experiences flooding approximately once every four years, primarily during the long rainfall seasons between March and May and, occasionally, during the short rains in October–December. The flood regime is influenced by intense convective rainfall, inadequate drainage infrastructure, and encroachment into flood-prone areas along Rivers Ramisi and Mwena. The combination of these drivers has resulted in frequent inundation of farmlands, homesteads, and ecosystems that support livelihoods.

Mabafueni and Mwandeo exhibit the highest exposure due to their location within the lower floodplains of the Ramisi and Mwena river systems. These areas experience recurrent overbank flooding and waterlogging during high rainfall events. The flat terrain and high groundwater

table limit infiltration, causing prolonged stagnation of floodwaters. Agricultural lands in these localities, dominated by cassava, maize, and horticultural crops, are highly vulnerable to submersion, leading to repeated yield losses and soil nutrient depletion. Infrastructure such as rural access roads and culverts are often rendered impassable, disrupting market linkages and emergency response.

Kiriku, characterized by its mixed land use of forest patches and smallholder farms, experiences flash floods and slope wash during heavy downpours. The adjacent Kiriku Forest, which plays a vital role in regulating runoff and stabilizing soils, has suffered degradation due to deforestation and charcoal production. Reduced canopy cover and compacted soils have diminished infiltration capacity, resulting in accelerated surface runoff. Consequently, farmlands at the forest edge are frequently eroded, while forest regeneration is suppressed by sediment deposition and uprooting of seedlings. These conditions make Kiriku one of the most ecologically exposed areas to flash flooding within the ward.

Tswaka and Mangawani, located along the coastal fringe, demonstrate dual exposure from both fluvial and coastal flooding. Heavy rainfall upstream leads to overflow of Rivers Mwena and Ramisi, while strong tidal surges coincide with river discharge, exacerbating backflow into low-lying settlements and mangrove zones. Mangrove ecosystems in these areas face sediment accumulation and salinity fluctuations due to alternating inflow of freshwater and tidal water. Prolonged inundation reduces mangrove seedling survival and alters species composition, thereby weakening their shoreline protection function. Similarly, seaweed farms operated by women's groups in the nearshore zones experience high exposure during flood periods when turbidity increases and water salinity is diluted, leading to poor growth and disease prevalence.

The exposure of the community is further aggravated by settlement patterns and inadequate drainage infrastructure. Many homesteads and community facilities in Mabafueni and Tswaka are built on low-lying floodplains or along seasonal streams without adequate embankments or stormwater channels. As a result, localized flooding frequently damages housing structures, contaminates shallow wells, and leads to outbreaks of waterborne diseases such as cholera and diarrhoea. Livestock watering points and grazing areas also become inaccessible, forcing herders to migrate temporarily or sell off weakened stock at low prices.

3.4.2.2 Socio-demographic vulnerability to floods

Socio-demographic vulnerability to floods in Pongwe–Kikoneni Ward is shaped by gender, age, income levels, education, and settlement patterns, with women and other marginalized groups disproportionately affected. The recurrent floods have exposed deep social and economic inequalities that determine the capacity of households and individuals to prepare for, respond to, and recover from flood events.

Women represent the most vulnerable demographic segment due to their central role in household management, food production, caregiving, and water collection. During flood seasons, the destruction of farmlands, contamination of water sources, and disruption of transport networks greatly increase their workload and psychological stress. Women are typically responsible for salvaging household items, caring for children and the elderly, and managing temporary relocations to safer grounds, all while maintaining domestic routines with limited resources. The flooding of Rivers Ramisi, and Mwena often damages shallow wells and boreholes, forcing women and girls to travel longer distances to fetch clean water or rely on unsafe sources, increasing exposure to waterborne diseases and gender-based risks during such journeys.

Economically, women's livelihoods, particularly in seaweed farming, small-scale agriculture, and small scale trade, are among the most severely impacted by floods. In coastal villages such as Mangawani and Tswaka, heavy runoff and saline water intrusion disrupt seaweed production, leading to loss of income for women's cooperatives. Similarly, flooding of agricultural plots in Mabafueni and Mwandeo destroys food crops and kitchen gardens that are key sources of household nutrition and income. The financial losses reduce women's ability to restock, repair damaged homes, or invest in livelihood recovery, pushing many households into deeper poverty.

Social norms and limited resource ownership exacerbate these vulnerabilities. Women's access to land and productive assets remains largely constrained under customary tenure systems, limiting their control over decision-making and recovery investments. In many households, male members migrate temporarily in search of income following flood events, leaving women to head households and manage recovery with minimal support. The lack of financial literacy, credit access, and insurance coverage further diminishes their adaptive capacity.

Health and safety risks also increase during floods. Congested temporary shelters expose women and children to poor sanitation, gender-based violence, and reproductive health challenges. Pregnant and lactating women face additional risks due to disrupted healthcare access and lack of clean water. In addition, the psychological burden of caregiving during crises coupled with loss of livelihoods and displacement contributes to emotional distress among women, who often lack access to psychosocial support systems.

Children, the elderly, and persons abled differently are also socially vulnerable, but women remain the primary caregivers for these groups during flood disasters. Their dual roles as income earners and caretakers intensify during recovery, yet their voices are often underrepresented in ward-level disaster planning and resource allocation processes.

3.4.3 Pests and Diseases

Pests and diseases present a recurrent hazard in Pongwe–Kikoneni Ward, occurring approximately once every two years and exerting significant pressure on both ecological and human systems. Historical data indicate periodic outbreaks since the early 1990s, with notable peaks in the mid-2000s and a resurgence between 2016 and 2025 (Figure 15). These cycles correspond with extended dry or humid periods that create favorable breeding conditions for crop pests, livestock diseases, and vector-borne human ailments. The frequency of these events has been linked to changing rainfall patterns, poor pest management practices, and ecosystem degradation, all of which weaken the natural resilience of both agroecosystems and forested landscapes.

Vulnerability analysis conducted under the 2025 PCVA shows that Kiwambale recorded the highest vulnerability score (50), followed by Anzuwani (32) and Barabarani (Majoreni) (31). Mkwiro exhibited relatively low vulnerability due to its small-scale subsistence farming and the presence of a well-equipped health facility capable of early disease surveillance and response (Figure 16). The varying degrees of vulnerability across villages are influenced by livelihood dependence, environmental conditions, and institutional support mechanisms.

The impacts of pests and diseases on natural resources have been profound. In Ganda and Kiriku forests, periodic outbreaks of defoliating insects, fungal infections, and invasive species have reduced tree vigor and hindered natural regeneration. Moisture stress during dry spells weakens native tree species such as *Brachystegia spiciformis* and *Afzelia quanzensis*, making them more susceptible to pathogens and borers. Recurrent infestations contribute to canopy thinning, loss of biodiversity, and increased fuel load, which in turn elevate forest fire risks. Similarly,

mangrove forests along the coastal belt have experienced dieback episodes attributed to pest infestations exacerbated by fluctuating salinity and temperature extremes. These stresses reduce the forests' ecological productivity, diminish fish nursery functions, and accelerate shoreline erosion.

Seaweed farms, have suffered periodic disease outbreaks, including "ice-ice" disease caused by environmental stress and bacterial infection. These outbreaks result in bleaching, reduced yield, and complete crop loss in severe cases. Since seaweed farming is predominantly undertaken by women's groups, these losses directly undermine their economic stability and food security. Moreover, increased runoff from upland areas following pest control activities has introduced pesticide residues into coastal waters, affecting marine biodiversity and compounding stress on seaweed ecosystems.

The hydrological systems, Rivers Ramisi and Mwena, also exhibit indirect vulnerability to pest and disease outbreaks. Excessive pesticide use and agrochemical runoff during pest control campaigns contaminate surface waters, altering aquatic species composition and reducing water quality. Algal blooms and fish kills have been reported during years of high pest infestation, particularly in sections of the Ramisi River downstream of cultivated zones. The decline in river health not only affects ecosystem services but also reduces the availability of clean water for domestic and agricultural use.

At the community level, recurrent pest and disease outbreaks undermine agricultural productivity and household resilience. Staple crops such as maize, cassava, and cowpeas experience repeated infestations from armyworms, cassava mosaic disease, and aphids, resulting in food shortages and reduced income. Livestock diseases such as foot-and-mouth and tick-borne infections contribute to loss of animal stock and reduced milk yields, further straining household nutrition. Human health is equally affected, with vector-borne diseases such as malaria and dengue fever increasing after heavy rains and during humid spells. The resulting medical costs and loss of labor productivity deepen poverty and limit the community's ability to recover from other concurrent hazards such as drought or floods.

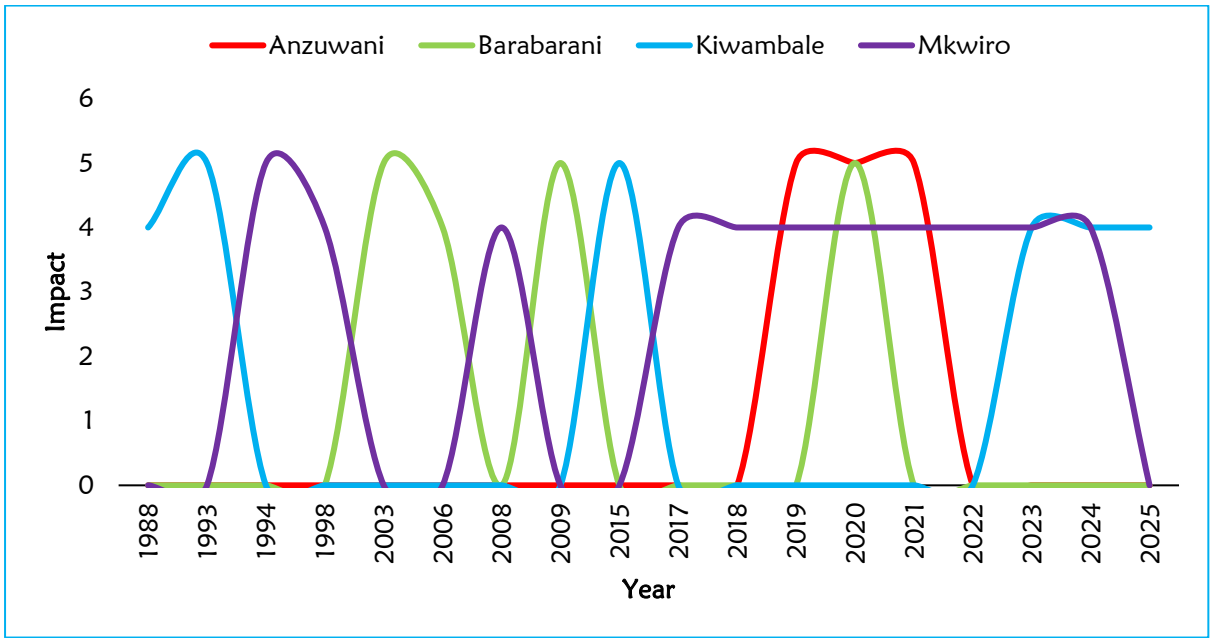

Figure 14: Pest and diseases prevalence historical trend (Source: PCVA,2025)

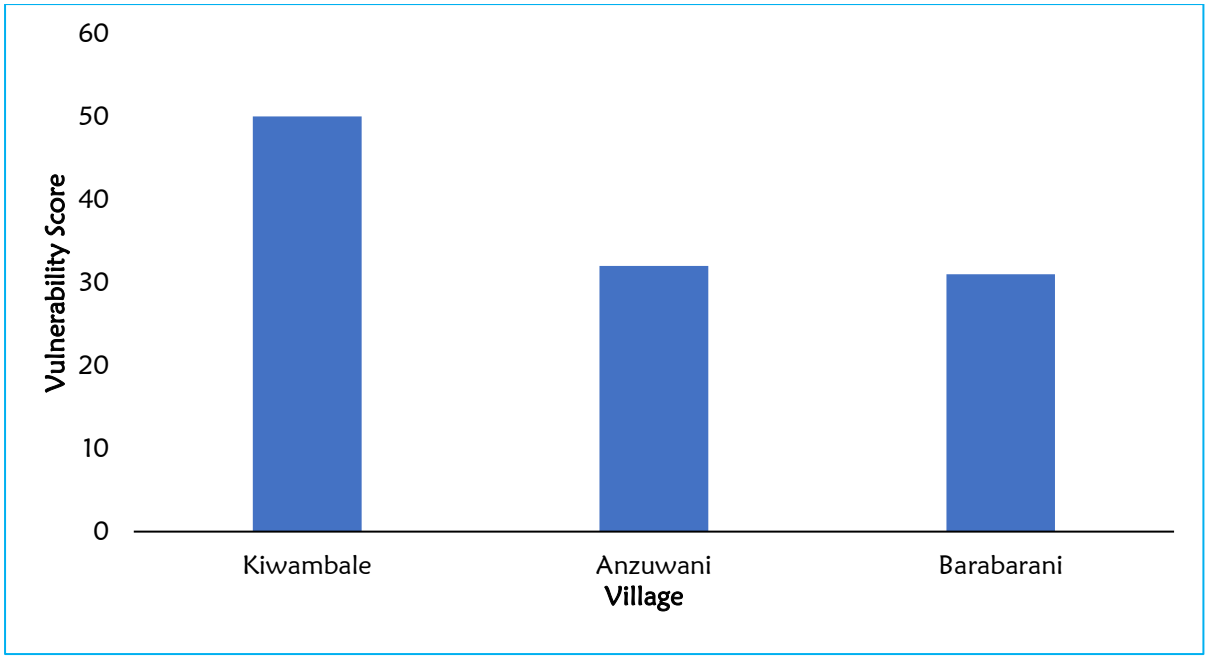

Figure 15: Pests and Diseases vulnerability score in Pongwe Kikoneni Ward (Source: PCVA,2025)

3.4.3.1 Exposure to Pests and Diseases

Exposure to pests and diseases in the ward is widespread and cross-cutting, affecting almost all ecosystems and livelihood zones. The ward experiences pest and disease outbreaks approximately once every two years, driven by fluctuating temperature, humidity, and rainfall conditions that favour the proliferation of crop, livestock, and human disease vectors. The spatial spread of these hazards extends across the entire ward, from agricultural areas in Kiwambale and Barabarani (Majoreni) to coastal and mangrove ecosystems in Anzuwani, Mkwiro and Kibuyuni indicating a broad ecological susceptibility.

Crops such as maize, cassava, cowpeas, and vegetables are particularly exposed to periodic infestations by armyworms, aphids, and cassava mosaic disease, which cause severe yield losses and food insecurity. Livestock herds, especially goats and cattle, face recurring tick-borne diseases, foot-and-mouth disease, and other parasitic infections, which intensify during warm and wet conditions. In the coastal zones, seaweed farms are highly exposed to "ice-ice" disease and epiphytic infestations, both of which spread rapidly during periods of elevated sea temperatures and freshwater inflows from rivers Ramisi and Mwena.

Forested ecosystems, including Ganda and Kiriku forests, experience outbreaks of defoliating insects and fungal pathogens that weaken native tree species and suppress regeneration. The interlinked exposure of forests, rivers, and farmlands demonstrates the ecological connectivity of pest and disease hazards, where disturbances in one system cascade into others. Overall, the widespread exposure highlights the need for integrated surveillance and ecosystem-based management to mitigate the cascading effects of pest and disease outbreaks across the ward.

3.4.3.2 Socio-Demographic Vulnerability to Pests and Diseases

Socio-demographic vulnerability to pests and diseases in Pongwe–Kikoneni Ward is deeply rooted in livelihood dependence, gender roles, poverty, and access to healthcare and agricultural extension services. The majority of residents rely on subsistence farming, livestock keeping, and seaweed cultivation, all highly sensitive to pest and disease outbreaks. Women, who constitute the main agricultural labour force and primary caregivers, are disproportionately affected. Their roles in food production, household nutrition, and caregiving

expose them to greater stress during outbreaks, as they manage declining crop yields, sick livestock, and increased household medical needs with limited resources.

Poverty amplifies vulnerability by restricting access to quality farm inputs, pesticides, veterinary care, and healthcare services. Households with low income are unable to afford timely pest control or treatment, leading to recurrent losses and food insecurity. Limited education and awareness also hinder early detection and management, allowing outbreaks to spread rapidly across villages.

Health vulnerabilities are particularly acute among children, the elderly, and persons abled differently, who face higher susceptibility to vector-borne diseases such as malaria and dengue fever during outbreak periods. The financial burden of treatment, combined with reduced labour productivity, undermines community resilience and perpetuates cycles of poverty and dependence on natural resource extraction.

3.4.4 Strong winds

Strong winds represent a recurrent but less frequent climatic hazard in Pongwe–Kikoneni Ward, occurring approximately once every four years. Although episodic, their localized impacts have been increasingly severe, particularly in the coastal villages of Mkwiro and Kibuyuni. Historical records from 2015 to 2025 indicate a consistent pattern of high-intensity wind events causing substantial environmental and infrastructural damage (Figure 17). According to data from the Kenya Meteorological Department, the average windspeed along the South Coast, ranges between 18–25 knots (approximately 33–46 km/h) during the monsoon seasons, occasionally surging above 30 knots (55 km/h) during extreme weather events (Kenya Meteorological Department, 2023). These conditions contribute to strong gusts and turbulence that significantly affect both terrestrial and marine ecosystems.

Vulnerability analysis from the PCVA (2025) assigned the highest vulnerability score to Mkwiro (72), followed closely by Kibuyuni (68), reflecting the severe physical and livelihood impacts associated with recurrent wind events. In contrast, inland villages such as Barabarani (Majoreni) (40), Anzuwani (38), and Kiwambale (35) recorded relatively lower scores due to their sheltered terrain and reduced exposure to coastal wind gusts (Figure 18).

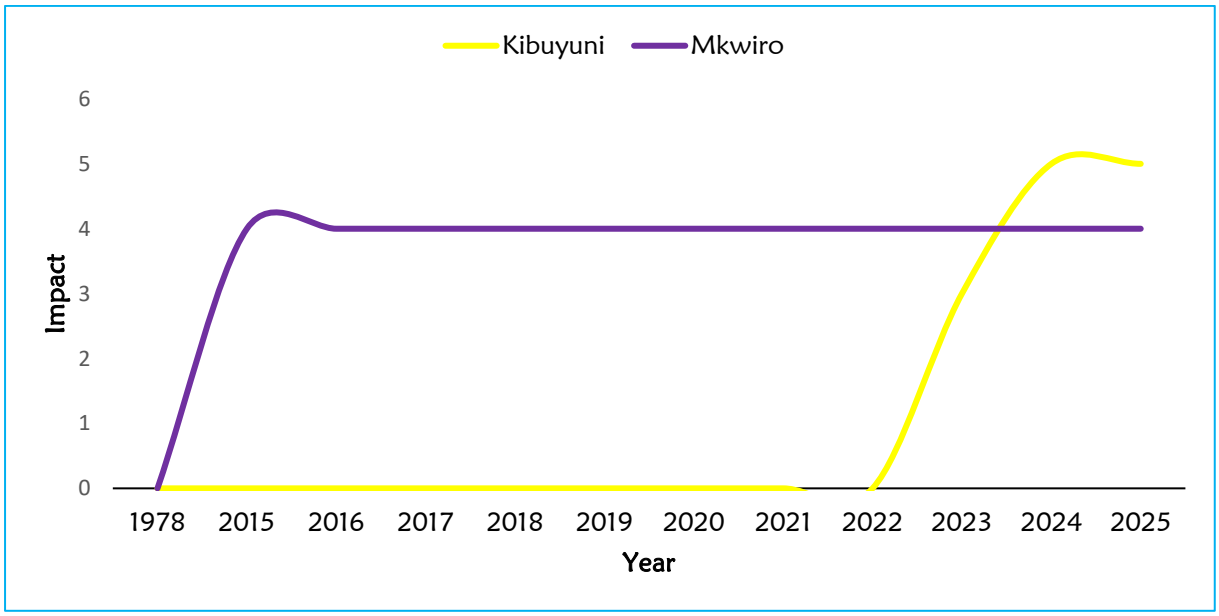

Figure 16: Strong winds prevalence historical trend (Source: PCVA, 2025)

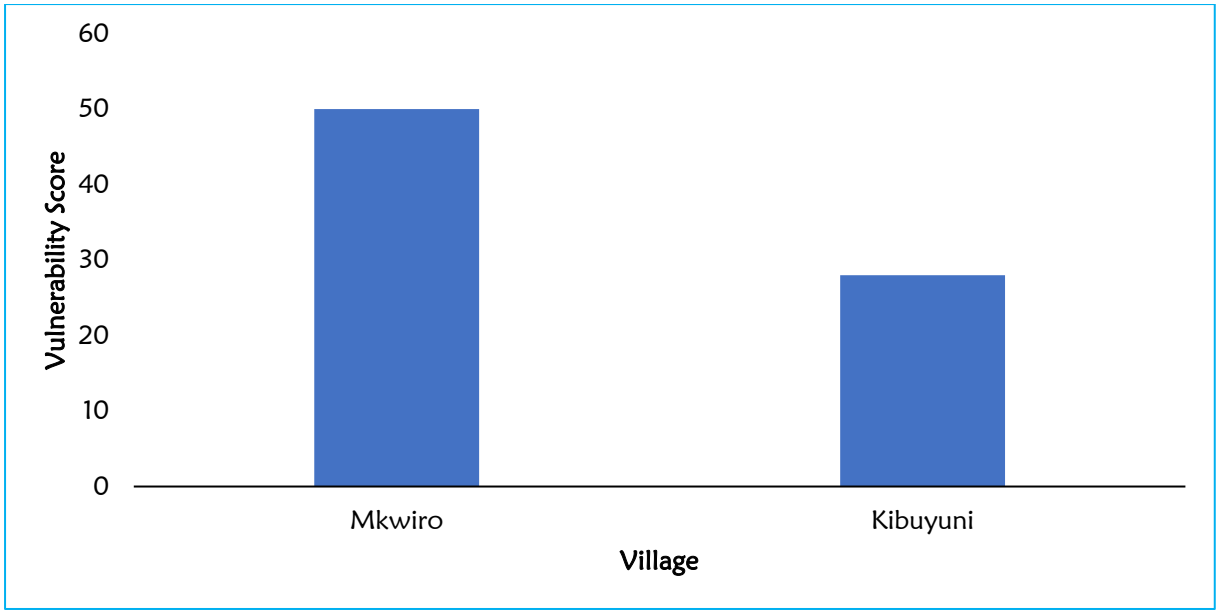

Figure 17: Strong winds vulnerability score in Pongwe Kikoneni Ward (Source: PCVA, 2025)

3.4.4.1 Exposure to Strong winds

Exposure to strong winds in Pongwe–Kikoneni Ward is spatially concentrated in the coastal hotspots of Mkwiro and Kibuyuni, where both environmental and livelihood systems are highly sensitive to the recurring four-year wind cycles. These areas lie along the open coastline facing the Indian Ocean, where seasonal monsoon winds (Kusi and Kaskazi) reach average speeds of 18–25 knots (33–46 km/h) and occasionally exceed 30 knots (55 km/h) around Shimoni (Kenya Meteorological Department, 2023). Their coastal orientation, limited vegetation buffers, and dependence on marine-based livelihoods heighten exposure to wind-related hazards.

In Mkwiro, exposure is driven by a combination of biophysical and socio-economic factors. The village is situated on a narrow coastal strip, exposing homes and community facilities directly to onshore gusts. The settlement pattern, characterized by closely spaced houses made of makuti, wooden frames, and corrugated iron sheets, creates minimal wind resistance. When strong winds occur, roofing materials are easily dislodged, walls collapse, and debris poses safety risks to residents. Public infrastructure, particularly schools, churches, and health outposts, frequently sustain structural damage, disrupting essential services. The artisanal fishing community in Mkwiro also faces heightened exposure; strong winds create turbulent sea conditions that limit access to fishing grounds, damage boats, and increase the risk of maritime accidents.

In Kibuyuni, exposure is primarily linked to the extensive seaweed farming areas and the adjacent mangrove-fringed shoreline. Strong onshore winds generate wave turbulence that breaks seaweed lines, dislodges floats, and increases sediment deposition, which hampers the growth of seaweed and causes heavy production losses. Since seaweed farming is dominated by women's groups, the exposure of these farms directly translates to livelihood instability and reduced household income.

The mangrove forests bordering Kibuyuni and Mkwiro are equally exposed. Strong wind gusts break branches, uproot young mangrove stands, and accelerate shoreline erosion by stripping protective vegetation. The reduced density of mangroves weakens their buffering capacity, exposing inland areas to storm surges and tidal intrusion. Inland forests such as Ganda and Kiriku experience moderate exposure when windstorms extend inland, often causing isolated tree falls and canopy opening that heighten erosion risks.

Exposure in these hotspots is further exacerbated by the lack of windbreak structures, poor roofing materials, and the community's reliance on marine and coastal livelihoods. The combination of physical exposure, inadequate infrastructure, and economic dependence on wind-sensitive resources makes Mkwiro and Kibuyuni the epicentres of wind-related risk within the ward.

3.4.4.2 Socio-Demographic Vulnerability to Strong Winds

Socio-demographic vulnerability to strong winds in Pongwe–Kikoneni Ward is deeply influenced by gender roles, livelihood patterns, income levels, and the structural fragility of housing and community infrastructure. Although strong winds occur approximately once every four years, their socio-economic repercussions are disproportionately felt by women, who form the backbone of household management, coastal livelihoods, and community resilience.

In Mkwiro and Kibuyuni, where exposure is highest, women bear the greatest burden due to their dual roles in sustaining family welfare and contributing to the local economy through seaweed farming, small-scale fish processing, and trading. Strong winds frequently destroy seaweed cultivation structures and drying racks, activities almost entirely run by women's groups. The resulting loss of income undermines household financial stability, food security, and women's ability to meet essential needs such as school fees, healthcare, and water procurement. The repetitive cycle of rebuilding damaged seaweed lines, replacing lost materials, and restoring farm structures imposes both economic and emotional strain on women.

Housing vulnerability further compounds the challenge. A large proportion of women-headed households occupy lightweight, makuti-roofed dwellings that are structurally weak against intense gusts. During strong wind events, roofs are often blown away, household assets are destroyed, and families are forced to seek temporary shelter in schools or churches—spaces that are also commonly damaged. In these situations, women become the primary caregivers managing children, the elderly, and persons abled differently, often without external support. The pressure to secure safety, food, and shelter during displacement heightens psychological stress and limits participation in post-disaster recovery and planning.

The social vulnerability of women is also linked to their limited access to financial capital, credit, and insurance schemes that could buffer against the economic shocks caused by wind-related damage. Cultural and structural barriers continue to restrict women's ownership of land and other productive assets, meaning that they cannot easily access formal financing to rebuild homes or invest in resilient infrastructure. In Kibuyuni, for instance, many women's groups rely on informal savings (chamas) to repair seaweed racks or replace lost equipment, a process that delays recovery and perpetuates cycles of poverty.

Health and safety risks are also more pronounced among women during and after strong wind events. The destruction of health facilities and disruption of water supply systems increase exposure to waterborne diseases and limit access to maternal and child health services. The physical strain of clearing debris, repairing damaged homes, and rebuilding livelihood structures further exposes women to injury and fatigue. Additionally, overcrowded temporary shelters can elevate the risk of gender-based violence and exploitation, particularly when privacy and security are compromised.

Despite their vulnerabilities, women play a critical role in community recovery. They lead informal early warning dissemination, organize collective rebuilding efforts, and participate in mangrove and coastal restoration activities. However, their contributions are often undervalued in formal disaster planning processes.

3.4.5 Sea level rise

Sea level rise has emerged as a persistent and escalating climatic hazard in Mkwiro Village, with water levels increasing steadily each year. The phenomenon is driven by global climate change, particularly ocean thermal expansion and glacial melt, compounded by local factors such as mangrove deforestation, coral reef degradation, and poorly managed shoreline development. The progressive rise in sea level has led to continuous coastal erosion, inundation of low-lying areas, and increased frequency of tidal flooding events, threatening both natural resources and human settlements.

Long-term monitoring along the South Coast indicates that the rate of sea level rise has intensified notably since 2015, consistent with national trends reported by the Kenya Meteorological Department (2023), which estimate an average annual increase of approximately 3.0–3.5 mm per year. In Mkwiro, this gradual but persistent elevation has resulted in visible shoreline retreat, saltwater intrusion, and loss of land previously used for habitation and small-scale infrastructure. Each year, the community records higher tidal marks, indicating that the hazard is no longer episodic but progressive and cumulative in its impacts (Figure 19).

The effects of sea level rise on key natural resources are far-reaching. The mangrove forests fringing Mkwiro Bay have experienced increased root submersion and salinity stress, leading to dieback of species such as *Rhizophora mucronata* and *Ceriops tagal.* The loss of mangrove density reduces their effectiveness as natural buffers against storm surges and wave energy, accelerating coastline erosion. Similarly, the Ganda and Kiriku forests, though inland, are indirectly affected through altered groundwater regimes and increased soil salinization in the lower catchment areas. This inhibits tree growth and regeneration, thereby reducing the overall ecological stability of the landscape.

The encroaching sea has also affected seaweed farms, a major livelihood source for women in Mkwiro and neighboring Kibuyuni. Rising sea levels and stronger tidal currents have altered the salinity and temperature balance of nearshore waters, increasing the risk of seaweed bleaching and disease outbreaks. These conditions cause structural damage to seaweed lines, reducing yields and profitability. Additionally, Rivers Ramisi and Mwena are experiencing saline water intrusion into their lower reaches, affecting freshwater ecosystems and reducing water quality for domestic and agricultural use. The degradation of these hydrological systems has further disrupted fish breeding grounds, threatening the productivity of small-scale fisheries.

At the community level, the impacts of sea level rise are severe and multidimensional. Homesteads along the coastline have been repeatedly inundated, forcing residents to relocate further inland. Traditional graveyards situated close to the shore have been eroded, exposing burial sites and undermining the community's cultural and spiritual heritage. The continuous loss of land has diminished available space for housing, small businesses, and public amenities, leading to increased land-use conflicts. Moreover, the fish landing site in Mkwiro has been heavily eroded by the advancing tides, disrupting the operations of artisanal fishers and traders who depend on it for daily income

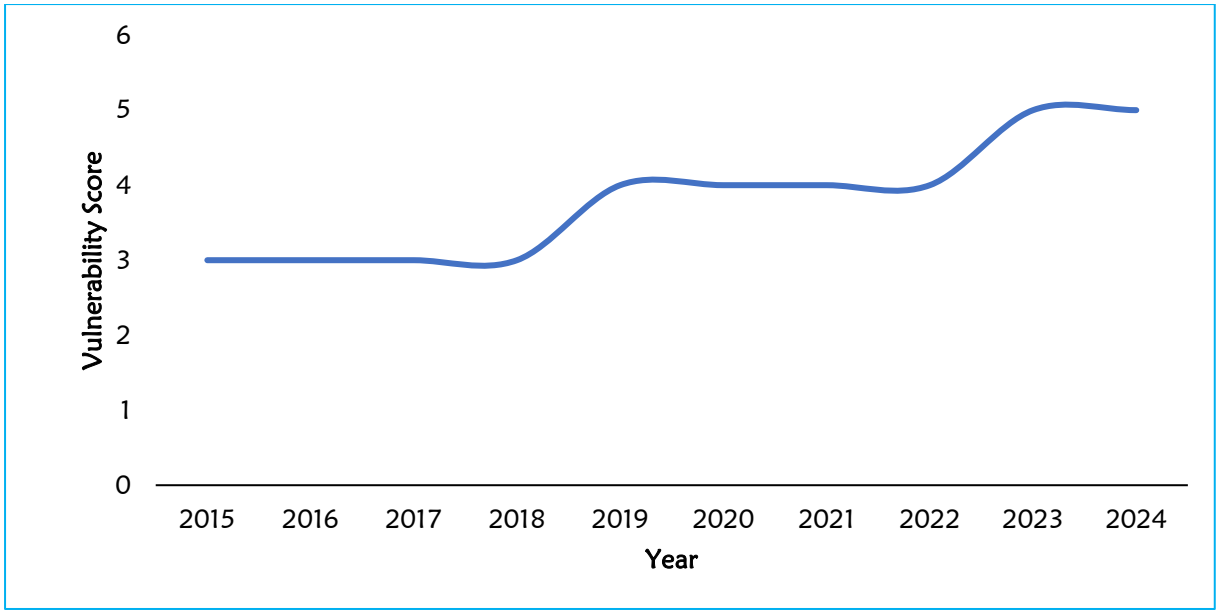

Figure 18: Sea level rise historical trend in Pongwe Kikoneni ward (Source: PCVA,2025)

3.4.5.1 Exposure to Sea level rise

Exposure to sea level rise in Pongwe–Kikoneni Ward is concentrated in the low-lying coastal settlements of Mkwiro and Kibuyuni, where the interaction between rising tidal levels, shoreline erosion, and degraded mangrove buffers has intensified vulnerability to coastal inundation. Over the past decade, sea level rise has become a progressively worsening phenomenon, occurring annually with measurable increments averaging 3.0–3.5 millimetres per year along Kenya's southern coastline (Kenya Meteorological Department, 2023). These changes correspond with both global climate trends and local ecosystem degradation, making Mkwiro and Kibuyuni among the most exposed villages within the ward.

In Mkwiro, exposure is driven by the village's direct proximity to the open ocean, low elevation, and erosion of natural shoreline barriers. The advancing sea has encroached onto residential areas and community spaces, submerging portions of homesteads and farmlands during high tides. The fish landing site, an essential hub for artisanal fishers, has faced repeated inundation, with waves eroding the landing zone and washing away support infrastructure such as drying racks, fish storage sheds, and access paths. Over the years, tidal surges have also reached graveyards located along the coastline, unearthing burial sites and causing significant cultural and psychological distress within the community. The loss of land to the sea has not only reduced habitable space but also displaced households, forcing informal resettlement inland and heightening competition for limited land resources.

In Kibuyuni, exposure to sea level rise is particularly evident in the intertidal and nearshore areas where extensive seaweed farms are located. The rising sea and intensifying wave energy have increased the frequency of farm submersion, disrupting cultivation cycles and reducing the stability of floating structures. As the sea level continues to rise, the boundary between arable and saline zones is shifting inland, causing saltwater intrusion that degrades coastal soils and reduces the productivity of food crops such as cassava and maize. The community's dependence on these coastal ecosystems for both subsistence and income amplifies their exposure to the hazard.

The mangrove forests that once provided a natural protective barrier have also been compromised by prolonged inundation and increased salinity. Species such as *Avicennia marina*

and *Rhizophora mucronata* exhibit stress symptoms, including stunted growth and reduced regeneration. The degradation of mangrove cover reduces the coastline's natural buffering capacity, exposing inland settlements and infrastructure to wave erosion and tidal flooding. The loss of these mangroves further exposes Rivers Ramisi and Mwena to saline water intrusion, which extends several kilometres upstream during high tides, affecting freshwater habitats and community water sources.

3.4.5.2 Socio-Demographic Vulnerability to Sea Level Rise

Socio-demographic vulnerability to sea level rise in Pongwe–Kikoneni Ward is shaped by the interaction of gender, poverty, livelihood dependence, and settlement patterns, with the most pronounced impacts observed in Mkwiro and Kibuyuni villages. These coastal communities are predominantly low-income and depend heavily on climate-sensitive livelihoods such as artisanal fishing, seaweed farming, and small-scale agriculture, all of which are directly threatened by the encroaching sea. The gradual annual increase in sea level, estimated at 3.0–3.5 mm per year (Kenya Meteorological Department, 2023), has translated into recurrent tidal flooding, shoreline erosion, and displacement, amplifying existing social and economic inequalities.

Women are disproportionately affected due to their socio-economic roles and limited access to adaptive resources. In Mkwiro and Kibuyuni, most women derive their livelihoods from seaweed farming, fish trading, and shell collection. Rising sea levels and stronger tidal currents frequently destroy seaweed lines, submerge drying racks, and reduce nearshore productivity. These losses directly affect women's income and financial independence, forcing them to rely more on male partners or informal credit systems. The increased salinity of coastal soils has also reduced the productivity of household kitchen gardens, undermining food security and women's capacity to provide for their families.

Displacement and loss of property have compounded women's vulnerability. As the sea continues to encroach inland, homesteads located near the shoreline, many of them belonging to female-headed households, are being eroded or flooded during high tides. Women are often the primary caregivers during such crises, responsible for evacuating children, the elderly, and persons abled differently, while safeguarding household assets under increasingly hazardous conditions. Temporary relocation leads to loss of privacy, heightened exposure to gender-based violence, and psychological stress, particularly where shelter facilities are inadequate.

Poverty and limited access to financial capital further intensify vulnerability. Most coastal women lack land ownership rights and collateral, restricting their ability to access loans or insurance for rebuilding or relocation. Female-led community cooperatives that previously supported seaweed production and savings have been destabilized by continuous damage to coastal farms, reducing their collective capacity to recover. Consequently, the economic empowerment of women, central to the local blue economy, has been progressively eroded by the effects of sea level rise.

The broader community also faces cascading socio-economic effects. The fish landing site in Mkwiro, a key economic and social hub, is under continuous threat from shoreline erosion, disrupting both fishing activities and post-harvest operations managed by women and youth. The loss of burial grounds and ancestral sites due to erosion has affected the community's cultural and spiritual identity, leading to emotional distress and a diminished sense of belonging. Meanwhile, saline intrusion into rivers Ramisi, and Mwena has reduced the availability of freshwater, forcing women and girls to travel longer distances inland to fetch potable water, thereby increasing their daily labour burden.

3.4.6 Human-Wildlife Conflicts

Human–wildlife conflicts (HWCs) in Kiwambale Village occur approximately once every eight years and have shown a gradual but notable increase in frequency and intensity over the past three decades. The first recorded incident dates back to 1993, marking the beginning of intermittent encounters between wildlife and local communities (Figure 20). However, since 2023, the occurrence of HWCs has become more pronounced, largely driven by habitat degradation, deforestation, and expanding human settlements into natural habitats. Prolonged droughts and declining water and food resources have further pushed wildlife closer to human settlements in search of sustenance, escalating the frequency of confrontations. The most affected areas include Kiwambale, Ganda, Mzizima, and Shimoni village units, where agricultural fields, homesteads, and community water points closely interface with forest and riparian ecosystems.

In Kiwambale, the primary species involved in these conflicts are baboons and crocodiles. Baboons and wild pigs which inhabit the nearby Ganda and Kiriku forests, frequently raid farms, destroying rice, maize, cassava, and fruit crops, leading to significant economic losses for smallholder farmers. Their incursions are most frequent during dry periods when natural food sources in the forest are scarce. Repeated crop losses not only undermine food security but also fuel hostility toward wildlife, prompting the use of traps or poisoning in retaliation, actions that further disrupt ecological balance. In some cases, baboons have also attacked goats and poultry, aggravating tensions between communities and conservation authorities.

Crocodile-related conflicts, on the other hand, are concentrated along Rivers Ramisi, where communities depend heavily on riverine water for domestic use, fishing, rice farming and small-scale irrigation. During prolonged dry spells, these rivers shrink, concentrating crocodiles in shallow pools that are shared with human users. Several incidents of attacks on livestock and humans have been reported, particularly when women and children fetch water or when fishers operate along overgrown riverbanks. These encounters have instilled fear among residents, reduced access to essential water sources, and disrupted livelihood activities such as fishing and riverine farming.

The impacts of HWCs on natural resources are equally significant. Retaliatory actions, such as the destruction of wildlife habitats, use of snares, and clearing of vegetation along riverbanks, have contributed to the degradation of ecosystems that support biodiversity and water regulation. The Ganda and Kiriku forests, which serve as habitats for primates and other small mammals, face increasing encroachment as communities expand farmland in an effort to avoid conflict zones. This land-use pressure diminishes forest cover, exacerbates soil erosion, and reduces the availability of forest products vital to household sustenance. In riparian zones, the fear of crocodile attacks has led to neglect of riverine management activities, resulting in overgrown reeds, reduced access to clean water, and increased siltation.

At the community level, the effects of HWCs extend beyond direct physical harm and property loss. Recurrent attacks and crop destruction have eroded household incomes and food security, pushing already vulnerable families into deeper poverty. Women, who are primarily responsible for water collection and crop cultivation, face heightened risk due to their frequent presence near rivers and farmlands. These conflicts also have psychosocial implications, as fear of attacks limits movement and participation in livelihood activities, especially for children and the elderly.

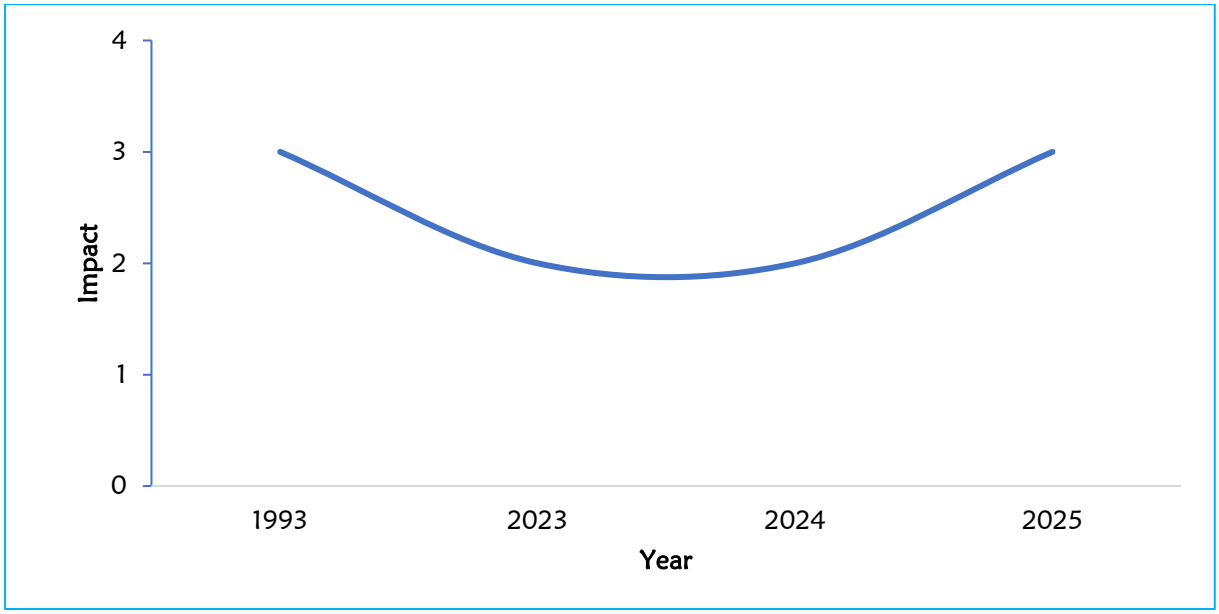

Figure 19: Human wildlife conflicts historical trend in Pongwe Kikoneni Ward (Source: PCVA, 2025)

3.4.6.1 Exposure to Human–Wildlife Conflicts

Exposure to human–wildlife conflicts (HWCs) in Pongwe–Kikoneni Ward is localized yet significant, with Kiwambale, Ganda, Mzizima, and Shimoni villages identified as the primary hotspots. These areas experience human–wildlife interactions approximately once every eight years, a frequency that has become more pronounced since 2023 due to rapid environmental degradation, expansion of human settlements into wildlife habitats, and recurrent drought conditions. The proximity of farmlands and water sources to natural habitats, particularly along Rivers Ramisi and within forested zones such as Ganda and Kiriku, has intensified exposure for households dependent on these ecosystems for food, water, and income.

In Kiwambale, exposure is highest because the community directly borders riparian zones that serve as habitats and watering areas for crocodiles. The local population relies heavily on these rivers for domestic use, livestock watering, fishing, and small-scale irrigation. During drought periods, when river levels drop and water pools contract, human and crocodile activity overlap significantly. This spatial overlap increases the risk of crocodile attacks, particularly on women and children who fetch water or wash clothes along the riverbanks. Recurrent incidents have led to restricted water access, reduced fishing activity, and heightened fear among residents, undermining both food security and community wellbeing.

In the upland areas of Ganda and Mzizima, exposure to baboon raids is the dominant form of human–wildlife conflict. These primates emerge from the adjacent Ganda and Kiriku forests to forage on nearby farms, particularly during dry spells when natural food sources are scarce. The close proximity of agricultural fields to degraded forest edges makes smallholder farmers highly exposed. Baboons frequently invade maize, cassava, and fruit farms, resulting in substantial losses for households that depend on subsistence farming. Their persistence and adaptability make conventional deterrence measures, such as scarecrows or fencing, largely ineffective. As a result, farmers must remain constantly vigilant, often sacrificing time for other income-generating activities.

In Shimoni, exposure stems from the village's adjacency to both marine and terrestrial habitats, where mangrove forests and tidal creeks support diverse wildlife populations. Occasional crocodile sightings in mangrove channels and along the lower Ramisi River increase community anxiety, particularly among artisanal fishers and seaweed farmers who operate in shallow tidal

zones. The destruction of mangrove buffers has intensified this exposure by reducing the natural separation between human settlements and wildlife movement corridors.

The widespread dependence on natural resources further amplifies exposure across these hotspots. The degradation of forest and river ecosystems, coupled with habitat fragmentation, forces wildlife to roam beyond their traditional ranges in search of food and water. Simultaneously, community activities such as charcoal burning, firewood collection, and riverbank cultivation extend human presence deeper into wildlife territories, heightening the likelihood of encounters.

3.4.6.2 Socio-Demographic Vulnerability to Human–Wildlife Conflicts

Socio-demographic vulnerability to human–wildlife conflicts (HWCs) in Pongwe–Kikoneni Ward is shaped by livelihood dependence, poverty levels, gender roles, and settlement patterns, with the most affected communities being those in Kiwambale, Ganda, Mzizima, and Shimoni villages. Although these incidents occur approximately once every eight years, their impacts are disproportionately borne by households with limited economic resilience and by women, who play central roles in farming, water collection, and household management.

In Kiwambale, where most HWC incidents involve crocodile attacks, women and children are at the greatest risk. Due to traditional gender roles, women are primarily responsible for fetching water, washing clothes, and managing household chores along riverbanks such as Ramisi and Mwena. During prolonged droughts, when water levels drop and human–crocodile contact zones increase, their exposure to danger rises significantly. Several reported attacks and near misses have instilled fear and restricted women's mobility, forcing them to rely on unsafe or distant water sources. This not only increases their daily labour burden but also heightens risks to health, sanitation, and personal safety.

In the upland villages where baboon raids on crops are recurrent, women again face heightened vulnerability. As the primary custodians of smallholder farms, they bear the direct consequences of crop destruction, food shortages, and economic losses. Repeated raids on maize, cassava, and fruit crops diminish household food reserves, reducing dietary diversity and forcing families to purchase food at higher prices during lean seasons. The constant need to guard farms against baboons often compels women to spend long hours in the fields, reducing time for income-generating activities and childcare. This unpaid and strenuous labour further exacerbates gender inequality and economic dependence.

Poverty and limited access to resources amplify these vulnerabilities. Most households in the affected villages are low-income and lack resources to invest in protective measures such as fencing, improved storage facilities, or alternative water access points. Women-headed households, which are increasingly common due to male outmigration for labour, are particularly disadvantaged as they lack both the manpower and financial means to mitigate risks or recover from losses. In addition, limited access to compensation mechanisms or wildlife insurance schemes leaves many families unable to recover from attacks or property destruction, perpetuating cycles of poverty and dependence on unsustainable resource use.

Children and the elderly also face social vulnerability due to their dependence on caregivers. School attendance drops temporarily during heightened conflict periods, especially when children are tasked with guarding crops or avoiding riverine zones. Elderly residents, often left to manage small farms or herds, are physically less able to deter wildlife or relocate livestock, making them more susceptible to economic and safety losses.

At the community level, HWCs erode social cohesion and trust in conservation institutions. The lack of timely compensation and perceived neglect by wildlife authorities have fuelled resentment among residents, particularly women's groups who feel excluded from formal decision-making and conflict-resolution mechanisms. This has weakened cooperation in wildlife conservation programs and increased retaliatory actions such as habitat clearing or animal trapping, which further degrade the ecosystem.

3.5 Classification of Exposure and Resilience through Livelihoods

The ward is characterised by livelihoods that are highly exposed to environmental, economic and institutional shocks, while its capacity for resilience remains constrained by limited income diversity, low margins and persistent poverty. The dominant cash-income sources are informal and closely tied to marine/coastal and transport services with modest supplementary incomes from micro-enterprises, small-scale agriculture and livestock or poultry. At the same time, county data indicate that Kwale's poverty headcount has been around 47 per cent, higher than the national average. These factors combine to yield a system with narrow buffers and high vulnerability, thereby demanding careful classification of exposure and resilience through the livelihood lens.

3.5.1 Livelihood Structure and Monthly Incomes

The livelihood profile across the five sampled villages of Pongwe Kikoneni Ward exhibits a diverse income structure influenced by ecological endowments, gendered labour divisions, and access to market opportunities. Fishing remains the dominant livelihood, generating monthly incomes of KES 15,000 in Kibuyuni and Mkwiro, KES 70,000 in Anzuwani, KES 30,000 in Kiwambale, and KES 75,000 in Barabarani (Majoreni), underscoring its economic centrality but also the spatial disparity in productivity and market reach (Table 4). Seaweed farming, practiced exclusively in Kibuyuni, yields about KES 3,000 per month, mostly supporting women's self-help groups engaged in marine-based enterprises. Livestock production contributes KES 4,000 in Kibuyuni, KES 9,000 in Kiwambale, and KES 18,000 in Barabarani (Majoreni), indicating its growing importance inland where grazing space is available. The sale of mangrove seedlings provides substantial conservation-linked income, amounting to KES 32,000 in Kibuyuni, KES 58,000 in Anzuwani, and KES 25,000 in Kiwambale. Motorbike transport, a youth-driven livelihood, generates KES 7,000 in Kibuyuni, KES 40,000 in Kiwambale, and KES 18,000 in Barabarani (Majoreni), reflecting expanding informal transport networks.

Government cash transfers (KES 2,000) and dowry payments (KES 100,000), both reported in Kibuyuni, serve as supplementary and culturally embedded income streams. Tourism yields high returns, with KES 60,000 in Mkwiro and KES 80,000 in Anzuwani, benefiting households near beach tourism corridors. Small-scale businesses provide variable earnings, KES 2,500 in Mkwiro, KES 20,000 in Anzuwani, KES 8,000 in Kiwambale, and KES 15,400 in Barabarani (Majoreni), while farming generates KES 20,000 each in Anzuwani and Kiwambale, and KES 4,000 in Barabarani (Majoreni), indicating moderate engagement in crop cultivation. Income from loans or chamas (KES 20,000 in Kiwambale and Barabarani (Majoreni)) and weaving (KES 3,000 in Kiwambale and KES 6,000 in Barabarani (Majoreni)) highlights the role of community savings groups and women's craft enterprises in household resilience. Casual labour contributes KES 10,000 in Mkwiro and KES 9,500 in Kiwambale, while beekeeping, recorded solely in Barabarani (Majoreni) at KES 83,000, stands out as a high-value climate-resilient livelihood. Collectively, these figures illustrate a multidimensional economic landscape where fishing, tourism, and emerging enterprises such as beekeeping and mangrove seedling production are central to income stability, while women-led initiatives and informal group savings mechanisms provide critical buffers against livelihood shocks.

Table 2:Monthly income earnings by livelihood per village (Source: PCVA, 2025)

Livelihood	Income (KES)				
	Kibuyuni	Mkwiro	Anzuwani	Kiwambale	Barabarani (Majoreni)
Fishing	15,000	15,000	70,000	30,000	75,000
Sea weed farming	3,000	-	-	-	-
Livestock production	4,000	-	-	9,000	18,000
Sale of Mangrove Seedlings	32,000	-	58,000	25,000	-
Motorbike transport	7,000	-	-	40,000	18,000
Government cash transfers	2,000	-	-	-	-
Dowry	100,000	-	-	-	-
Tourism	-	60,000	80,000	-	-
Small scale businesses	-	2,500	20,000	8,000	15,400
Farming	-	-	20,000	20,000	4,000
Loans/Chamas	-	-	-	20,000	20,000
Weaving mats	-	-	-	3,000	6,000
Casual labour	-	10,000	-	9,500	-
Beekeeping	-	-	-	-	83,000

The analysis reveals a strong correlation between women's financial constraints and the income disparities observed across the five villages in Pongwe Kikoneni Ward. In low-income villages such as Anzuwani and Kibuyuni, where household earnings are heavily reliant on irregular sources like fishing and mangrove seedling sales, 100% of women lacked money for food, clothing, and medication, reflecting severe economic vulnerability and dependence on unstable natural resource-based livelihoods. In Barabarani (Majoreni), despite higher household incomes from beekeeping and fishing, 83.3% of women still lacked money for food and 50% for medication, underscoring unequal intra-household income control. Kiwambale and Mkwiro, with more diversified income streams from transport, tourism, and casual work, showed relatively lower deprivation levels, particularly for food (50% and 33.3%, respectively), though access to medication remained critical, affecting all women in Kiwambale and two-thirds in Mkwiro (Figure 21).

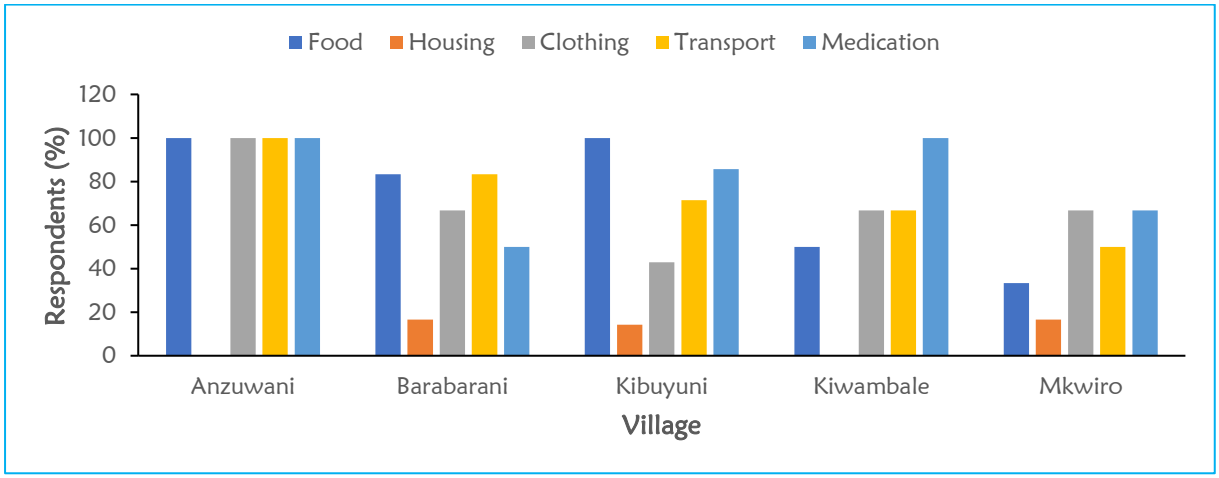

Figure 20: Respondents who reported to have lacked money for basic needs in the last 12 Months (Source: PCVA,2025)

3.5.2 Vulnerabilities of livelihood options

The livelihoods across Pongwe Kikoneni Ward are highly climate-sensitive, with varying degrees of exposure and adaptive capacity depending on their dependence on natural resources and

environmental stability. Fishing, the primary livelihood for most households, is acutely vulnerable to rising sea surface temperatures, erratic rainfall patterns, and increasing sea levels that disrupt fish breeding grounds and alter migratory routes. Strong winds, ocean currents, and tidal surges affect fishing days and increase the risk of accidents at sea, while saltwater intrusion and coral bleaching reduce fish stock diversity and productivity. Such climatic changes directly threaten household income stability and food security, particularly for small-scale fishers using traditional gear.

Seaweed farming, mainly practiced by women in Kibuyuni, is equally sensitive to ocean warming and salinity fluctuations. Rising sea temperatures trigger seaweed bleaching, while heavy rainfall and sediment runoff reduce water clarity and hinder photosynthesis. These changes lower yields and compromise quality, limiting access to premium export markets. The lack of alternative sites for relocation, coupled with damage to drying racks during storms, intensifies vulnerability, particularly for women who depend on this livelihood for supplemental income.

Livestock production in Kiwambale and Barabarani (Majoreni) faces increasing threats from prolonged droughts that cause water scarcity, pasture depletion, and livestock mortality. Reduced rainfall affects fodder availability and triggers outbreaks of climate-related diseases such as foot-and-mouth and tick-borne infections. The dependence on rain-fed grazing systems and limited access to water points heighten household vulnerability, especially during dry spells when women and children must walk longer distances to fetch water for both domestic and livestock use.

The sale of mangrove seedlings, though promoting ecological restoration, is indirectly climate-dependent. Extended dry periods affect mangrove seed germination and survival rates, while flooding and coastal erosion from sea-level rise wash away nursery sites. Additionally, the availability of propagules fluctuates with seasonal rainfall and tidal conditions, making this livelihood unpredictable under changing climate regimes.

Motorbike transport is susceptible to climate extremes such as heavy rains and flooding that render roads impassable, isolate villages, and damage infrastructure. Riders lose income during prolonged wet seasons, while dust and heat during drought periods increase operational costs and mechanical wear. Similarly, tourism in Mkwiro and Anzuwani is highly vulnerable to seasonal weather variability and climate-induced environmental degradation. Coral bleaching, beach erosion, and storm surges affect aesthetic and recreational value, leading to reduced visitor arrivals and income losses for households reliant on tourism-related services.

Small-scale businesses and casual labour are indirectly affected by climate change through disruptions in supply chains, reduced consumer purchasing power, and inflation during prolonged droughts or floods. Extreme weather limits market accessibility, damages goods, and constrains daily earnings, particularly for women traders who operate roadside stalls. Farming activities are among the most climate-exposed, given the reliance on rain-fed systems. Irregular rainfall, droughts, and pest infestations lead to frequent crop failures, reduced soil fertility, and lower yields of staple crops such as maize, cassava, and sorghum. Flooding during heavy rains also causes erosion and nutrient loss, further undermining productivity and threatening household food security.

While beekeeping is generally more climate-resilient, it still faces risks from rising temperatures and prolonged dry spells that reduce flowering cycles and nectar availability. Deforestation and changes in vegetation composition driven by climate shifts further threaten honey yields and colony health. Weaving, on the other hand, is indirectly vulnerable as the availability of raw

materials like palm fronds depends on local vegetation health, which is affected by erratic rainfall and prolonged dry periods.

Even seemingly less climate-dependent income sources such as chamas, dowry, and cash transfers are affected indirectly. During prolonged droughts or floods, reduced household incomes lower savings and repayment capacity within chamas, weakening group resilience. Similarly, when climate impacts reduce household assets or increase dependency, the value of dowry and the effectiveness of cash transfers diminish, offering limited protection against livelihood shocks.

3.5.3 Overall Classification and Implications

The cumulative assessment classifies Pongwe Kikoneni Ward as a high-exposure and low-to-moderate-resilience area, where livelihoods and household welfare are strongly influenced by climate variability and environmental degradation. The community's heavy dependence on climate-sensitive sectors such as fishing, seaweed farming, small-scale farming, and livestock production exposes households to frequent income shocks during droughts, floods, and coastal storms. High poverty levels, limited access to productive assets, and the disproportionate burden of unpaid domestic and care work among women further constrain adaptive capacity. To reduce these vulnerabilities, the ward requires integrated adaptation strategies that enhance income stability across key sectors, particularly through value addition in fisheries, beekeeping, and mangrove-based enterprises. Promoting livelihood diversification into climate-resilient and higher-value chains, expanding access to microfinance and cooperative savings platforms, and introducing labour-saving technologies to reduce women's time poverty will strengthen economic resilience. In parallel, investment in ecosystem-based adaptation measures—including mangrove restoration, coastal forest protection, and sustainable fisheries management—should be prioritized. Complementary improvements in infrastructure, such as road access, cold storage facilities, and water supply systems, will be crucial to transition Pongwe Kikoneni Ward from reactive coping mechanisms toward proactive, community-driven climate adaptation and resilience building.

3.6 Analyzing Causes of Vulnerability

3.6.1 Classification of Severity of the Vulnerabilities and Their Causes

The ward exhibits varying levels of vulnerability across its five sampled villages; Kibuyuni, Mkwiro, Anzuwani, Kiwambale, and Barabarani (Majoreni), driven by their ecological settings, livelihood dependence, and adaptive capacities. The classification was derived from participatory scoring during the PCVA (2025) and spatial mapping of hazard exposure. Kibuyuni and Mkwiro recorded the highest vulnerability due to their proximity to the coastline and dependence on marine-based livelihoods such as fishing, seaweed farming, and tourism, which are increasingly affected by rising sea levels, saline intrusion, and coastal erosion. Anzuwani and Barabarani (Majoreni) were categorized as moderately vulnerable, facing drought stress, flooding in low-lying areas, and declining productivity in farming and beekeeping. Kiwambale, located further inland, was assessed as moderately to highly vulnerable due to recurrent droughts, water scarcity, and pasture depletion affecting livestock and small-scale agriculture.

The severity of vulnerability in Pongwe Kikoneni is further compounded by limited livelihood diversification, widespread poverty, and inadequate institutional and technical support. Households dependent on marine resources such as fish and mangroves are highly exposed to oceanic and climatic fluctuations, while those reliant on rain-fed farming face recurrent crop failures. Women remain disproportionately affected due to their concentration in low-return livelihoods such as weaving, poultry rearing, and petty trade, coupled with restricted access to financial services and increased unpaid care work during climate shocks. The classification

highlights that vulnerability in Pongwe Kikoneni is multidimensional, shaped by a combination of biophysical exposure, socio-economic inequality, and gendered constraints.

3.6.2 Identification of Unsafe Conditions

Unsafe conditions across Pongwe Kikoneni Ward are closely linked to settlement patterns, ecological degradation, and inadequate infrastructure. Communities living along the Kibuyuni–Mkwiro coastal belt face recurrent flooding, storm surges, and saltwater intrusion that damage houses, contaminate water sources, and destroy fishing gear and seaweed farms. Poor drainage systems and unregulated waste disposal aggravate these impacts by blocking stormwater flow, while the degradation of mangrove forests along the Shimoni and Gazi creeks exposes the shoreline to intensified erosion and tidal surges. In Kiwambale and Barabarani (Majoreni), the absence of water pans, desilted boreholes, and climate-resilient roads limits access to markets and emergency services during floods and droughts.

From a gender perspective, unsafe conditions also manifest through the increased workload placed on women who must walk longer distances to fetch water and fuelwood during dry spells, often through degraded forested and mangrove areas, exposing them to both physical and health risks. Limited access to climate-resilient facilities such as raised storage sheds, secure fish-landing sites, and wind-resistant housing further increases vulnerability, particularly among female-headed households, widows, and persons abled differently.

3.6.3 Identification of Dynamic Pressures or Determinants of Vulnerability

Dynamic pressures in Pongwe Kikoneni arise from ongoing socio-economic and environmental processes that intensify vulnerability. Rapid population growth and unregulated settlement expansion have encroached on coastal and forest ecosystems, while overdependence on natural resources has accelerated degradation of the Ganda and Kiriku Forests and mangrove belts in Mkwiro and Kibuyuni. In Barabarani (Majoreni) and Kiwambale, uncontrolled livestock grazing and slash-and-burn cultivation inhibit natural forest regeneration, while clearing for small-scale agriculture has fragmented forest corridors. The degradation of these ecosystems undermines their capacity to regulate microclimate, store carbon, and buffer against floods and strong winds.

Mangrove degradation in Mkwiro and Kibuyuni has similarly been driven by increased pole harvesting, sedimentation from upland erosion, and poor enforcement of conservation regulations. Declining fish stocks and reduced seaweed productivity have prompted households to adopt unsustainable coping mechanisms such as charcoal burning and sand harvesting, perpetuating a cycle of environmental decline and economic vulnerability. Weak institutional coordination among the Ward Climate Change Planning Committee (WCCPC), county departments, and agencies such as KFS, KMFRI, and KWS further limits effective resource management. Insufficient early warning systems, irregular enforcement of by-laws, and limited access to microfinance constrain both preparedness and adaptive response. Socially, women and youth remain underrepresented in mangrove restoration and fisheries governance, undermining inclusive and sustained adaptation planning.

3.6.4 Identification of Underlying Causes of Vulnerability

The underlying causes of vulnerability in Pongwe Kikoneni are rooted in structural poverty, environmental degradation, and institutional weaknesses. High poverty levels limit access to diversified income sources, productive assets, and adaptive technologies, leaving many households dependent on fishing, seaweed farming, and rain-fed agriculture, all highly climate-sensitive. With an average household size of 11 persons, dependency ratios are high, straining food, water, and energy resources. Environmental degradation, particularly in Ganda and Kiriku Forests and along mangrove belts in Mkwiro and Kibuyuni, has reduced the capacity of

local ecosystems to regulate microclimates, stabilize soils, and provide essential goods such as fuelwood and construction materials. Loss of vegetation has also led to declining groundwater recharge, worsening drought impacts in inland villages such as Kiwambale and Barabarani (Majoreni).

Institutional constraints, including inadequate coordination between county departments and local committees, insufficient funding for climate adaptation programs, and weak enforcement of environmental laws, exacerbate vulnerability. The absence of localized early warning systems and limited dissemination of climate information further hinder proactive response. Gender inequality compounds these issues: patriarchal norms and customary land tenure systems restrict women's ownership of assets and participation in decision-making. Women's concentration in unpaid care work and low-return economic activities reduces their adaptive capacity and overall household resilience.

3.6.5 Prioritization of Causes

Participatory ranking during the PCVA exercises identified poverty, ecosystem degradation, and weak institutional coordination as the top three drivers of vulnerability in Pongwe Kikoneni Ward. Poverty limits investment in adaptive technologies, while degradation of Ganda and Kiriku Forests and the Mkwiro–Kibuyuni mangrove belt diminishes ecosystem services crucial for buffering climate impacts. Weak institutional linkages between community structures, county departments, and national agencies slow implementation of adaptation interventions.

Secondary but equally important factors include gender inequality, rapid population growth, and limited access to climate information and microcredit. Addressing these causes requires integrated, gender-responsive climate adaptation strategies. Key interventions include restoration of degraded forest and mangrove ecosystems, promotion of climate-smart agriculture and beekeeping, investment in water harvesting and irrigation infrastructure, and strengthening of early warning and response systems. Empowering women and youth through targeted capacity-building, access to financial resources, and inclusion in natural resource governance will be essential to enhancing resilience and ensuring sustainable climate adaptation across all five villages of Pongwe Kikoneni Ward.

4 Chapter 4: Gender dimensions, Unpaid work and Vulnerability Analysis for Climate Hazards

Gender and age dynamics strongly shape patterns of exposure, sensitivity, and resilience to climate hazards in Pongwe Kikoneni Ward. The Ward experiences recurrent droughts, seasonal floods, widespread crop pests and livestock diseases, strong winds, advancing sea level rise in coastal fringe areas, and persistent human–wildlife conflicts, particularly crocodile attacks in Kiwambale village due to its proximity to the Ramisi River. These hazards intersect with entrenched gender inequalities, socio-economic disparities, and livelihood dependence on climate-sensitive sectors such as subsistence farming, livestock keeping, artisanal fishing, and small-scale trade (Table 3). As in much of Kwale County, women and girls shoulder disproportionate burdens arising from socially assigned domestic roles, limited access to productive assets, and restricted decision-making power, which collectively heighten their vulnerability to climate shocks (UN Women, 2023; IPCC, 2022).

In Pongwe Kikoneni, women's and girls' vulnerability is amplified by extensive unpaid domestic and care responsibilities, including cooking, laundry, childcare, fetching water, collecting firewood, and supporting farm work. During the PCVA, this daily workload was economically valued at KES 1,800, underscoring its substantial contribution to household functioning despite its invisibility in formal economic systems. Climate hazards, particularly drought, greatly intensify this unpaid labour. As water sources such as shallow wells and seasonal streams dry up, women and girls walk longer distances, sometimes exceeding 4–6 kilometres, to access water and fuelwood. This contributes to severe fatigue, heightened psychosocial stress, musculoskeletal strain, and reduced time for income-generating activities or participation in ward-level adaptation planning.

Drought remains the most severe hazard in the Ward, driving crop failure, livestock mortality, food shortages, and water scarcity. Women face increased burdens in securing food and water, while female-headed households, common in parts of the ward struggle with limited labour support and reduced adaptive capacity. Men experience reduced farm productivity, declining livestock value, and income loss, increasing household financial pressure. Youth (18–35) experience loss of casual farm work, growing idleness, and heightened inclination towards risky coping strategies. Elderly individuals and Persons Abled Differently (PADs) face reduced mobility and heavy dependence on caregivers during prolonged droughts, limiting their access to water points, relief supplies, and health services.

Flooding, particularly in low-lying and riverine zones, disrupts household livelihoods by damaging farms, homes, and social infrastructure. Women's caregiving workload escalates as they support sick family members, salvage household assets, and secure food and water during market disruptions. Flooding frequently contaminates shallow wells and boreholes, forcing women and girls to travel farther for safe water. Men, especially those depending on fishing, transport, and casual labour, lose income when access roads and market routes are cut off. Youth experience disrupted schooling, increased idleness, and higher vulnerability to exploitation. Elderly persons and PADs face acute challenges evacuating or accessing safe shelter due to mobility constraints.

Pests and livestock diseases, including fall armyworm, tick-borne diseases, and poultry infections, significantly impact households dependent on farming and livestock rearing. Women, who manage subsistence plots and small livestock, experience reduced food security and income. Their limited access to extension services, certified seed, and pest control technologies lowers their adaptive capacity (FAO, 2021). Men incur financial losses from livestock mortality and reduced crop viability, while youth lose casual labour opportunities.

Elderly individuals and PADs face heightened dependency and increased nutritional vulnerability when household food stocks decline.
Human–wildlife conflict represents a unique and severe hazard in Pongwe Kikoneni Ward, particularly crocodile attacks in Kiwambale village, where households rely on the Ramisi River for water, laundry, fishing, and small-scale irrigation. Women and girls are at increased risk when fetching water or washing along the riverbanks, restricting their mobility and reducing productivity. Men guarding farms near riverine zones also face injury risks, while youth helping with grazing or water collection encounter frequent danger. Elderly persons and PADs are extremely vulnerable due to mobility limitations and inability to flee during wildlife encounters, increasing dependence on family members for safety and reporting. Limited access to compensation and bureaucratic delays further deepen household vulnerability.

Environmental stress linked to sea level rise affects coastal households, particularly those engaged in small-scale fishing, shoreline trade, and marine-based livelihoods. Women involved in beachside trade or seaweed drying experience disruptions when rising tides or storms destroy drying racks, market stalls, or storage areas. Men incur losses from damaged boats and submerged landing sites. Youth lose emerging opportunities in marine tourism and fishing support services. Elderly populations and PADs face significant constraints relocating to safer areas, increasing their physical and emotional vulnerability during tidal surges.

Strong winds compound existing vulnerabilities by increasing risks of structural damage to homes, disrupting household routines, and elevating caregiving burdens for women. Fishermen face higher risks at sea due to capsizing and gear loss. Youth face injury risks from collapsing trees and structures, while elderly persons and PADs experience severe challenges securing safe shelter quickly.

Gender-based violence (GBV) further compounds these gendered vulnerabilities, functioning both as a climate-induced risk and as a structural barrier to resilience in Pongwe Kikoneni Ward. Emerging global evidence demonstrates a strong association between climate stressors and rising GBV incidence, with every 1°C increase in temperature linked to a 4.7% rise in intimate partner violence (UN Spotlight Initiative, 2024). Heatwaves have been associated with a 28% increase in femicide, while climate-driven displacement heightens risks of child marriage, sexual exploitation, and trafficking (UN Spotlight Initiative, 2024). In Kenya, rising temperatures and socioeconomic stressors have coincided with escalating femicides with over 930 cases documented between 2016 and 2024, highlighting how climate stress intersects with entrenched gender inequalities (Kenya Femicide Data Study, 2024). These dynamics are most acute in fragile, resource-dependent communities such as Pongwe Kikoneni, where drought, floods, and livelihood loss amplify household tensions, reduce economic security, and disproportionately expose women and girls to physical, psychological, and sexual violence. GBV undermines climate resilience by restricting women's mobility, decision-making autonomy, access to productive assets, and participation in ward-level adaptation processes (UN Women, 2022). As such, GBV must be integrated into anticipatory climate action, beyond response and preparedness, through early warning systems, forecast-based triggers, community surveillance, and pre-positioned support services coordinated with mandated county structures such as the County Gender Department, Social Services, child protection units, and GBV coordination mechanisms that programmes like COSME have actively engaged. Embedding GBV mitigation within resilience planning ensures that climate adaptation measures safeguard rights, safety, and justice, which are foundational to sustainable and inclusive resilience building.

Table 3: Gender/Age Vulnerability Analysis for Climate Hazards

Women	Men	Youth	Elderly	PADs
Drought				
• Long distances (4–6 km) to fetch water and firewood from depleted sources, causing fatigue and back strain. • Increased burden of securing food and water for households. • Reduced time for income-generating activities. • High psychosocial stress during food shortages.	• Loss of income from declining crop yields, livestock deaths, and failed irrigation along the Ramisi belt. • • Heightened financial pressure as food prices rise. • • Increased stress linked to reduced livelihood options.	• Reduced productivity and income from casual labour losses • High idleness → increased risk of crime, drugs, and unsafe coping strategies • Higher incidence of early marriages and pregnancies among girls	• Mobility limitations restrict access to water and food • Increased susceptibility to heat stress, dehydration, and hunger • Dependence on caregivers worsens during drought periods	• High dependency for water, food, and healthcare access • Mobility constraints limit ability to queue for relief supplies • Increased risk of neglect during prolonged shortages
Floods				
• Increased caregiving demands for sick or displaced family members. • Damage to farm produce and informal trading spaces reduces income. • Contaminated wells force longer	• Loss of fishing, transport, and casual labour opportunities • Psychological stress from inability to provide for families • Damage to tools and assets disrupts livelihoods	• School closures disrupt learning and training • Loss of casual labour -increased idleness, crime, and drug use • Girls particularly vulnerable to early marriage and pregnancy	• High susceptibility to waterborne diseases and cold exposure • Limited mobility restricts evacuation • Greater dependence on others for shelter, food, and medicine	• Severe mobility barriers during evacuation and accessing aid • High risk of injury and neglect in shelters • Limited access to emergency services

Women	Men	Youth	Elderly	PADs
walking distances for safe water. • Heightened emotional stress and safety concerns.				
Pests and diseases				
• Reduced food security due to crop pests and poultry/livestock diseases. • • Limited access to certified seeds and extension services constrains adaptation.	• Income loss from crop and livestock declines • Increased stress due to reduced labour demand and food shortages • Higher pressure to diversify livelihoods	• Loss of agricultural casual labour opportunities • Increased unemployment and financial vulnerability • Greater exposure to negative coping behaviours	• Increased dependence for food and care during outbreaks • Reduced capacity to protect crops and livestock • Nutritional deficits worsen chronic illnesses	• High dependency on family for food and adaptive support • Limited access to adapted technologies and extension services • Increased vulnerability from household food shortages
Human wildlife conflicts				
• Fear and restricted mobility when fetching water, washing clothes, or farming along the Ramisi River. • Higher risk of attacks, especially during water collection.	• High risk of injury when guarding farms and livestock, often at night • Economic strain from repeated crop and livestock losses • Pressure to seek alternative income sources	• Increased risk when helping to guard farms • Reduced productivity and fewer livelihood opportunities • Higher vulnerability to migration and exploitation	• Very limited capacity to respond to wildlife encounters • Increased dependence after livestock or crop losses • Heightened food insecurity	• Inability to flee or protect themselves during wildlife incidents • Extremely high dependence on caregivers for safety and recovery • High vulnerability during

Women	Men	Youth	Elderly	PADs
• • Limited access to compensation mechanisms.				compensation or reporting processes
Sea level rise				
• Loss of seaweed drying spaces and shoreline trading spots reduces income (in coastal fringe households). • Increased time spent relocating seaweed lines or market stalls.	• Submerged landing sites and damaged gear reduce catch • Higher risk during offshore fishing • Increased household financial pressure	• Fewer opportunities in fishing, seaweed farming, and tourism • Increased economic migration • Higher stress and vulnerability during relocation	• Very limited ability to relocate • High health risks from flooding and contamination • Emotional distress from loss of ancestral land	• Extreme difficulty relocating to safer areas • Heightened exposure to contamination and displacement • Increased reliance on caregivers
Strong winds				
• Increased domestic burdens when homes or granaries are damaged. • Risk of injury when securing household assets.	• High danger at sea and gear damage • Increased risk of capsizing during strong winds	• High injury risk from collapsing structures or debris • Limited ability to seek safe shelter	• Difficulty accessing safe shelter quickly • High injury and exposure risk	• Severe challenges evacuating safely • High exposure to falling debris and structural collapse

4.1 Children's Vulnerability to Climate Hazards

Children in Pongwe Kikoneni Ward face heightened vulnerability to climate hazards due to their developmental needs, high dependency on caregivers, and limited physical capacity to cope with environmental stress. The recurrent droughts, seasonal floods, widespread crop and livestock diseases, strong winds, sea level rise, and persistent crocodile conflicts in Kiwambale collectively undermine children's health, safety, nutrition, learning, and emotional well-being.

During droughts, children are among the first to experience nutritional deficits, especially in households dependent on subsistence farming and livestock. Crop failure and livestock mortality reduce access to protein and micronutrient-rich foods, increasing the risk of stunting, acute malnutrition, anaemia, and dehydration. Water scarcity often forces families to rely on unsafe sources, exposing children to diarrhoea, typhoid, and skin infections. In many homesteads, water collection shifts to children, particularly girls, who walk long distances under high temperatures. This results in fatigue, heat stress, early morning school lateness, and reduced class concentration. Prolonged droughts contribute to increased school absenteeism and, in extreme cases, early involvement of children in household labour to supplement family survival.

Floods pose direct physical risks to children, especially in villages adjacent to seasonal rivers and low-lying farms. Floodwaters destroy homes, disrupt Early Childhood Development (ECD) centres, wash away learning materials, and damage WASH facilities. Contaminated floodwater increases exposure to cholera, dysentery, and respiratory infections. Schools are occasionally used as evacuation shelters, halting learning, straining feeding programmes, and weakening psychosocial stability. Children experience anxiety, disrupted routines, sleep disturbances, and loss of a sense of safety when displaced.

Human–wildlife conflict is a distinct threat in Pongwe Kikoneni, particularly crocodile attacks along the Ramisi River in Kiwambale. Children accompanying caregivers to fetch water, wash clothes, or graze livestock face direct physical danger. Fear of wildlife encounters restricts children's movement, reduces time for play and learning, and heightens emotional stress. Injuries or loss of a caregiver during wildlife encounters can create long-term trauma and social instability for children. Pests and diseases affecting crops and livestock exacerbate household food shortages, directly impacting children's diets. Reduced meal frequency and diminished dietary diversity weaken immunity, making children more susceptible to infections. Strong winds and sea level rise further threaten child safety through collapsing structures, falling trees, storm surges, and repeated displacement for coastal households.

Across all hazards, poverty magnifies child vulnerability. Reduced household income leads families to adopt harmful coping strategies, including withdrawing children from school, increasing child labour, reducing meal portions, or marrying off girls to ease financial strain. These coping mechanisms undermine children's rights, perpetuate intergenerational vulnerability, and reduce long-term resilience.

Protecting children in Pongwe Kikoneni Ward requires a child-centred adaptation approach that integrates early warning systems for households and schools, strengthened nutrition surveillance, and climate-resilient WASH systems. Upgrading schools with storm-resistant roofing, elevated floors, shaded play areas, and reliable water supply will safeguard learning continuity. Establishing psychosocial support and child-friendly spaces during and after climate shocks will help reduce emotional distress. Expanding social protection measures, such as school feeding programmes and targeted cash transfers, will stabilise child well-being during climate-induced livelihood disruptions. Building the capacity of teachers, community health volunteers, religious leaders, and caregivers is essential to ensure that children remain protected before, during, and after climate hazards.

5 Chapter 5: Adaptation Actions

5.1 Introduction

Building on the findings from the hazard and vulnerability assessment, this chapter presents the adaptation actions for Pongwe Kikoneni Ward, aimed at enhancing climate resilience across its five villages; Kibuyuni, Mkwiro, Anzuwani, Kiwambale, and Barabarani (Majoreni). These actions were formulated through extensive community consultations using Participatory Climate Vulnerability Assessment (PCVA) tools, participatory ranking, and impact chain analysis to ensure that interventions are grounded in the ward's unique ecological and socio-economic realities. The proposed measures respond to both immediate climatic stresses, including droughts, floods, sea-level rise, strong winds, pest and disease outbreaks, and human–wildlife conflicts, and the underlying structural challenges of poverty, gender inequality, and ecosystem degradation.

The adaptation actions are structured along short-, medium-, and long-term priorities. Short-term measures focus on community sensitization, institutional strengthening, and capacity building for local actors, including the Ward Climate Change Planning Committee (WCCPC), Beach Management Units (BMUs), and forest user associations. Medium- and long-term strategies emphasize ecosystem restoration, climate-smart infrastructure, and the promotion of nature-based and technological solutions to reduce livelihood vulnerability. For instance, solarized boreholes and rainwater harvesting systems will address recurrent water shortages, while mangrove restoration in Mkwiro and Kibuyuni, and reforestation of Ganda and Kiriku forests, will mitigate flooding, soil erosion, and coastal degradation.

The formulation of these actions draws from both local knowledge systems and scientific best practices, aligning with national and county frameworks such as Kenya's National Adaptation Plan (2015–2030), the National Climate Change Action Plan (2023–2027), and the Kwale County Climate Change Action Plan (2023–2027). Gender equality and social inclusion are mainstreamed throughout to ensure that adaptation benefits are equitably shared. Women, who bear the heaviest burden of climate stress through unpaid domestic work and resource scarcity, are prioritized in livelihood diversification, financial empowerment, and leadership programs. Youth engagement is equally emphasized through digital climate information systems, eco-enterprises, and community monitoring initiatives focused on resource conservation.

Currently, Pongwe Kikoneni Ward already hosts community-led initiatives that complement these adaptation priorities, including mangrove rehabilitation in Mkwiro and Kibuyuni, seaweed farming groups, beekeeping enterprises in Barabarani (Majoreni), and village-based savings and loan associations (VSLAs) that finance small-scale, climate-resilient businesses. These ongoing efforts form a solid foundation for scaling up coordinated adaptation interventions.

Implementation will occur at multiple levels i.e., household, community, ward, county, and national, under the coordination of the Ward Climate Change Planning Committee (WCCPC). Strong collaboration among local institutions such as BMUs, WRUAs, Community Forest Associations (CFAs), and relevant county departments will be essential to ensure that adaptation efforts in Pongwe Kikoneni are inclusive, synergized, and sustainable, effectively transitioning the ward from reactive responses to proactive climate resilience building.

5.2 Existing climate-resilient and livelihood-enhancing projects

Pongwe Kikoneni Ward hosts a range of ongoing projects and initiatives implemented by government agencies, development partners, and community-based organizations, all geared toward strengthening climate resilience, enhancing livelihoods, and promoting sustainable natural resource management (Table 4). These interventions target both marine and terrestrial

ecosystems, integrating environmental conservation with social empowerment, gender equity, and economic diversification to address the ward's climate vulnerabilities.

Plan International, through the Conservation and Sustainable Management of Coastal and Marine Ecosystems (COSME) project, is among the leading partners supporting community-driven adaptation in Pongwe Kikoneni Ward. The initiative promotes Village Savings and Loan Associations (VSLAs), table banking, and gender justice programs, enabling women and youth to access microfinance for climate-resilient livelihoods such as seaweed farming, small-scale trade, and poultry keeping. By improving financial inclusion and gender equality, the project enhances household adaptive capacity and supports equitable participation in resource management.

The Kenya Fisheries Service (KeFS), under the Kenya Marine Fisheries and Socio-Economic Development (KEMFSED) project, focuses on coastal community empowerment and sustainable fisheries management. Through capacity building and the promotion of ecosystem-based fisheries practices, the project supports local Beach Management Units (BMUs) in Kibuyuni and Mkwiro to reduce overfishing and adopt environmentally friendly methods. This intervention is vital in sustaining marine biodiversity and strengthening the livelihoods of households dependent on fishing, one of the most climate-sensitive sectors in the ward.

The County Government of Kwale has also played a central role in promoting seaweed farming by facilitating the formation of producer groups and providing financial and technical support to farmers, particularly in Kibuyuni and Mkwiro villages. These initiatives empower women and youth engaged in seaweed production and contribute to income diversification while enhancing the resilience of marine-based livelihoods to climate variability.

World Vision Kenya, through the Restore Africa Project, complements terrestrial ecosystem restoration by supporting tree planting and forest restoration in degraded areas around Ganda and Kiriku forests. The organization provides seedlings and promotes community awareness on sustainable forest management, directly addressing deforestation and enhancing local microclimates. This initiative contributes to long-term carbon sequestration and improved water regulation, which are critical in mitigating drought impacts.

Similarly, the World Bank supports the ward through the Kenya Coastal Development Project (KCDP), which provides financial grants for alternative livelihoods targeting smallholder farmers, fishers, and women entrepreneurs. The project enhances adaptive capacity by reducing dependence on climate-sensitive resources through the promotion of alternative enterprises such as beekeeping, eco-tourism, and small-scale agribusiness.

The Kenya Wildlife Service (KWS) leads interventions related to wildlife management and human–wildlife conflict mitigation, particularly in areas bordering forest patches and grazing zones. The agency's efforts aim to harmonize community–wildlife coexistence through awareness creation, conflict reporting mechanisms, and the establishment of buffer zones that reduce crop and livestock losses during dry seasons when wildlife migrates in search of water and forage.

Kenya Agricultural and Livestock Research Organization (KALRO) supports farmers by supplying certified seedlings and grants, promoting drought-tolerant crops and sustainable agricultural practices. This enhances food security and resilience to erratic rainfall, especially in inland villages like Kiwambale and Barabarani (Majoreni). Kenya Forest Service (KFS) complements these efforts by leading the conservation and protection of forest ecosystems in Ganda and Kiriku, coupled with community sensitization on afforestation and the impacts of

deforestation. Through collaboration with community forest associations, KFS supports the restoration of degraded sites and the integration of tree-based livelihoods within farmlands.

Private sector actors such as KOMAZA are also active in the ward, promoting commercial tree planting and providing quality seedlings to farmers for woodlot establishment. This partnership not only increases forest cover but also provides long-term income opportunities through sustainable timber and carbon markets.

On the social front, Kids Care Kenya implements programs that integrate climate resilience with social protection. The organization focuses on awareness creation on gender-based violence (GBV) and the provision of physiotherapy and psychosocial support to Persons Abled Differently (PADs). These activities ensure inclusivity in climate adaptation by addressing the social dimensions of vulnerability that often limit participation of marginalized groups in community development.

Table 4: Existing climate-resilient and livelihood-enhancing projects in Pongwe Kikoneni Ward

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
Conservation and Sustainable Management of Coastal and Marine Ecosystems (COSME) / VSLAs, Table Banking, and Gender Justice	Plan International	Kibuyuni, Mkwiro, Anzuwani	Climate-Resilient Livelihoods and Gender Empowerment	Promotes financial inclusion through VSLAs; strengthens women's adaptive capacity; aligns with the Kwale County Climate Change Action Plan (2023–2027) on inclusive climate finance.
Kenya Marine Fisheries and Socio-Economic Development (KEMFSED)	Kenya Fisheries Service (KeFS)	Kibuyuni, Mkwiro	Sustainable Marine Resource Management	Enhances fisheries governance through BMUs; builds resilience of fishing-dependent households; supports ecosystem-based management as outlined in Kenya's Blue Economy Strategy.
Seaweed Farming Development and Support	County Government of Kwale	Kibuyuni, Mkwiro	Livelihood Diversification and Marine Ecosystem Restoration	Promotes seaweed farming as a climate-resilient livelihood; provides start-up capital and training; supports gender-inclusive economic empowerment in line with the County Blue Economy Framework.
Restore Africa Project – Tree Planting and Forest Restoration	World Vision Kenya	Kiwambale, Barabarani (Majoreni), Anzuwani	Ecosystem Restoration and Climate Mitigation	Promotes tree planting and rehabilitation of degraded sites within Ganda and Kiriku forests; enhances carbon sequestration and

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
				watershed protection under the Kwale Forest and Landscape Restoration Plan.
Kenya Coastal Development Project (KCDP)	World Bank	Ward-wide (All Villages)	Sustainable Livelihoods and Climate Adaptation	Provides grants for alternative livelihoods (e.g., beekeeping, eco-tourism); supports income diversification and resilience building; aligns with Kenya's National Adaptation Plan (2015–2030).
Human–Wildlife Conflict Management and Biodiversity Protection	Kenya Wildlife Service (KWS)	Barabarani (Majoreni), Kiwambale	Ecosystem Protection and Safety	Promotes human–wildlife coexistence and mitigates climate-induced conflicts; supports the National Wildlife Conservation and Management Strategy.
Drought-Tolerant Crop Production and Farmer Support	Kenya Agricultural and Livestock Research Organization (KALRO)	Kiwambale, Barabarani (Majoreni), Anzuwani	Climate-Smart Agriculture	Provides drought-tolerant seedlings and grants; promotes sustainable crop production; supports Kenya Climate Smart Agriculture Strategy (2017–2026).
Forest Conservation and Awareness Creation	Kenya Forest Service (KFS)	Ganda Forest, Kiriku Forest, Barabarani (Majoreni)	Forest Conservation and Restoration	Leads afforestation and community sensitization on deforestation; contributes to the County

Project / Activity	Implementing Partner(s)	Location / Target Villages	Adaptation Theme / Focus Area	Key Contributions / Linkages to Existing Strategies
				Forest and Landscape Restoration Implementation Plan (FOLAREP 2022–2027).
Commercial Tree Planting and Seedling Supply	KOMAZA	Kiwambale, Barabarani (Majoreni)	Green Enterprise and Sustainable Forestry	Provides seedlings for woodlot establishment; supports income generation and forest restoration; complements county afforestation targets.
Community Awareness on GBV and Physiotherapy Support for PADs	Kids Care Kenya	Ward-wide (All Villages)	Social Inclusion and Resilience	Integrates gender and disability inclusion in adaptation efforts; promotes psychosocial resilience among vulnerable groups; aligns with the County Gender Mainstreaming Strategy (2023).

5.3 Adaptation actions

Adaptation actions in Pongwe Kikoneni Ward are designed as an integrated and gender-responsive climate resilience framework that protects livelihoods, strengthens ecosystems, and safeguards the well-being of all population groups across the five villages; Anzuwani, Kiwambale, Barabarani (Majoreni), Kibuyuni, and Mkwiro (Table 5). These actions respond to six priority climate hazards; drought, floods, pests and diseases, strong winds, sea-level rise, and human–wildlife conflicts and are anchored in the Kenya Climate Change Act (2023), the National Adaptation Plan (2015–2030), the Kwale County Climate Change Act (2021), and national frameworks on gender equality, child protection, and sexual and reproductive health and rights (SRHR). Implementation follows a phased approach comprising short-term interventions (≤1 year), medium-term measures (≤3 years), and long-term interventions (>3 years), all anchored in gender equity, social inclusion, SRHR continuity, and community ownership to ensure climate benefits are distributed fairly and sustainably.

The first thematic area emphasizes gender-responsive water security, recognizing that women, girls and PADs bear a disproportionate burden of water collection, exposure to wildlife risks and increased domestic care demands during droughts. In the short term, proposed actions include expanding household-level rainwater harvesting through the installation of gutters and storage tanks across Anzuwani, Kibuyuni, Kiwambale, Majoreni- Barabarani and Mkwiro, together with increasing water storage capacities in schools to safeguard learning continuity, particularly for girls, who are often withdrawn from school when water scarcity intensifies. The use of solar-powered pumps on existing shallow wells is also proposed as a means to reduce the physical strain and long queuing times experienced by women and children during dry periods. In the medium term, the ward may enhance resilience through drilling and solarising community boreholes and establishing water pans to buffer household consumption, livestock and smallholder farming needs. Strategic solar pumping systems along River Ramisi could further improve access to safe water while reducing women's exposure to crocodile attacks at unsafe abstraction points. Long-term actions point toward developing an integrated water reticulation system, including expanding storage through a majibao in Mkwiro and constructing water pipelines and kiosks that could eventually supply Mkwiro and Wasini Island, thereby reducing the gendered labor burden associated with water collection over time.

The second thematic area focuses on food security, biodiversity, ecosystem restoration and livelihoods, recognizing that women and youth farmers face structural constraints in land access, farm labor, extension services and climate information. In the short term, proposed actions involve training farmers, especially women and youth, on Resilience Design techniques such as soil fertility enhancement, zai pits and labour-efficient farm layouts that reduce production costs and increase adaptive capacity. Additional proposals include training on climate-smart agriculture practices, drought-tolerant crop varieties and fodder conservation methods to stabilize food and livestock systems, alongside sensitizing women and youth groups on crop and livestock insurance products to cushion them from climate-related shocks. Medium-term priorities include supporting selected women and youth lead farmers to establish demonstration plots on CSA and Resilience Design, strengthening extension support, improving access to certified seed and fertilizer, and establishing fodder seed banks and managed pastures that could reduce seasonal feed shortages. For marine-dependent households, particularly women in Kibuyuni and Mkwiro, proposals include providing improved seaweed farming gear, drying racks and training on advanced production systems such as bamboo rafts. Long-term adaptation points toward institutionalizing continuous agricultural learning through the establishment of a Farmer Field School and improving market competitiveness by aggregating produce from women's and youth groups for value addition and enterprise development.

In addition, the thematic area recognizes the role women and youth play in natural resource stewardship and seeks to enhance restoration benefits for livelihoods, biodiversity and shoreline protection. In the short term, proposed actions include supporting women's and youth groups to establish and manage tree nurseries for agroforestry, boundary planting and woodlots across all five villages, as well as providing training on mangrove nursery establishment, transplanting and monitoring in the four coastal-facing villages. Medium-term proposals involve assessing existing mangrove nurseries to improve seedling quality, nursery management and technical capacity, thereby strengthening community participation, especially among women engaged in mangrove harvesting and restoration. In the long term, an integrated mangrove restoration programme across Anzuwani, Kibuyuni, Kiwambale and Majoreni Barabarani is envisioned to enhance shoreline stability, fish nursery habitats, carbon storage potential and overall ecological resilience, with gender-responsive benefit-sharing mechanisms incorporated into community agreements.

The third thematic area prioritizes the diversification of climate-resilient livelihoods, especially for women and youth whose income streams are highly sensitive to climatic disruptions such as ice-ice disease in seaweed farming. Short-term proposals include introducing alternative livelihood options such as kitchen gardening, beekeeping, poultry production and eco-tourism (particularly in Mkwiro), alongside offering swimming and snorkelling training for women seaweed farmers to enhance safety in changing marine environments. In the medium term, communities may receive support through provision of equipment and materials such as horticultural seeds, organic fertilizer, modern beehives, egg incubators, eco-tourism boats and snorkelling gear to strengthen emerging enterprises. Over the long term, the ward could explore initiating a women- and youth-led mangrove blue carbon project aligned with Plan Vivo standards and Kenya's Climate Change (Carbon Markets) Regulations (2024), as well as relocating seaweed farms to deeper waters where productivity may be more stable, accompanied by appropriate vessels and safety equipment to ensure the inclusion and protection of women farmers.

The forth thematic area addresses the vulnerability of households, particularly women-headed households, children, elderly persons and PADs, to floods, strong winds and unsafe settlements. Short-term adaptation options include community sensitization on resilient construction techniques such as reinforcing rooftops and strengthening weak structures in Kibuyuni and Mkwiro, alongside training community emergency teams in Anzuwani and Majoreni Barabarani to support inclusive evacuation and first-aid actions. Youth engagement in clearing drainage channels and maintaining culverts in Anzuwani is proposed to improve stormwater flow, while tree planting in Kibuyuni and Mkwiro could reduce wind damage and enhance microclimate regulation. Medium-term options extend to operationalizing early warning systems for floods, implementing labor-based public works (for example, food-for-work programmes) that create temporary income opportunities for youth and women, and designating safe assembly points and temporary shelters that consider the needs of pregnant women, lactating mothers and PADs. Long-term proposals include constructing raised footbridges and improving road drainage in Majoreni Barabarani and equipping community response teams with appropriate gear such as PPEs, boats and first-aid kits to ensure gender-responsive disaster preparedness.

The fifth thematic area recognizes that women and children are disproportionately exposed to crocodile risks along River Ramisi due to their daily roles in water collection, washing and caregiving. Short-term actions therefore propose sensitizing Kiwambale residents, especially women's groups, on wildlife hotspots, safe behaviours, reporting protocols and compensation mechanisms through KWS. Medium-term measures may include strengthening community reporting structures and maintaining surveillance to minimize risky encounters. In the long term,

installing sectional solar-powered fencing in identified crocodile hotspots is proposed as a means of reducing attacks, safeguarding livelihoods and decreasing the time women spend navigating unsafe riverbanks.

The final thematic area mainstreams gender equality, PAD inclusion, youth empowerment and SRHR continuity across all adaptation actions. Short-term proposals include designing GBV anticipatory actions for drought and floods, assessing existing early warning systems for their capacity to communicate risks to women, girls and PADs, and strengthening awareness on rights, helplines and GBV referral pathways through established structures such as VSLAs. Youth-focused boot camps may address drugs, antisocial behaviour and reproductive health, while PADs could be linked with organizations providing mobility aids, specialized training and entrepreneurship support. Medium-term actions aim to strengthen gender-responsive governance by increasing the participation of women, youth and PADs in ward-level committees and climate decision-making, while equipping local institutions with first-aid kits, rape kits and transport mechanisms to improve GBV response outcomes. In the long term, the ward may consider the establishment of a GBV recovery centre and a youth drug rehabilitation facility at county level, complemented by expanded school feeding programmes, scholarships and vaccination drives to protect the wellbeing of children and adolescents during climate-induced shocks.

Table 5: Pongwe Kikoneni Ward Adaptation Actions

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium term	Long-term		
Thematic area 1: Water Security and Management				
Rainwater harvesting through Installation of gutters on house roofs and water tanks at household level in Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages	Rainwater harvesting installation of gutters and high-capacity water tanks (10m³) in schools – one school in each village to enhance water access by women and children	Drilling and solarizing boreholes in Kiwambale and Majoreni Barabarani villages	Kwale County Government-FLLoCA, NGOs and Private Sector Possible partnership with Kenya Ports Authority (KPA) Shimoni Port on Mkwiro Water Pipeline	50,000/HH 400,000/ school 5,000,000/ Borehole, 3,000,000/ Water pan 70,000/ water pump BOQ for Majibao & water pipeline in Mkwiro
	Construction of a new majibao storage at Mkwiro to reduce water scarcity and workload among women and girls Construction of water pans in	Construction of water pipeline and kiosks to supply Mkwiro – Wasini Island		
	Installation of solar powered water pumps on existing wells to reduce women workload (turn-around time) in Anzuwani, Kibuyuni , Kiwambale, Majoreni and Barabarani villages	Strategic (where embankment for water intake is viable) installation of solar powered water pump at River Ramisi and elevated storage tanks to provide water domestic and irrigation water sources for women and girls as well as reduce their exposure to wildlife threats		
Thematic area 2: Food security, ecosystem restoration and livelihoods				
Training and supporting women and youth on Resilience Design techniques from homestead to watershed level (Soil fertility improvement, water conservation/retention (e.g. Zai Pits), farm design- to maximize on time and use of local inputs to reduce farming costs)	Supporting select women and youth lead farmers to fully implement resilient design techniques including climate smart agriculture to serve as demonstration farms for continuing learning by other farmers to enhance sustainability of the interventions	Establishing a Farmer Field School in the Ward	Kwale County Government, KARLO, NGOs and Private Sector	Farmer Field Schoos (FFFs) 10,000 per farmer for a 12-month cycle

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium term	Long-term		
Training women and youth on Climate Smart Agriculture (CSA) (drought tolerant crops such as katumani maize, cassava, sunflower, groundnuts, cashewnuts, sorghum), silage/hay making to address livestock feed challenges during drought in Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages	Provision of farm inputs (certified seed varieties for drought tolerant crops, organic and inorganic fertilizers, pesticides) and extension services across Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages	Aggregating women and youth farmers' produce to enhance farm produce marketability and enhance value addition and agribusiness opportunities	Kwale County Government, KARLO, NGOs and Private Sector	Production costs: Maize 55,000 per acre and Cassava 40,000 per acre 10,000 for livestock farmers per year or crop cycle (Free if it is provided by the government)
Sensitize and facilitate access to information on crop and livestock insurance (index-based drought coverage) targeting women and youth groups involved in CSA	Establish livestock fodder seed banks, pastures, and introduce drought-tolerant livestock breeds e.g., Gala goats/Dairy cows for women and youth groups in Anzuwani, Kibuyuni, Kiwambale	Support seaweed farmers with improved farming gear, drying racks, and capacity building on advanced farming techniques (bamboo rafts, tubular nets) in Mkwiro and Kibuyuni	Kwale County Government, KARLO, NGOs and Private Sector	Village barazas at 20,000, 55,000 per acre for pasture, 15,000 per Gala goat, 100,000 per dairy cow 50,000 per farm
Facilitate/train women and youth to establish tree nurseries of priority species (Annex 1) for on-farm tree production (Agroforestry, woodlots and boundary planting) in Anzuwani, Kibuyuni, Kiwambale, Majoreni Barabarani and Mkwiro villages			Kwale County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee (50 pax from the five villages) 2,000 per HH (20 seedlings per HH)
Sensitization and training women and youth on mangrove restoration (nursery establishment, transplanting, tending and monitoring) in Anzuani, Kibuyuni, Kiwambale and Majoreni Barabarani	Assessment and provision of technical support to the existing mangrove nurseries to ensure quality and sustainable seedlings supply for restoration interventions	Implement an integrated mangrove restoration programme in Anzuani, Kibuyuni, Kiwambale and Majoreni Barabarani villages to enhance shoreline	Kwale County Government, KFS, KEFRI, CFAs, NGOs, Private Sector	5,000 per day per trainee (50 pax from the five villages) 1,000,000 per acre-mangroves

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium term	Long-term		
		protection and ecosystem resilience		
Thematic area 3: Diversified Livelihoods and Social Protection				
Training women, youth on alternative livelihoods such as kitchen gardening (horticultural crops), eco-tourism (Mkwiro), beekeeping and poultry farming to address seaweed farming vulnerability, in Kibuyuni and Mkwiro villages Training women and youth involved in seaweed farming on swimming and snorkelling	Training women, youth on alternative livelihoods such as kitchen gardening (horticultural crops), eco-tourism (Mkwiro), beekeeping and poultry farming to address seaweed farming vulnerability, in Kibuyuni and Mkwiro villages Training women and youth involved in seaweed farming on swimming and snorkelling	Training women, youth on alternative livelihoods such as kitchen gardening (horticultural crops), eco-tourism (Mkwiro), beekeeping and poultry farming to address seaweed farming vulnerability, in Kibuyuni and Mkwiro villages Training women and youth involved in seaweed farming on swimming and snorkelling	Training women, youth on alternative livelihoods such as kitchen gardening (horticultural crops), eco-tourism (Mkwiro), beekeeping and poultry farming to address seaweed farming vulnerability, in Kibuyuni and Mkwiro villages Training women and youth involved in seaweed farming on swimming and snorkelling	Training women, youth on alternative livelihoods such as kitchen gardening (horticultural crops), eco-tourism (Mkwiro), beekeeping and poultry farming to address seaweed farming vulnerability, in Kibuyuni and Mkwiro villages Training women and youth involved in seaweed farming on swimming and snorkelling
Thematic area 4: Community infrastructure and disaster management				
Conduct community training on resilient building practices to strengthen structures (use of nails instead of strings on rooftops) in Kibuyuni and Mkwiro to address strong winds risks	Design and implement a food for work programme during drought (this has been done before in the county) targeting the youth to restore drainage channels, clean culverts, construct gabions and plant trees for erosion and flood control	Construct raised footbridges/culverts at flood-prone crossing points in Majoreni Barabarani Improve road drainage and install culverts to channel floodwaters in Majoreni Barabarani	Kwale County Government, KERRA, NGOs and Private Sector	5,000 per day per trainee (20 pax from the two villages Bill of quantities required

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium term	Long-term		
Train selected community emergency response teams in both Anzuwani and Majoreni Barabarani villages	Develop and operationalize Early Warning Systems (EWS) for floods (forecasts via chiefs, barazas, media) in both Anzuwani and Majoreni Barabarani villages	Designate flood disaster assembly areas and temporary camps for vulnerable communities	Kwale County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	5,000 per day per trainee 70,000 per village for EWS
Undertake tree planting to address strong winds in Kibuyuni and Mkwiro villages Mobilize the youth to desilt drainage channels and maintain culverts to improve stormwater flow in Anzuwani village	Equip community emergency response teams (motorized boat, PPEs, first aid kits)		Kwale County Government, Ministry of Interior and National Administration Kenya Red Cross, NGOs and Private Sector	Village barazas at 20,000 BOQ required for equipping community emergency response teams
Thematic area 5: Conflicts resolution (Human Wildlife Conflicts)				
Sensitize Kiwambale community on potential conflict hotspots, minimizing human-wildlife interactions and reporting to KWS (hotline) as well as compensation procedures		Installing sectional solar powered fences in crocodile hotspot areas in Kiwambale	KWS, Ministry of Interior and National Administration, Kwale County Government	5,000,000- Death, 3,000,000- Permanent Disability, 2,000,000- Any other injury, Crop/Property destruction- compensation based on market rates, 1,000,000/ km for electric fence
Thematic area 6: Gender and social inclusion to enhance adaptation action				
Design targeted GBV anticipatory actions for drought and floods climate hazards	Develop and implement a Ward GBV anticipatory action plan in collaboration with relevant partners	Establish GBV recovery center (this could be at the county level)	National Government, Kwale County Government and NGOs, , National Police Service, NGOs, GVRC, COVAW, HAK	1,000,000 for consultancy services Construction of GBV recovery center subject to a detailed Bill of Quantities

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium term	Long-term		
Assess the existing early warning systems to produce and transmit reliable and timely information and forecasts for the implementation of GBV anticipatory actions ahead of climate hazards	Strengthen gender-responsive governance- inclusion of women, youth and PADs through capacity building across Anzuwani, Kiwambale, Kibuyuni Majoreni Barabarani, and Mkwiro villages		Kwale County Government and NGOs	5,000 per person per day for training
Use of existing women associations such as VSLAs to train women and girls on their rights and GBV response strategies, helplines and recovery centers for women and girls as well as the role of men in peaceful conflict resolution	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		Kwale County Government, Civil Society, NGOs, Private Sector	30,000 per VSLA BOQ for GBV response mechanisms
Create awareness on existing GBV response strategies, helplines and recovery centers for women and girls	Procure/provide GBV response mechanisms such as transfer vehicles to health facilities and specialized first aid and rape kits to treat and preserve evidence for prosecution respectively		National Government Department of Gender, Kwale County Government, National Police Service, NGOs, Gender Violence Recovery Centre (GVRC), Coalition on Violence Against Women (COVAW) Healthcare Assistance Kenya (HAK)	KES 10,000 for first aid kits in placed in local dispensaries Village barazas at KES 20,000 per village
Organize and hold youth boot camps on guidance and counselling focusing on drugs, antisocial behavior and reproductive health	Promote vaccination of children and youth against diseases such as cholera which could arise and spread on account of climate hazards Support school feeding programmes and scholarships	Construct drug rehabilitation centre targeting the youth (this can be done at the county level)	National Government, NACADA, Kwale County Government, NGOs, Private Sector	100,000- Boot camp, Vaccination at local medical facilities, BOQ required for drug rehabilitation centre

Adaptation Actions			Potential implementing partners	Budget (KES)
Short-term	Medium term	Long-term		
				School feeding and scholarships needs assessment
Facilitate PADs to link with organizations that can provide infrastructural assistance such as wheelchairs, specialized training and entrepreneur skills			Kwale County Government, NGOs, Private Sector	50,000 for meetings

5.4 Resource Mobilisation

Implementation of the Ward Adaptation Plan will rely on a diversified mix of financial, human, and technical resources drawn from county, national, donor, and community-based funding streams. The primary sources include county budget allocations, where relevant departments such as Water, Agriculture, and Infrastructure will integrate adaptation priorities into their Annual Development Plans and sector budgets, supported by at least KES 2–3 million annually earmarked for ward-level climate resilience as required under the Climate Change Act, 2023. Additional financing will be accessed through the Kwale County Climate Change Fund (CCCF) established under the County Climate Change Act, 2022, which channels grants from development partners such as the World Bank's FLLoCA programme to locally led adaptation initiatives. At the national level, Pongwe Kikoneni Ward will benefit from the National Drought Emergency Fund and other climate financing mechanisms such as the Kenya Adaptation Fund Programme, coordinated through the National Treasury's Climate Finance Unit. Development partners and NGOs, including Plan International, WWF, World Vision, and COMRED, will continue to support ecosystem restoration, capacity building, and livelihood diversification projects, while community contributions through in-kind labour, savings groups, and user fees will promote ownership and sustainability. The private sector will be engaged through partnerships for water infrastructure, crop and livestock insurance, and digital early warning systems, complemented by corporate social responsibility initiatives. Overall, the ward expects to mobilize approximately KES 20–30 million over five years from this blended financing framework, ensuring continuity and long-term sustainability of adaptation interventions.

6 Chapter 6: Adaptation Monitoring and Evaluation

6.1 Introduction

Monitoring, Evaluation, and Learning (MEL) form an integral component of the Ward Climate Adaptation Plan, providing a structured mechanism for systematic performance tracking, evidence-based decision-making, and adaptive management throughout the implementation cycle. This chapter outlines the framework through which Pongwe Kikoneni Ward will monitor the implementation of adaptation interventions (Monitoring), assess their effectiveness, efficiency, and impact (Evaluation), and institutionalize continuous knowledge generation and feedback loops to enhance future climate resilience planning (Learning).A technically sound MEL framework will be established to ensure accountability, transparency, and data-driven insights for both the community and development partners. The framework will support adaptive management by enabling the Ward Climate Change Planning Committee (WCCPC) to periodically refine strategies, resource allocation, and implementation approaches in response to empirical evidence and evolving climatic conditions.

Recognizing the long-term and dynamic nature of climate change, the MEL system will prioritize longitudinal tracking of resilience outcomes that reflect gradual improvements in ecosystem stability, livelihood diversification, and adaptive capacity. The approach will apply participatory monitoring methodologies, integrating community-based data collection and indigenous knowledge with scientifically derived indicators. This integrated model ensures that both quantitative performance metrics and qualitative community insights inform performance assessments and decision-making.

The MEL framework will be aligned with the County Climate Change Monitoring and Evaluation System and the National Climate Change Monitoring, Reporting, and Verification (MRV) Framework, in accordance with the Climate Change Act (2023). Such alignment guarantees methodological consistency, facilitates upward data integration, and ensures the Ward contributes effectively to national climate reporting and accountability requirements.
Ultimately, the MEL system will operate as a dynamic learning platform, capturing lessons learned, best practices, and innovations from implementation processes. These insights will inform iterative updates of the Ward Climate Adaptation Plan, strengthen institutional capacity, and enhance overall climate governance. By embedding participatory, evidence-based, and adaptive MEL practices, Pongwe Kikoneni Ward will ensure that resilience-building interventions remain measurable, transparent, and continuously optimized to achieve sustainable climate adaptation outcomes.

6.2 The Adaptation Theory of Change

The theory of change is a conceptual model that explains how the planned interventions will lead to the desired climate resilience outcomes. In simplified form:

- **Inputs/Activities:** Financial resources, machinery and equipment, Institutional support, Training and awareness sessions, Extension services Policy formulation and enforcement, Health and social protection, Water infrastructure, seedlings (these are the actions from Chapter 4).

- **Outputs:** boreholes drilled and water pans constructed, number of farmers trained and empowered with drought tolerant seedlings, policies set and implemented, health centres and veterinary centres constructed, tanks provided, inputs and seedlings provided

- **Outcomes:** The short to medium-term changes due to those outputs – e.g., increased water access during dry season, reduced environmental degradation, adoption of drought-tolerant crops on farms, reduced time spent fetching water (especially for women, decreased crop losses from pests.

- **Impact (Goal):** The higher-level objective is enhanced climate resilience and adaptive capacity of the Pongwe Kikoneni Ward community. This means the community can better withstand droughts, floods, and other hazards without suffering devastating losses, and can recover more quickly. It also includes improved livelihoods and food security under a changing climate, and a reduction in climate-induced emergencies over time.

The theory of change for Pongwe Kikoneni can be summarized as: "IF we provide reliable water sources, promote resilient farming and livelihoods, strengthen natural resource management, enhance environmental restoration and enhance local institutions' capacity, it is expected the community's exposure and sensitivity to climate hazards will decrease and their adaptive capacity will increase, because they will have the necessary resources, knowledge, and organization to cope with climate stresses." This will lead to the ultimate impact of a more climate-resilient Pongwe Kikoneni Ward with secure livelihoods and wellbeing despite climate change.

This logical flow will guide M&E. Each level (input -> output -> outcome -> impact) will have specific indicators to check. For instance, training farmers (input) should lead to output of farmers implementing new techniques, which yields outcome of yield stability in drought, contributing to impact of overall food security. If a link is failing

6.3 Performance Indicators for Adaptation

Based on the theory of change, a set of Performance Indicators will be used to measure progress and effectiveness. These indicators will be both quantitative and qualitative, and they align with national adaptation M&E frameworks where possible (Kenya's NAP/NCCAP indicators) as well as community-defined measures of success. The performance indicators are structured per thematic areas;

Input/Output Indicators:

1. WATER SECURITY AND MANAGEMENT
   - Number of household rainwater harvesting systems installed (gutters + tanks).
   - Number of schools equipped with high-capacity rainwater harvesting systems.
   - Number of water pans constructed or desilted.
   - Length/coverage of pipelines and kiosks constructed (e.g., Mkwiro–Wasini).
   - Number of manual and solar-powered pumps installed on existing wells.
   - Number of women, youth, and PADs using solarised/manual hand pumps.
   - Number and percentage of women, youth, and PADs in water management committees.
   - Number of village water storage structures constructed (Majibao, elevated tanks).
   - Amount of budget allocated and utilised for water infrastructure.
   - Number of boreholes drilled, equipped, and solarised.
   - Number of boreholes fully operational and supplying water year-round.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

   - Number of farmers trained on Resilience Design techniques.

- Number of farmers trained on CSA practices (land prep, planting, pest control, post-harvest).
- Acreage under drought-tolerant crops supported with certified seeds.
- Number of Farmer Field Schools established and functional.
- Number of demonstration farms established by women and youth lead farmers.
- Number of farmers receiving inputs (certified seeds, fertilizers, pesticides).
- Number of households supported with organic manure production and composting.
- Number of farmers adopting terracing, zai pits, swales, and other soil/water conservation measures.
- Number of livestock fodder seed banks and pasture plots established.
- Number of households receiving improved livestock breeds (Gala goats, dairy cows).
- Number of farmers sensitised on crop/livestock insurance.
- Number of households supported with grain storage equipment (hermetic bags, grain banks).
- Number of village pest and disease surveillance teams established and trained.
- Number of households receiving indigenous/fruit seedlings.
- Number of community seedling distribution events held.
- Number of on-farm tree planting sessions conducted.
- Number of mangrove nurseries supported and maintained.
- Number of acres of mangrove restored through enrichment planting.
- Number of community trainings on mangrove protection and sustainable harvesting.
- Number of NBS windbreaks (Neem, Casuarina) planted across all villages.
- Number of environmental conservation awareness sessions held.
- Number of households adopting sustainable charcoal alternatives or clean energy.
- Budget allocated and utilised for restoration activities.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Number of women and youth trained in alternative livelihoods (eco-tourism, beekeeping, poultry).
- Number of modern beehives, incubators, or eco-tourism boats distributed.
- Number of households supported with livelihood start-up kits.
- Number of community groups formed for value addition and market linkages.
- Number of training sessions on business development, value addition, and financial literacy.
- Number of social protection initiatives implemented (food distribution, cash transfer mobilizations).
- Number of school feeding programmes supported.
- Budget allocations for livelihood diversification implemented.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Number of culverts reconstructed or improved with proper drainage.
- Number of flood-prone crossing points fitted with raised footbridges.
- Number of youth mobilised in food-for-work programmes for drainage, gabions, and erosion control.
- Number of villages with functional flood Early Warning Systems (EWS) linked to KMD.
- Number of community dissemination channels established (chief barazas, noticeboards, SMS).

- Number of community emergency response teams trained.
- Number of community emergency response teams equipped (boats, PPE, first-aid kits).
- Number of designated disaster assembly points established and labelled.
- Number of community trainings on resilient building practices conducted.
- Number of households implementing safer roofing techniques (replacing strings with nails).
- Budget allocated and utilised for infrastructure upgrades (roads, culverts, drainage).

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)
   - Number of community sensitization sessions on human–wildlife conflict hotspots.
   - Number of households supported with HWC reporting mechanisms (hotline awareness).
   - Number of kilometres of solar-powered fencing installed in hotspot areas (e.g., Kiwambale).
   - Number of community trainings on safe behaviour in crocodile, baboon, and snake-risk zones.
   - Number of reported HWC cases logged and submitted to KWS for follow-up.
   - Funding allocated and utilised for HWC mitigation (fencing, awareness, surveillance).

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

   - Number of gender-responsive governance trainings conducted.
   - Number and percentage of women, youth, and PADs participating in ward committees.
   - Number of VSLAs supported with GBV awareness and rights training.
   - Number of GBV anticipatory action sessions conducted for drought, floods, and strong winds.
   - Number of GBV response kits (first-aid kits, rape kits) distributed to health facilities.
   - Number of early warning systems integrating GBV alert messaging.
   - Number of PADs supported with assistive devices, rehabilitation linkages, or livelihood equipment.
   - Number of youth bootcamps conducted on drugs, behaviour change, and reproductive health.
   - Number of awareness campaigns on GBV helplines and safe reporting mechanisms.
   - Number of meetings with PAD support organizations facilitated.

Outcome Indicators:

1. WATER SECURITY AND MANAGEMENT

   - Reduction in average time spent collecting water (minutes/day), disaggregated by women, youth, and PADs.
   - Increase in household water storage capacity (litres/HH).
   - Percentage of water facilities functional year-round.
   - Increase in number/percentage of women, youth, and PADs with independent access to water sources (boreholes, solarised pumps, kiosks).
   - Increased reliability of water supply during drought periods (days/months of service continuity).

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Stability of crop yields over 3–5 seasons among supported farmers.
- Improvement in soil health indicators, including:
- Increase in soil organic matter (%)
- Increase in soil moisture retention capacity
- Improvement in soil nutrient status (N, P, K)
- Improved aggregate stability and soil structure
- Percentage of farmers adopting integrated soil and water conservation (terracing, zai pits, swales).
- Increase in acreage under diversified and climate-resilient farm systems.
- Increase in household-level adoption of post-harvest management technologies (grain banks, hermetic bags).
- Percentage of women, youth, and PADs participating in CSA and Resilience Design training cycles.
- Increase in on-farm and community tree cover (%) over baseline.
- Increase in mangrove seedling survival rates and total acreage restored.
- Reduction in incidents of illegal harvesting, charcoal burning, and mangrove cutting.
- Increase in households adopting windbreak and shelterbelt systems.
- Increased community participation in environmental conservation and sustainable harvesting practices.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Percentage increase in household income from diversified livelihoods (eco-tourism, poultry, beekeeping, seaweed value addition).
- Increase in the number of households with multiple income streams across seasons.
- Percentage of women, youth, and PADs accessing new livelihood opportunities.
- Increased financial literacy and business skills among trained groups.
- Increase in households benefiting from school feeding, cash transfers, or food-for-work schemes.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Reduction in households exposed to flood-prone or erosion hotspots.
- Percentage of villages receiving timely Early Warning System (EWS) alerts.
- Community response rate to EWS alerts (attendance at assembly points, evacuation compliance).
- Reduction in flood-related damages to dwellings and access roads.
- Level of preparedness and readiness of community emergency response teams (measured via drills).
- Increased adoption of wind-resilient building practices in households.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Reduction in frequency of human–wildlife conflict incidents (crocodile attacks, baboon raids).
- Increase in the number of households protected by solar-powered fences.
- Increase in community capacity to report and respond to HWC events (via KWS hotline).
- Increased community compliance with safe-behaviour protocols in HWC hotspots.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Increase in the number and proportion of women, youth, and PADs participating in ward-level governance and adaptation committees.
- Reduction in drudgery for women, youth, and PADs due to improved water access and resilient infrastructure.
- Percentage increase in women's and PADs' access to credit, extension services, and productive resources.
- Increase in women's ownership of productive assets (livestock, seaweed farms, equipment).
- Improvement in community knowledge and awareness of GBV risk mitigation during climate hazards.
- Increased utilisation of GBV response services (safe shelters, first-aid kits, referral pathways).

Impact (Goal) Indicators:

1. WATER SECURITY AND MANAGEMENT

- Percentage of the ward population with year-round access to safe and reliable water.
- Increase in average litres of water available per person per day across villages.
- Long-term reduction in water scarcity–related vulnerabilities for women, youth, and PADs.
- Sustained decrease in waterborne diseases due to improved water quality and availability.
- Community-wide resilience to prolonged drought through diversified water sources and storage systems.

2. FOOD SECURITY, ECOSYSTEM RESTORATION AND LIVELIHOODS

- Sustained increase in crop yields above baseline across 3–5 year periods.
- Long-term reduction in seasonal yield fluctuations linked to droughts, floods, and pest outbreaks.
- Measurable improvements in soil health at landscape level, including:
- Higher soil organic carbon (SOC)
- Increased vegetation cover (NDVI index)
- Reduced erosion and sediment loss
- Enhanced household food security across all seasons.
- Increased biodiversity within integrated farming systems (trees, fodder, pollinator presence).
- Increases in forest cover, on-farm trees, and mangrove extent compared to baseline.
- Restoration of degraded landscapes evidenced by improvements in NDVI and tree density.
- Reduction in illegal harvesting, mangrove cutting, and charcoal burning.
- Enhanced ecosystem services, including shoreline protection, soil stability, and microclimate regulation.
- Long-term improvement in habitat quality for biodiversity and fisheries productivity.

3. DIVERSIFIED LIVELIHOODS AND SOCIAL PROTECTION

- Sustained increases in household incomes from diversified livelihoods (beekeeping, eco-tourism, seaweed, poultry).
- Improved household resilience to income shocks from climate hazards.
- Decreased poverty levels due to stable and multi-stream livelihoods.

- Increased school attendance and nutritional outcomes for children due to social protection schemes.
- Greater financial stability among women, youth, and PAD households.

4. COMMUNITY INFRASTRUCTURE AND DISASTER MANAGEMENT

- Significant reduction in climate-related damage to homes, roads, and community facilities.
- Communities demonstrating effective disaster preparedness and response during floods and storms.
- Long-term reduction in displacement, injuries, and asset losses during climate hazards.
- Increased structural safety and durability of settlements in strong-wind and flood-prone areas.

5. CONFLICTS RESOLUTION (HUMAN WILDLIFE CONFLICTS)

- Long-term reduction in human injuries, livestock losses, and crop destruction from wildlife.
- Improved coexistence between communities and wildlife due to effective mitigation (fencing, reporting systems).
- Reduced economic losses from HWC-related disruptions.
- Increased safety and mobility for women, children, and PADs in high-risk zones.

6. GENDER AND SOCIAL INCLUSION TO ENHANCE ADAPTATION ACTIONS

- Percentage increase in women's, youth's, and PADs' participation in climate governance and local decision-making structures.
- Strengthened continuity of SRHR, maternal health, and GBV response services during climate hazards.
- Increased leadership roles held by women, youth, and PADs in adaptation implementation.
- Growth in the number of marginalised and vulnerable groups leading or influencing climate adaptation actions.
- Reduction in GBV risks during droughts, floods, and strong winds due to anticipatory action systems.

For each indicator, we will establish baselines (where we are starting from, using historical data or initial surveys) and set targets for 1-year, 3-year, 5-year intervals. For example: evaluate the current number of women equipped in seaweed farming and the projected number in 5 years period. This will be evaluated to project the rate at which adaptation actions are being effective and assimilated into the society.

Data for these indicators will be obtained from various sources: routine project monitoring (engineer reports for infrastructure, training attendance sheets), surveys and focus groups for outcomes (e.g., seasonal household surveys to gauge water access, yields, or time saved), and existing data (health clinic records for disease incidence, acreage of mangroves restored).

All indicators will be disaggregated by gender where applicable to monitor differential benefits. Also, where feasible, by village to see if some areas lag. Measuring adaptation success is complex because it's about avoided losses and capacity built, but these indicators together provide a robust

picture. The Ward Committee will use a simple M&E matrix listing each indicator, baseline, target, data source, frequency of data collection, and responsible person.

6.4 Adaptation Reporting and Learning

Reporting: Pongwe Kikoneni Ward will establish a structured and transparent reporting system to facilitate timely information flow among all stakeholders, including community members, county authorities, and development partners. The Ward Climate Change Planning Committee (WCCPC) will prepare quarterly progress reports detailing implemented activities, achieved outputs, encountered challenges, and corresponding expenditures. These reports will be presented in public barazas to promote accountability and enable community feedback and subsequently submitted to the County Climate Change Unit for integration into county-level monitoring processes. In addition, an annual Ward Climate Adaptation Report will be developed to comprehensively assess progress against planned targets and performance indicators. This annual report will align with the requirements of the Climate Change Act (2023), ensuring that Pongwe Kikoneni Ward's results and lessons contribute to the counties and nation's broader climate change monitoring, reporting, and verification (MRV) systems.

The annual report will highlight case studies or success stories (e.g., "Farmer groups in Mkwiro doubled their harvest despite the 2025 drought by using new techniques a success attributed to our adaptation training"). It will also not shy away from discussing failures or shortfalls, as these provide learning.

At community level, participatory monitoring meetings will be held. For instance, after each rainy season, community members can gather to discuss amount mangrove seedlings planted. Did the new seeds perform better in this season? Such qualitative feedback will be recorded and included in reports.

The Ward Administrator and WCCPC Chair will also report in county forums (e.g., County Adaptation Planning Committee meetings) on progress, championing their ward and possibly leveraging more support by showcasing results.

Learning and Reflective Review: Adaptation is a long journey with uncertainty, so fostering a culture of learning is vital. Twice during the implementation period (e.g., mid-term in year 3, and end-term in year 5), the ward will conduct an outcome evaluation and learning workshop. This involves bringing together community representatives, implementing partners, and possibly external evaluators to systematically review what outcomes have been achieved and why. Key questions: Which interventions are most effective? Are there unintended impacts? What obstacles are reducing effectiveness (maybe maintenance issues, or cultural barriers to adoption)? For example, maybe the uptake of an improved cookstove program is low – through reflection the group might learn women prefer a different design, or fuel is an issue, leading to adjusting that action. These learning workshops will use tools like community scorecards, ranking of interventions by community benefit, and problem tree analysis for any persistent challenges (similar to the PCVA style but now focusing on project performance). Lessons learned will be documented and used to refine the adaptation plan. The Ward Climate Committee is empowered to tweak or reprioritize actions based on evidence. For instance, if a particular crop variety fails, they will note it and try an alternative next season.

Knowledge Sharing: Pongwe Kikoneni Ward will also engage in horizontal learning i.e., exchanging experiences with other wards or counties. If something worked really well here (say the VSLA

approach to fund water tanks), we can host visitors from nearby wards to showcase it, and similarly we can learn from others (maybe a neighbouring ward piloted a solar irrigation scheme that we could replicate). The county climate change unit can facilitate such inter-ward learning exchange visits or meetings.

Feedback to Policy: Through reporting channels, the ward's experiences may also inform higher level policies. For example, if the ward finds that insurance uptake is low because of certain policy constraints, this feedback can go to the national level via NDMA or the National Climate Change Council to refine such programs. The 2023 Act calls for progressive improvement of climate response, our ward's data and stories become part of that national learning loop.

Sustainability of M&E: The Ward Committee will assign an M&E focal person (or sub-committee) to drive these activities. But monitoring won't be just "external evaluation"; it will be integrated into community life e.g., water committees will track water usage and downtime of boreholes as part of their routine, farmer groups will collectively observe and record crop performance. By embedding monitoring tasks into existing structures, we ensure continuity even if specific projects end.

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Annexure

1. Priority tree and fruit species

Scientific Name	Common Name
Azadirachta indica	Neem
Afzelia quanzensis	Pod Mahogany / Mkongo
Casuarina equisetifolia	Whistling Pine / Casuarina
Milicia excelsa	African Teak / Iroko / Mvule
Gmelina arborea	Gmelina / White Teak
Tectona grandis	Teak
Grevillea robusta	Silk Oak / Grevillea
Senna spp.	Senna / Cassia
Moringa oleifera	Moringa / Drumstick Tree
Azalea spp.	Azalea
Citrus sinensis	Sweet Orange
Anacardium occidentale	Cashew
Mangifera indica	Mango
Cocos nucifera	Coconut
Carica papaya	Papaya